

2.4.1 Act Utilitarianism

We noted previously that utilitarians look at the expected outcomes or consequences of an act to determine whether or not that act is morally permissible. However, some critics point out that because utilitarianism tends to focus simply on the roles that individual acts and policies play in producing the overall social good (the greatest good for the greatest number), it is conceptually flawed. Consider a hypothetical scenario in which a new controversial policy is being debated.

► SCENARIO 2-3: A Controversial Policy in Newmerica

A policy is under consideration in a legislative body in the nation of Newmerica, where 1 % of the population would be forced to work as slave laborers in a manufacturing facility to produce inexpensive computer chips. Proponents of this policy argue that, if enacted into law, it would result in lower prices for electronic devices for consumers in Newmerica. They argue that it would also likely result in more overall happiness for the nation's citizens because the remaining 99 % of the population, who are not enslaved, would be able to purchase electronic devices and other computer-based products at a much lower price. Hence, 99 % of Newmerica's population benefit at the expense of the remaining 1 %. This policy clearly seems consistent with the principle of producing the greatest good for the greatest number of Newmerica's population, but should it be enacted into law? ■

The above scenario illustrates a major flaw in at least one version of utilitarianism, namely, *act utilitarianism*. According to act utilitarians,

An act, X, is morally permissible if the consequences produced by doing X result in the greatest good for the greatest number of persons affected by Act X.

All things being equal, actions that produce the greatest good (happiness) for the greatest number of people seem desirable. However, policies and practices based solely on this principle can also have significant negative implications for those who are not in the majority (i.e., the greatest number). Consider the plight of the unfortunate few who are enslaved in the computer chip-processing plant in the above scenario. Because of the possibility that such bizarre cases could occur, some critics who embrace the goals of utilitarianism in general reject act utilitarianism.

Critics who reject the emphasis on the consequences of individual acts point out that in our day-to-day activities, we tend not to deliberate on each individual action as if that action were unique. Rather, we are inclined to deliberate on the basis of certain principles or general rules that guide our behavior. For example, consider some principles that may guide your behavior as a consumer. Each time that you enter a computer store, do you ask yourself, "Shall I steal this particular software game in this particular store at this particular time?" Or have you already formulated certain general principles that guide your individual actions, such as, "It is never morally permissible to steal"? In the latter case, you are operating at the level of a rule or principle rather than deliberating at the level of individual actions.

2.4.2 Rule Utilitarianism

Some utilitarians argue that the consequences that result from following *rules* or principles, not the consequences of individual acts, ultimately matter in determining whether or not a certain practice is morally permissible. This version of utilitarian theory, called *rule utilitarianism*, can be formulated in the following way:

An act, X, is morally permissible if the consequences of following the general rule, Y, of which act X is an instance, would bring about the greatest good for the greatest number.

Note that here we are looking at the consequences that result from following certain kinds of rules as opposed to consequences resulting from performing individual acts. Rule utilitarianism eliminates as morally permissible those cases in which 1% of the population is enslaved so that the majority (the remaining 99%) can prosper. Rule utilitarians believe that policies that permit the unjust exploitation of the minority by the majority will also likely have overall negative social consequences and thus will not be consistent with the principal criterion of utilitarian ethical theory.

How would a rule utilitarian reason in the case of the trolley accident involving five victims (described in the preceding section), each of whom needs an organ transplant to survive? For an (extreme) act utilitarian, the decision might be quite simple: Remove the five organs from the one healthy patient (even though he will die) so that five humans who otherwise would die could now live. But would a rule utilitarian see this particular action as justifiable on rule-utilitarian grounds—that is, could it form the basis for an acceptable policy (in general) for hospitals and medical facilities?

Imagine a society in which it is possible for a person to report to a medical center for a routine physical exam only to discover that his or her vital organs could be removed in order to save a greater number of people. Would anyone be willing to submit to a routine physical exam in such a society? Of course, a rule utilitarian could easily reject such a practice on the following grounds: Policies that can intentionally cause the death of an innocent individual ought not to be allowed, even if the net result of following such policies meant that more human lives would be saved. For one thing, such a policy would seem unfair to all who are adversely affected. But perhaps more importantly from a rule utilitarian's perspective, adopting such a policy would not result in the greatest good for society.

Rule utilitarianism would seem to be a more plausible ethical theory than act utilitarianism. However, some critics reject all versions of utilitarianism because they believe that no matter how this theory is expressed, utilitarianism is fundamentally flawed. These critics tend to attack one or both of the following aspects of utilitarian theory:

- I. Morality is basically tied to the production of happiness or pleasure.
- II. Morality can ultimately be decided by consequences (of either acts or policies).

Critics of utilitarianism argue that morality can be grounded neither in consequences nor in happiness. Hence, they argue that some alternative criterion or standard is needed.

► 2.5 DUTY-BASED ETHICAL THEORIES

Immanuel Kant (1724–1804) argued that morality must ultimately be grounded in the concept of duty, or obligations that humans have to one another, and never in the consequences of human actions. As such, morality has nothing to do with the promotion of happiness or the achievement of desirable consequences. Thus, Kant rejects utilitarianism in particular and all consequentialist ethical theories in general. He points out that, in some instances, performing our duties may result in our being unhappy and may not necessarily lead to consequences that are considered desirable. Theories in which the notion of duty, or obligation, serves as the foundation for morality are called *deontological* theories because they derive their meaning from the Greek root *deon*, which means duty. How can a deontological theory avoid the problems that plague consequentialist theories such as utilitarianism? Kant provides two answers to this question, one based on our nature as rational creatures and the other based on the notion that human beings are ends in themselves. We briefly consider each of Kant's arguments.

What does Kant mean when he says that humans have a rational nature? Kant argues that what separates us from other kinds of creatures and what binds us morally is our rational capacity. Unlike animals who may be motivated only by sensory pleasure, humans have the ability to reason and deliberate. So Kant reasons that if our primary nature were such that we

merely seek happiness or pleasure, as utilitarians suggest, then we would not be distinguishable from other creatures in morally relevant ways. But because we have a rational capacity, we are able to reflect upon situations and make moral choices in a way that other kinds of (non-rational) creatures cannot. Kant argues that our rational nature reveals to us that we have certain duties or obligations to each other as “rational beings” in a moral community.

We can next examine Kant’s second argument, which concerns the roles of human beings as ends in themselves. We have seen that in focusing on criteria involving the happiness of the majority, utilitarians allow, even if unintentionally, that the interests and well-being of some humans can be sacrificed for the ends of the greatest number. Kant argues that a genuinely moral system would never permit some humans to be treated simply as means to the ends of others. He also believes that if we are willing to use a standard based on consequences (such as social utility) to ground our moral system, then that system will ultimately fail to be a moral system. Kant argues that each individual, regardless of his or her wealth, intelligence, privilege, or circumstance, has the same moral worth. From this, Kant infers that each individual is an end in himself or herself and, therefore, should never be treated merely as a means to some end. Thus, we have a duty to treat fellow humans as ends.

2.5.1 Rule Deontology

Is there a rule or principle that can be used in an objective and impartial way to determine the basis for our moral obligations? For Kant, there is such a standard or objective test which can be formulated in a principle that he calls the *categorical imperative*. Kant’s imperative has a number of variations, and we will briefly examine two of them. One variation of his imperative directs us to:

Act always on that maxim or principle (or rule) that ensures that all individuals will be treated as ends-in-themselves and never merely as a means to an end.

Another variation of the categorical imperative can be expressed in the following way:

Act always on that maxim or principle (or rule) that can be universally binding, without exception, for all human beings.⁹

Kant believed that if everyone followed the categorical imperative, we would have a genuinely moral system. It would be a system based on two essential principles: universality and impartiality. In such a system, every individual would be treated fairly since the same rules would apply universally to all persons. And because Kant’s imperative observes the principle of impartiality, it does not allow for one individual or group to be privileged or favored over another. In other words, if it is morally wrong for you to engage in a certain action, then it is also morally wrong for all persons like you—that is, all rational creatures (or moral agents)—to engage in that action. And if you are obligated to perform a certain action, then every moral agent is likewise obligated to perform that action. To illustrate Kant’s points about the role that universal principles play in a moral system, consider the following scenario.

► SCENARIO 2-4: Making an Exception for Oneself

Bill, a student at Technical University, approaches his philosophy instructor, Professor Kanting, after class one day to turn in a paper that is past due. Professor Kanting informs Bill that since the paper is late, he is not sure that he will accept it. But Bill replies to Kanting in a way that suggests that he is actually doing his professor a favor by turning in the paper late. Bill reasons that if he had turned in the paper when it was due, Kanting would have been swamped with papers. Now, however, Kanting will be able to read Bill’s paper in a much more leisurely manner, without having the stress of so many papers to grade at once. Kanting then tells Bill that he appreciates his concern about his professor’s well-being, but he asks Bill to reflect a bit on his rationale in this incident. Specifically, Kanting asks Bill to imagine a case in which all of the students in his class, fearing that their professor would be overwhelmed with papers arriving at the same time, decided to turn their papers in one week late. ■

On deontological grounds, Bill can only make an exception for himself if everyone else (in this case, every other student in Bill's class) had the right to make exceptions for himself or herself as well. But if everyone did that, then what would happen to the very notion of following rules in a society? Kant believed that if everyone decided that he or she could make an exception for himself or herself whenever it was convenient to do so, we couldn't even have practices such as promise keeping and truth telling. For those practices to work, they must be universalizable (i.e., apply to all persons equally) and impartial. When we make exceptions for ourselves, we violate the principle of impartiality, and we treat others as means to our ends.

In Kant's deontological scheme, we do not consider the potential consequences of a certain action or of a certain rule to determine whether that act is morally permissible. Rather, the objective rule to be followed—that is, the litmus test for determining when an action will have moral worth—is whether the act complies with the categorical imperative.

For a deontologist such as Kant, enslaving humans would always be immoral, regardless of whether the practice of having slaves might result in greater social utility for the majority (e.g., being able to purchase consumer products at a lower price) than the practice of not allowing slavery. The practice of slavery is immoral, not because it might have negative social consequences in the long term but because (i) it allows some humans to be used only as a means to an end and (ii) a practice such as slavery could not be consistently applied in an objective, impartial, and universally binding way.

Kant would ask, for example, whether we could consistently impose a universal maxim that would allow slavery. He believed that we could not consistently (in a logically coherent sense) formulate such a principle that would apply to all humans, unless we also were willing to be subject to slavery. If we allow for the practice that some individuals can be enslaved but not others, then we would be allowing for exceptions to the moral rule. We would also allow some individuals to be used merely as a means to the ends of others rather than having a system in which all humans are treated as ends in themselves.

Although Kant's version of deontological ethics avoids many of the difficulties of utilitarianism, it, too, has been criticized as an inadequate ethical theory. Critics point out, for example, that even if Kant's categorical imperative provides us with the ultimate test for determining when some particular course of action is our duty, it will not help us in cases where we have two or more conflicting duties. Consider that, in Kant's system, we have duties both to keep promises and tell the truth. Thus, acts such as telling a lie or breaking a promise can never be morally permissible. However, Kant's critics point out that sometimes we encounter situations in which we are required *either* to tell the truth and break a promise *or* to keep a promise and tell a lie. In these cases, we encounter genuine moral dilemmas. Kant's deontological theory does not provide us with a mechanism for resolving such conflicts.

2.5.2 Act Deontology

Although Kant's version of deontology has at least one significant flaw, some philosophers believe that a deontological account of morality is nonetheless the correct kind of ethical theory. They also believe that a deontological ethical theory can be formulated in a way that avoids the charges of Kant's critics. One attempt at reformulating this theory was made by David Ross (1877–1971). Ross rejected utilitarianism for many of the same reasons that Kant did. However, Ross also believed that Kant's version of deontology is not fully adequate.

Ross argues that when two or more moral duties clash, we have to look at individual situations in order to determine which duty will override another. Like act utilitarians, then Ross stresses the importance of analyzing individual situations to determine the morally appropriate course of action to take. Unlike utilitarians, however, Ross believes that we must not consider the consequences of those actions in deliberating over which course of action morally trumps, or outweighs, another. Like Kant, Ross believes that the notion of duty is the ultimate

criterion for determining morality. But unlike Kant, Ross does not believe that blind adherence to certain maxims or rules can work in every case for determining which duties we must ultimately carry out.

Ross believes that we have certain *prima facie* (or self-evident) *duties*, which, all things being equal, we must follow. He provides a list of *prima facie* duties such as honesty, benevolence, justice, and so forth. For example, each of us has a *prima facie* duty not to lie and a *prima facie* duty to keep a promise. And if there are no conflicts in a given situation, then each *prima facie* duty is also what he calls an *actual duty*. But how are we to determine what our actual duty is in situations where two or more *prima facie* duties conflict with one another? Ross believes that our ability to determine what our actual duty will be in a particular situation is made possible through a process of “rational intuitionism” (similar to the one used in mathematics).¹⁰

We saw that for Kant, every *prima facie* duty is, in effect, an absolute duty because it applies to every human being without exception. We also saw that Kant’s scheme does not provide a procedure for deciding what we should do when two or more duties conflict. However, Ross believes that we can determine what our overriding duty is in such situations by using a deliberative process that requires two steps:

- a. Reflect on the competing *prima facie* duties.
- b. Weigh the evidence at hand to determine which course of action would be required in a particular circumstance.

The following scenario illustrates how Ross’s procedure can be carried out.

► SCENARIO 2-5: A Dilemma Involving Conflicting Duties

You have promised to meet a classmate this evening at 7:00 in the college library to study together for a midterm exam for a computer science course you are taking. While driving in your car to the library, you receive a call on your cell phone informing you that your grandmother has been taken to the hospital and that you should go immediately to the hospital. You consider calling your classmate from your car, but you realize that you don’t have his phone number. You also realize that you don’t have time to try to reach your classmate by e-mail. What should you do in this case? ■

All things being equal, you have a moral obligation to keep your promise to your friend. You also have a moral obligation to visit your grandmother in the hospital. On both counts, Kant and Ross are in agreement. But what should we do when the two obligations conflict? For a rule deontologist like Kant, the answer is unclear as to what you should do in this scenario, since you have two absolute duties. For Ross, however, the following procedure for deliberation is used. You would have to weigh between the two *prima facie* duties in question to determine which will be your actual duty in this particular circumstance. In weighing between the two conflicting duties, your actual duty in this situation would be to visit your grandmother, which means, of course, that you would have to break your promise to your friend. However, in a different kind of situation involving a conflict of the same two duties, your actual duty might be to keep the promise made to your friend and not visit your grandmother in the hospital.

Notice that in cases of weighing between conflicting duties, Ross places the emphasis of deliberation on certain aspects of the particular situation or context rather than on mere deliberation about the general rules themselves. Unlike utilitarians, however, Ross does not appeal to the consequences of either actions or rules in determining whether a particular course of action is morally acceptable. For one thing, Ross argues that he would have to be omniscient to know what consequences would result from his actions. So, like all deontologists, Ross rejects the criteria of consequences as a viable one for resolving ethical dilemmas.

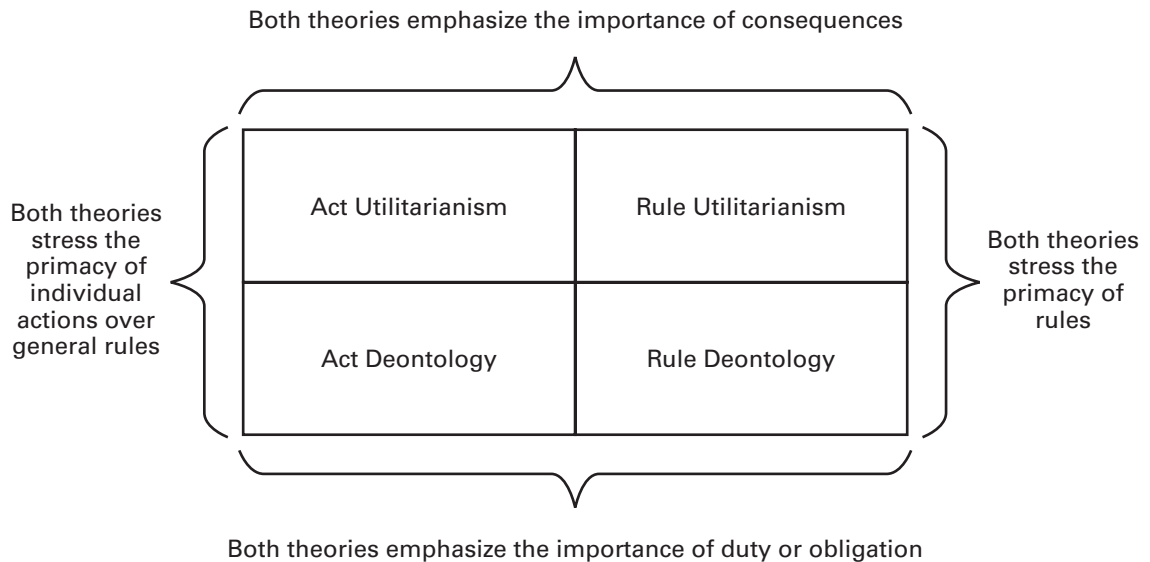


Figure 2-3 Acts vs. rules and consequences vs. duties.

One difficulty for Ross's position is that, as noted earlier, it uses a process called "rational intuitionism." Appealing to the intuitive process used in mathematics to justify certain basic mathematical concepts and axioms, Ross believes that the same process can be used in morality. However, his position on moral intuitionism is controversial and has not been widely accepted by contemporary ethicists. And since intuitionism is an important component in Ross's theory of act deontology, many ethicists who otherwise might be inclined to adopt Ross's theory have been skeptical of it. Nevertheless, variations of that theory have been adopted by contemporary deontologists.

Figure 2-3 summarizes key features that differentiate act and rule utilitarianism and act and rule deontology.

► 2.6 CONTRACT-BASED ETHICAL THEORIES

During the past two centuries, consequence-based and duty-based ethical theories have tended to receive the most attention from philosophers and ethicists. However, other kinds of ethical theories, such as those that emphasize criteria involving social contracts and individual rights, have recently begun to receive some serious attention as well.

From the perspective of some social contract theories, a moral system comes into being by virtue of certain contractual agreements between individuals. One of the earliest formal versions of a contract-based ethical theory can be found in the writings of Thomas Hobbes (1588–1679). In his classic work *Leviathan*, Hobbes describes an original "pre-moral" state that he calls the "state of nature." It is pre-moral because there are no moral (or legal) rules yet in existence. In this state, each individual is free to act in ways that satisfy his or her own natural desires. According to Hobbes, our natural (or physical) constitution is such that in the state of nature, we act in ways that will enable us to satisfy our desires (or appetites) and to avoid what Hobbes calls our "aversions." While there is a sense of freedom in this natural state, the condition of our day-to-day existence is hardly ideal. In this state, each person must continually fend for herself, and, as a result, each must also avoid the constant threats of others, who are inclined to pursue their own interests and desires.

Hobbes describes this state of nature as one in which life is “solitary, poor, nasty, brutish, and short.” Because we are rational creatures and because we see that it would be in our best interests to band together, Hobbes notes that we eventually establish a formal legal code. In doing this, Hobbes believes that we are willing to surrender some of our “absolute” freedoms to a sovereign. In return, we receive many benefits, including a system of rules and laws that are designed and enforced to protect individuals from being harmed by other members of the system.

One virtue of the social contract model of ethics is that it gives us a motivation for being moral. We see that it is in our individual self-interest to develop a moral system with rules. This type of motivation for establishing a moral system is conspicuously absent in both the utilitarian and deontological theories.¹¹ So a contract-based ethical theory might seem to have one advantage over them.

2.6.1 Some Criticisms of Contract-Based Theories

Some critics, such as Pojman (2006), point out that contract-based theories provide the foundation for only a minimalist morality. They are minimalist in the sense that we are obligated to behave morally only where an explicit or formal contract exists. So if I have no express contract with you or if a country such as the United States has no explicit contract with a developing nation, there is no moral obligation for me to help you or for the United States to come to the aid of that developing nation. Of course, we can think of many situations involving morality where there are no express contracts or explicit laws describing our obligations to each other. Yet we also tend to believe that in at least some of these cases, we are morally obligated to help others when it is in our power to do so.

Consider the case of Kitty Genovese who was murdered outside her apartment building in Queens, New York, in 1964, as 38 neighbors in her apartment building watched. During the incident, none of Genovese’s neighbors came to her rescue or called the police. When interviewed after the fact, some of her neighbors responded that they did nothing wrong.¹² In one sense, they were correct, since there was no explicit law requiring that they do anything at all. So technically, these neighbors were correct, at least from a legal perspective. But we can certainly ask whether her neighbors had a moral obligation to do something rather than simply be indifferent. It is in this sense, then, that social contract theory can be seen as being minimalist and legalistic and not a robust theory of morality.

Another way to think about minimalist morality is to think of the difference between two principles: (i) doing no harm and (ii) doing good. A minimalist morality would insist merely that we do not harm others. As such, it does not require that we come to the aid of others. But is that an adequate view of morality? Should we accept such a moral system as complete? If you happen to see a child drowning in water that is four feet deep and it is in your power to rescue the child, are you not morally obligated to do so? Are you under no moral obligation to assist simply because you may have no explicit legal contract requiring you to rescue that particular child?

According to a minimalist account of morality, you are not *required* to make any effort to save the child. All that is required is that you not actively harm the child (or anyone else). But some argue that a moral system demands much more of us than simply doing no harm. That is, it may also obligate us to do good when it is in our power to do so. According to the latter view of morality, then, if we could rescue the child without any significant inconvenience to ourselves, we would be morally obligated to do so (even if we have no explicit contract).

2.6.2 Rights-Based Contract Theories

Closely associated with social contract ethical theories are rights-based theories of morality. Some philosophers have argued that independent of whether individuals happen to have any legal rights, all humans have certain moral rights or natural rights. Philosophers such as Thomas Aquinas (1225–1274), as well as several of the Founders of the United States, believed that humans possess some natural rights. In the Declaration of Independence, for example, Thomas Jefferson asserted that all humans are entitled to life, liberty, and the pursuit of happiness because these rights are “inalienable” and “self-evident.”

Of course, it is one thing for philosophers and legal scholars to assert that humans are endowed with natural or moral rights; and it is something altogether different to ensure that such rights are guaranteed and protected by the state—hence, the need for explicit legal rights identified in a governing charter or constitution. Legal rights are grounded in “positive law,” or civil law, whereas moral rights or natural rights are not. However, some argue that moral rights are derived from natural law; and because of this, they further argue that these rights exist independently of any legal rights that might happen to be granted to citizens of a certain nation via that nation’s system of positive laws.¹³

Philosophers and legal scholars often differentiate between two kinds of legal rights: *positive rights* and *negative rights*. Having a negative right to something simply means that one has the right not to be interfered with in carrying out the privileges associated with that right. For example, your right to vote and your right to own a computer are both negative rights. They are negative rights in the sense that as a holder of those rights, you have the right (and the expectation) not to be interfered with in exercising your right to go to polls to cast your vote in a particular election or your right to purchase a computer. However, as the holder of a negative right, you cannot demand (or even expect) that others must either physically transport you to the voting polls if you are unable to travel there on your own or provide you with a computer if you cannot afford to purchase one.

Positive rights, it turns out, are very rare. And since those rights tend to be far more controversial than negative rights, philosophers and legal scholars have had a much more difficult time justifying them. In the United States, one’s right to receive an education (through the twelfth grade of high school) is a positive right. All American citizens are entitled to such an education; thus, they must be provided with a free public education through the 12th grade. An interesting question, for our purposes, is, what would happen in the event that our formal education process requires that each student own a computer and that he or she has access at home to the Internet? In that case, would students also have to be provided with a home computer and free Internet access? (We take up the question of universal access issues and the “digital divide” in Chapter 10.)

Some would argue that access to adequate healthcare should also be a positive right as well, because they believe healthcare is something that citizens have a right to be provided (even if they cannot afford to pay for it). In Canada as well as in many European countries, universal healthcare is viewed as a positive right. In the United States, however, this view is still being debated. Although the Patient Protection and Affordable Care Act, also informally known as “Obamacare,” was enacted into law in March 2010, it has since come under severe criticism and serious challenges from opponents. Depending on the outcome of the fall 2016 presidential elections, this law could be repealed, in which case healthcare in the United States would not be a positive right.

Discussion about the nature of rights can be both confusing and controversial. In the United States, many conservative political and religious leaders believe that in recent years, far too much emphasis has been placed on individual rights. As a result, they believe that we have not paid enough attention to corresponding responsibilities that individuals also

have by virtue of possessing those rights. However, we will not pursue that line of controversy here.

► 2.7 CHARACTER-BASED ETHICAL THEORIES

A fourth type of ethical theory that must be considered, especially in light of the recent attention it has received, is *virtue ethics* (also sometimes described as “character ethics”). This ethical theory ignores the special roles that consequences, duties, and social contracts play in moral systems, especially with respect to determining the appropriate standard for evaluating moral behavior. Rather, it focuses on criteria having to do with the character development of individuals and their acquisition of good character traits from the kinds of habits they develop. The fundamental principles of virtue ethics were introduced in the writings of Plato and Aristotle nearly 2,500 years ago. In more recent times, virtue ethics has gained respect among ethicists as a viable contemporary ethical theory, in part, through the influential work of Philippa Foot, Alasdair MacIntyre, and others.¹⁴

2.7.1 Being a Moral Person vs. Following Moral Rules

Aristotle believed that ethics was something not merely to be studied, but rather to be lived or practiced. In fact, Aristotle (384–322 BC) thought of ethics as a “practical science,” like politics. To become an ethical person, in Aristotle’s view, one is required to do more than simply memorize and deliberate on certain kinds of rules. What is also needed, Aristotle argued, is that people develop certain *virtues*. The Greek word for virtue is *arete*, which means excellence. Aristotle believed that to be a moral person, one had to acquire the right virtues (strengths or excellences). Through the proper training and acquisition of good habits and character traits, Aristotle believed that one could achieve moral virtues such as temperance and courage that are needed to “live well.”

Because virtue ethics focuses primarily on character development and moral education, it does not need to rely on a system of formal rules. Consider that both utilitarians and deontologists depend on having a system of rules when they ask a question such as, What should we do in such and such a case or situation? For utilitarians, the answer could be found by measuring the anticipated outcomes of following a general rule or principle. And for deontologists, the answer can be determined by using a formal rule such as the categorical imperative as a principle for determining which duties you have. For contract theorists, questions involving moral obligation ultimately rest on the principle or rule: What is the nature of my contract, if any, in this particular situation? Virtue ethicists take a very different tack. Instead of asking, “What should I *do* in such and such a situation?” a virtue ethicist asks, *What kind of person should I be?*—hence, the emphasis on *being a moral person* and not simply on understanding what moral rules are and how they apply in certain situations. Whereas deontological and utilitarian theories are action oriented and rule oriented, virtue ethics is “agent-oriented” because it is centered on the moral development and character of the agent himself/herself.

Virtue ethicists believe that a moral person is one who is necessarily disposed to do the right thing. They correctly point out that when we engage in routine acts in our daily lives, including many of our nonnormative actions, we do not deliberate by asking ourselves, What ought I to do in such and such a case? In our earlier criticism of act utilitarianism, we considered a situation in which an individual would be required to deliberate over whether or not to steal an item each time he or she entered a store. A virtue ethicist would point out that if that person had developed the right kind of moral character (through the acquisition of the “correct” moral habits), he or she would not be in a position that required such deliberation. That is, the moral person is already disposed not to steal items from stores (or from fellow human beings) because of the kinds of character traits that he or she has previously developed. And in the example involving the drowning child, considered in our criticism of contract-based ethical theory, a virtue ethicist would also

likely point out that a moral person would not have to deliberate. Regardless of whether someone had an explicit legal contract to help rescue the child, the virtue ethicist would point out that a moral person is predisposed to attempt to rescue the child if it were in his or her power to do so.

2.7.2 Acquiring the “Correct” Habits

Consider the following illustration of a disposition to behave in a certain way. When you woke up this morning and began to prepare for your day’s events, did you ask yourself the question, Should I brush my teeth today? Most likely, this question never crossed your mind. Why not? The answer, of course, is that you have already developed certain habits such that you are disposed to brush your teeth in the morning without having to question it or even think about it. Of course, the act of brushing one’s teeth is not an act that has any moral significance. But it is the process of character formation, especially the development of moral habits, that is crucial to becoming a fully moral person, from the perspective of virtue ethics.

As noted earlier, Aristotle believed that ethics was something to be lived and practiced, not simply studied. Thus, some philosophers and ethicists believe that to teach ethics, one must first be an ethical person. The teacher who instructs students on the virtues but who himself lacks them would be a poor model for aspiring students. Moor (2002) suggests that virtue ethics instruction is the “first level” in teaching (computer) ethics. He believes that building habits of character such as kindness, truthfulness, honesty, trustworthiness, helpfulness, generosity, and justice is an important prerequisite in preparing for the second level of instruction. Once students have mastered the virtues, they can then move to the second level where they learn the established rules of a moral system.

Some instructors have argued that their students are better able to relate to classroom examples that involve virtue ethics than to those illustrating other traditional theories. For this reason, Grodzinsky (2001) has suggested that aspects of virtue ethics should be incorporated into the ethics training for computing professionals. Grodzinsky believes that aspiring computer professionals who wish to develop an appropriate level of sensitivity to ethical aspects of their profession often find the principles of virtue ethics far more useful than the kinds of rigid rules required in ethical theories such as utilitarianism and deontology. She notes that action-guiding rules associated with utilitarian and deontological theories often tend to be perceived by students as too abstract and formal. On the contrary, however, many of those students are able to grasp what it means to develop certain character traits and thus become (or be) a certain kind of person.

It would seem that the reemergence of virtue ethics, despite the fact that its origins can be traced back to classical Greece, has provided ethicists with some fresh insights. However, we should also note that virtue ethics is not without its critics. One of the chief drawbacks of taking virtue ethics as a complete theory of ethics is that it neither helps resolve conflicts that can arise among the competing virtues nor encourages examination of consequences. Some critics point out that a virtue- or character-based ethics would seem to have a better chance of taking hold in a society that is homogeneous rather than in one that is heterogeneous or pluralistic. The ancient Greek society could be considered fairly homogeneous in the sense that the world that Plato and Aristotle inhabited included a consensus as to what the ideal values, including the moral education of the young, were. In contemporary America, which is much more heterogeneous than classical Greek society, we have a diversity of views about which ideals and values are most important.

It is also worth pointing out that character-based ethical systems would most likely flourish in cultures where the emphasis placed on community life is stronger than that accorded to the role of individuals themselves. Beginning with the Enlightenment period in the West in the seventeenth and eighteenth centuries, considerable emphasis has been placed on the importance of individual autonomy and individual rights. As you might already have suspected, aspects of

utilitarianism, deontological ethics, and contractualist ethics are strongly tied to the notions of individual rights and responsibilities. In the ancient Greek world of Aristotle’s time, the notion of community was paramount. Thus, virtue ethics faces certain challenges in contemporary Western society that it would not have had to endure in the classical Greek *polis*, or city-state.

► 2.8 INTEGRATING ASPECTS OF CLASSICAL ETHICAL THEORIES INTO A SINGLE COMPREHENSIVE THEORY

We have completed our examination of the four main types of ethical theories, and we have noted some of the strengths and weaknesses of each theory. Consequentialist theories such as utilitarianism are useful because they aim at promoting happiness and the social good. Yet, we also saw that utilitarians tend to ignore the importance of justice and fairness in their preoccupation with promoting social utility for the majority. Deontologists, on the contrary, stress the importance of obligation and respect for all persons and thus emphasize the principles of fairness and justice. However, we saw that deontologists fail to pay sufficient attention to the promotion of happiness and the social good.

Contract theory seems useful in that it provides a motivation for being moral and it enables us to articulate which explicit moral obligations we have and do not have, both as individuals and as a society. However, the weakness of the social contract view is that it provides us with only a minimalist theory of morality. Virtue ethics stresses character development and the acquisition of good habits on the part of individuals, but its disadvantage is that it depends on homogeneous community standards for determining the correct virtues. Thus, each theory has its weakness, despite its strengths. Table 2-3 summarizes the advantages and disadvantages of each of the four ethical theories we examined.

Because of problems with the four types of traditional ethical theories that we considered, some have advocated for alternative ethical theories including feminist ethics. Adam (2008) has drawn from some of the insights of Carol Gilligan,¹⁵ as well as from other authors who have contributed to the literature on feminist ethics, in making her case for why at least some computer ethics issues would be better understood if they were analyzed from the perspective of feminist ethical theory. We briefly examine Adam’s arguments for a “gender-informed” computer ethics in Chapter 10, where we consider gender issues affecting cybertechnology.

Others have proposed ways in which elements of two or more traditional theories can be integrated into a single, more comprehensive framework. For example, Gert (2005, 2007) has integrated aspects of two theories by incorporating Kant’s insights on the importance of impartiality with the claims of utilitarians about consequences, but he thinks that each theory, in itself, is inadequate. Gert has also shown how his moral system, which he calls “common morality,” can be directly applied to issues involving computer ethics such as copying proprietary

TABLE 2-3 Four Types of Ethical Theory

Type of Theory	Advantages	Disadvantages
Consequence based (utilitarian)	Stresses promotion of happiness and utility	Ignores concerns of justice for the minority population
Duty based (deontology)	Stresses the role of duty and respect for persons	Underestimates the importance of happiness and social utility
Contract based (rights)	Provides a motivation for morality	Offers only a minimal morality
Character based (virtue)	Stresses character development and moral education	Depends on homogeneous community standards for morality

Copyright © 2015, Wiley. All rights reserved.

software.¹⁶ Recall our discussion of Gert's notion of the moral system and its corresponding "moral rules" in Section 2.1.1.

Influenced by the work of Gert and others, Moor (2004) has proposed a scheme that integrates aspects of utilitarian and deontological theories into a framework he calls "just consequentialism." We next examine some key aspects of this theory.

2.8.1 Moor's Just-Consequentialist Theory and Its Application to Cybertechnology

Moor believes that only an ethical approach that combines considerations of *consequences* of action with more traditional deontological considerations of duties, rights, and *justice* can provide us with a defensible ethical theory—namely, *just consequentialism*—that yields a useful framework for applied ethics. Moor begins by considering what kind of conduct we want ethics to regulate. He believes first and foremost everyone wants to be protected against suffering unnecessary harms. We don't want to be killed or suffer great pain or have our freedom taken away. Human nature is such that people value the same kind of basic goods (life, happiness, abilities, security, knowledge, freedom, opportunities, and resources). The specifics of these may manifest somewhat differently in different cultures (e.g., some kinds of freedom may be more important in some cultures than others), but the general set of goods, which Moor calls "core values" (see Section 2.1.2), is shared by all. Losing any of these goods counts as harm, and all of us want ethics to protect us from others causing us harm. This point is captured by the familiar ethical maxim "Do no harm," described earlier. Stealing someone's computer causes a loss of resources to that person, and lying about software bugs undermines the purchaser's knowledge. Thus, it is not surprising that we regard stealing and lying as unethical activities in light of their harmful consequences.

Another desirable objective of ethics, according to Moor, is to support justice, rights, and duties. We want others to keep their promises and agreements, to obey the law, and to fulfill their duties in whatever roles they play. These specific obligations are generated within societies, and to the extent that they spring from just agreements, laws, and social situations, we justifiably expect others to fulfill their duties toward us. For example, we want a software engineer to produce reliable software. We believe it is her duty as a professional to develop effective and safe software and that we have a right to expect good quality when we buy it. Another familiar maxim of ethics is "Do your duty," where "duty" here designates specific duties people acquire by their roles in society such as a signer of contract, a citizen, a parent, an employer, or an employee. Violating one's just duty, such as knowingly designing defective software for later production and sales, in the absence of contravening considerations, is clearly unethical.

Moor believes that if all we had to do to be ethical were to do no harm and perform our duties, ethics would be challenging but at least easy to understand. But, as Moor argues, the ethical life is not nearly so simple. Often, actions involve a mixture of goods and evils as well as conflicts among duties. Sometimes, we need to make exceptions to our general policies for action. How do we decide what to do? His answer involves two steps: the deliberation stage and the selection stage. First, at the *deliberation stage*, we should consider the various possible policies for action from an impartial point of view. Impartial does not mean that everyone is treated the same but that the policy is regarded as a rule governing the situation without consideration of the particular individuals who happen to be involved. This is what Gert has in mind by his "blindfold of justice" (see Section 2.1.1) or what Rawls suggests with his "veil of ignorance." This is a technique to establish the justice of a policy—it will not be just if one will not accept the policy as a general rule of conduct, not knowing who plays which roles in the situation.

For example, consider the scenario involving the hacking of nude photos of celebrities from their mobile phones, discussed in Chapter 1. Let us assume that the hacker was obsessed with his victims and got significant gratification out of his deeds. If we consider a policy for justifying such an action impartially, we will clearly reject it. We will not endorse a policy of

allowing people to hack into someone's electronic device to make controversial photos available online to the public even if people get significant pleasure from it. It is easy to reject such a policy as unjust and unethical when considered from an impartial point of view.

However, many policies will pass the impartiality test, and we will still need to consider whether we should adopt them. We need to move to the second step in the decision-making process, the *selection stage*, and carefully weigh the good consequences and the bad consequences of the remaining policies. In this second step, it may be less of a choice between ethical and unethical policies than between better and worse policies. Although we may be able to at least partially rank policies, legitimate disagreements about the rankings often exist.

For instance, consider the controversial issues as to whether we should adjust or even have a policy of intellectual property protection. For many years in many places, there were no laws protecting intellectual property. It is far from clear that this situation was unjust or even bad. A culture might maintain that sharing information and invention is more valuable to the society's members' welfare and the society's cohesiveness than trying to protect intellectual property. Witness the rationale given for the "open source movement" in software development. Critics of this movement, however, might maintain that having intellectual property protection laws is important to protect creators and to produce innovative products for everyone's benefit.

According to Moor, it is important to keep in mind that although we may disagree about the merits of various policies and how to rank them, rational discussion of the relevant policies is very possible and highly desirable. People may overlook values embedded in a situation and may change their rankings once informed. People may not be fully aware of the consequences of various policies. Moor does not believe that complete agreement on controversial policies can or necessarily should be reached, as people may ultimately rank benefits and harms differently. Nevertheless, considerable consensus about some policies being better than others can often be generated. Moor points out that frequently much of the disagreement hinges on differences about the facts of the case than on value differences. (Recall our early analysis of differences involving "disagreements about principles" and "disagreements about facts" in Section 2.2.1, in our discussion of discussion stoppers in ethics.) It would radically change much of the debate about the need for protecting MP3 files, for example, if it could be demonstrated that, *as a matter of fact*, downloading MP3 files to preview them dramatically increases sales or if it could be demonstrated that, *as a matter of fact*, downloading MP3 files to preview them dramatically decreased the quality of music that was produced.

2.8.2 Key Elements in Moor's Just-Consequentialist Framework

Moor's ethical framework of just consequentialism can be summarized in terms of a strategy that includes the following steps:

1. *Deliberate* over various policies from an impartial point of view to determine whether they meet the criteria for being ethical policies. A policy is ethical, if it:
 - a. Does not cause any unnecessary harms to individuals and groups and
 - b. Supports individual rights, the fulfilling of duties, etc.
2. *Select* the best policy from the set of just policies arrived at in the deliberation stage by ranking ethical policies in terms of benefits and (justifiable) harms. In doing this, be sure to:
 - a. Weigh carefully between the good consequences and bad consequences in the ethical policies and
 - b. Distinguish between disagreements about facts and disagreements about principles and values, when deciding which particular ethical policy should be adopted. (Knowledge about the facts surrounding a particular case should inform the decision-making process.)

As we noted in our discussion of virtue ethics in Section 2.7.2, Moor points out that developing the appropriate habits of character such as kindness, truthfulness, honesty, trustworthiness, helpfulness, generosity, and justice is an important prerequisite in moral behavior. So if one has not already developed the “correct” habits required for moral behavior, it may be difficult for an individual to successfully carry out the steps in Moor’s just-consequentialist model. In this sense, elements of virtue ethics or character-based ethics are also presupposed in Moor’s framework.

We apply the just-consequentialist framework, wherever appropriate, in suggesting policies in response to moral issues that arise from specific cyberethics issues examined in Chapters 4–12 of this textbook.

► 2.9 CHAPTER SUMMARY

In this chapter, we defined ethics as the study of morality. In elaborating on that definition, we drew some useful distinctions between morality (as a system of rules and principles) and ethics (as the study of that system). Acknowledging the distinction between normative and descriptive studies of morality, we saw that normative investigations into morality can be conducted from the perspectives of religion and law as well as from philosophy. We also noted that only philosophical ethics offers a method to analyze moral issues based exclusively on the application of ethical theory and logical argumentation. We briefly identified and analyzed some common “discussion stoppers” that are frequently invoked in ways that prematurely close down, even if unintentionally, the possibility of constructive ethical dialogue.

We also examined the roles that ethical theories ideally play in guiding us in our moral deliberations about cyberethics issues. We saw that consequence-based, duty-based, contract-based, and character-based theories each had certain strengths and weaknesses. Finally, we examined James Moor’s proposal for a framework that incorporates aspects of consequence-based and duty-based theories (and to some extent character-based theories) into one unified, comprehensive theory, called “just consequentialism.” We summarized Moor’s framework into a two-step process that we will use, wherever possible, in our analysis of the cyberethics issues examined in this textbook.

► REVIEW QUESTIONS

1. What is *ethics*, and how can it be distinguished from *morality*?
2. What is meant by a *moral system*?
3. What are the “rules of conduct” that comprise a moral system? Give some examples of these (moral) rules.
4. Describe the key differences between rules of conduct that are individual “directives” and those that are “social policies.” Provide an example of each.
5. What does Bernard Gert mean when he describes morality in terms of a system that is both “public” and “informal”?
6. Describe how the ideals of “rationality” and “impartiality” function in Gert’s moral system.
7. What are *values*, and what criteria differentiate *intrinsic* values from *instrumental* values?
8. What does James Moor mean by “core values,” and how are they both similar to and different from intrinsic values and instrumental values?
9. What factors differentiate *moral values* from *non-moral values*? Provide examples of each.
10. How do religion, law, and philosophy each provide different grounds for justifying a moral principle? How can each perspective be applied in analyzing the moral principle “Stealing is wrong?”
11. What are the basic differences separating *ethicists* from *moralists*?
12. Summarize each of the four different kinds of “discussion stoppers” that often arise in ethical discourse.
13. Why are these “discussion stoppers” problematic for the advancement of dialogue and debate about ethical issues?
14. What is *moral relativism*? How is it different from *cultural relativism*?
15. What is *moral objectivism*, and how is it different from *moral absolutism*?
16. What is ethical theory, and what important functions do ethical theories play in the analysis of moral issues?

17. What are the distinguishing features of consequence-based ethical theories?
18. Describe some of the key differences between act utilitarianism and rule utilitarianism.
19. Which features distinguish duty-based ethical theories from alternative types of theories?
20. Describe some of the main differences between act deontology and rule deontology.
21. What is meant by the expression “contract-based” ethical theories?
22. What features distinguish “character-based” (or “virtue-based”) ethical theories from alternative schemes of morality?

► DISCUSSION QUESTIONS

23. Why does Gert believe that the notion of “personal morality” is an oxymoron? For Gert, how is a moral system both similar to and different from a game? Apply Gert’s notion of a moral system to the analysis of a contemporary ethical issue affecting cybertechnology. Analyze that issue in terms of the four features that comprise a moral system for Gert.
24. How does James Moor’s “just-consequentialist” theory incorporate aspects of utilitarian and deontological theories into one comprehensive ethical framework? Describe the strategies used in the two different stages of Moor’s theory: the deliberation and the selection stage. Identify a contemporary moral issue affecting cybertechnology, and apply Moor’s just-consequentialist theory to it.
25. Recall the four types of “discussion stoppers” that we examined in this chapter. Is that collection of “stoppers” complete? Can you think of any additional discussion stoppers that might also block or shut down moral discourse? Why is it so easy to fall victim to one or more of those stoppers when discussing moral issues in general, as well as moral issues involving the use of cybertechnology in particular?
26. Are any of the four traditional ethical theories we examined—that is, consequence based, duty based, contract based, and character based—adequate to handle moral issues that arise as a result of cybertechnology? If not, is an alternative kind of ethical theory needed, as some have argued (e.g., Adam 2008)? Or, can a comprehensive, integrated theory, such as the one proposed by James Moor (i.e., his theory of “just consequentialism”) be used successfully to resolve moral issues involving cybertechnology?

Scenarios for Analysis

1. In analyzing the following scenario, describe how an *act utilitarian*, a *rule utilitarian*, a *rule deontologist*, and an *act deontologist* would each reach a solution to this dilemma. Which solution seems most plausible? Finally, apply Moor’s just-consequentialism framework in your analysis of this scenario.

You have just been appointed to the board of directors of XYZ.com. Unfortunately, the dot-com company has been experiencing some difficult financial times, resulting in revenue losses in three of the last four quarters. As you assume your new position, you discover that two proposals are on the table. Each proposal has been put forth as a means for dealing with XYZ’s immediate financial problems. Proposal #1 recommends that all employees be retained but that an immediate freeze on salary increases (raises) for all employees be imposed for the next six months. (Employees may even be asked to take a 5% cut in pay if things do not improve by the end of that period.) Proposal #2 recommends that wages not be frozen but that 5% of the XYZ’s

workforce be laid off. (One piece of reasoning behind this proposal is that taking more drastic measures will “protect” 95% of XYZ’s workers and will send a message to Wall Street and local investors that XYZ is serious about improving its financial position and that it will soon be a stable company once again.) The board is evenly split, seven members favoring proposal #1 and seven favoring proposal #2. Yours will be the tiebreaking vote.

2. Analyze the dilemma in the following scenario from the vantage point of both utilitarian and deontological ethical theories. In particular, how might Ross’s theory of act deontology apply?

The U.S. government, with the approval of the majority of Americans, has decided to round up all Arab-Americans and relocate them into internment camps. You have a friend who is an American citizen of Arab descent. She asks you to protect her from the authorities. You have known this person all of your life, and you are convinced that she is a loyal American. So you agree to hide her in the

third floor of your house. Next, imagine that a U.S. federal agent knocks on your door and asks if you know the whereabouts of the person you are hiding. How would you respond to that agent?

You realize that you cannot both keep your promise to your friend and tell the truth to the federal agent. Initially, your gut reaction might suggest that the solution to your dilemma is really quite simple: A far greater good will be served by lying to the federal agent than by breaking your promise to your

friend. However, to embrace the moral principle underlying that line of reasoning is to embrace a form of utilitarianism. And we have already seen some of the difficulties that can result from trying to be a consistent and thoroughgoing utilitarian. Furthermore, could you consistently universalize a moral principle that states: Whenever you must choose between telling the truth to authorities and breaking a promise to a friend, you should always keep your promise? Will that principle always work?

► ENDNOTES

1. Analyses of moral dilemmas based on examples using the (now classic) “trolley problem” have proliferated since this “thought experiment” was introduced by philosopher Philippa Foot in 1967. For an interesting variation of this dilemma (in the context of an autonomous computer-driven trolley), see Wallach and Allen (2009).
2. This classic definition of ethics has been used by many philosophers. See, for example, Paul W. Taylor’s *Principles of Ethics: An Introduction* (Belmont, CA: Wadsworth, 1980).
3. Bernard Gert’s “blindfold of justice” is similar in some ways to John Rawls’ well-known “veil of ignorance,” articulated in Rawls’ class work *A Theory of Justice* (rev. ed. 1999). However, the two notions also differ in key respects.
4. See, for example, Pojman (2006).
5. For example, some critics point out that even though there appear to be significant differences in the moral beliefs held by various groups and cultures, when examined at the surface level, a closer analysis will suggest that there are also some (“deep”) universal or core moral beliefs that lie under the surface.
6. Gert suggests that his ten “moral rules” are objective in nature. However, he does not use the label “objectivism” to describe his moral system.
7. Although Gert does not call himself a moral objectivist, I interpret his position to be compatible with the view I describe as moral objectivism.
8. Wallach and Allen (2009) also consider a variation of the “trolley case” in which the trolley’s driver has been replaced by a computerized system.
9. The variations of Kant’s Categorical Imperative expressed here closely follow the original formulation in Kant’s writings.
10. For more detail on this strategy, see Ross (1930).
11. Pojman (2006) explains in more detail how this theory provides a “motivation” for someone or some group to behave morally.
12. For a classic account of this incident, see “Queens Woman Is Stabbed to Death in Front of Home,” *New York Times*, March 14, 1964.
13. This is a basic distinction in Natural Law, a theory that we are not able to examine in detail in this chapter.
14. See, for example, Foot’s *Theories of Ethics* (Oxford University Press, 1967) and MacIntyre’s *After Virtue* (Notre Dame, IN: University of Notre Dame Press, 1981).
15. Gilligan’s classic book *In a Different Voice* (Cambridge, MA: Harvard University Press, 1982) has influenced many authors writing on topics in feminist ethics.
16. See Gert (2004).

► REFERENCES

- Adam, Alison. 2008. “The Gender Agenda in Computer Ethics.” In K. E. Himma and H. T. Tavani, eds. *The Handbook of Information and Computer Ethics*. Hoboken, NJ: John Wiley and Sons, pp. 589–619.
- Aristotle. 1962. *Nicomachean Ethics*. Trans. M. Oswald. New York: Bobbs-Merrill.
- Bentham, Jeremy. 1948. In W. Harrison, ed. *Introduction to the Principles of Morals and Legislation*. London: Oxford University Press.
- De George, Richard T. 2010. *Business Ethics*. 7th ed. Upper Saddle River NJ: Prentice Hall.
- Ess, Charles. 2006. “Ethical Pluralism and Global Information Ethics.” *Ethics and Information Technology* 8, no. 4: 215–66. p. 63–82.
- Ess, Charles. 2014. *Digital Media Ethics*. 2nd ed. London, UK: Polity Press.
- Gert, Bernard. 2004. “Common Morality and Computing.” In R. A. Spinello and H. T. Tavani, eds. *Readings in CyberEthics*. 2nd ed. Sudbury, MA: Jones and Bartlett, pp. 96–106. Reprinted from *Ethics and Information Technology* 1, no. 1 (1999): 37–56.
- Gert, Bernard. 2005. *Morality: Its Nature and Justification*. Rev. ed. New York: Oxford University Press.
- Gert, Bernard. 2007. *Common Morality: Deciding What to Do*. New York: Oxford University Press.
- Grodzinsky, Frances S. 2001. “The Practitioner from Within: Revisiting the Virtues.” In R. A. Spinello and H. T. Tavani, eds. *Readings in CyberEthics*. Sudbury, MA: Jones and Bartlett, pp. 580–91. Reprinted from *Computers and Society* 29, no. 1 (1999): 9–15.

- Hobbes, Thomas. 1962. *Leviathan*. New York: Collier Books.
- Kant, Immanuel. 1965. *Fundamental Principles of the Metaphysics of Morals*. Trans. T. K. Abbott. London: Longman's.
- Mill, John Stuart. 1965. *Utilitarianism*. New York: Bobbs-Merrill.
- Moor, James H. 2002. "The Importance of Virtue in Teaching Computer Ethics." In M. Ochi, et al., eds. *Proceedings of the Foundations of Information Ethics*. Hiroshima, Japan: Hiroshima University Press, pp. 29–38.
- Moor, James H. 2004. "Just Consequentialism and Computing." In R. A. Spinello and H. T. Tavani, eds. *Readings in CyberEthics*. 2nd ed. Sudbury, MA: Jones and Bartlett, pp. 107–13.
- Pojman, Louis P. 2006. *Ethics: Discovering Right and Wrong*. 5th ed. Belmont, CA: Wadsworth.
- Rawls, John. 1999. *A Theory of Justice*. Rev. ed. New York: Oxford University Press.
- Ross, W. D. 1930. *The Right and the Good*. London: Oxford University Press.
- Wallach, Wendell and Colin Allen. 2009. *Moral Machines: Teaching Robots Right from Wrong*. New York: Oxford University Press.

► FURTHER READINGS

- Anderson, Michael and Susan Leigh Anderson, eds. 2011. *Machine Ethics*. Cambridge: Cambridge University Press.
- Cahn, Steven and Peter Markie, eds. 2012. *Ethics: History, Theory, and Contemporary Issues*. 5th ed. New York: Oxford University Press.
- Hinman, Lawrence M. 2013. *Ethics—A Pluralistic Approach to Moral Theory*. 5th ed. Belmont, CA: Wadsworth Publishing.
- Rachels, James and Stuart Rachels. 2012. *The Elements of Moral Philosophy*. 7th ed. New York: McGraw-Hill.
- Scalet, Steven and John Arthur, eds. 2013. *Morality and Moral Controversies: Readings in Moral, Social, and Political philosophy*. 9th ed. Upper Saddle River, NJ: Prentice Hall.
- Sterba, James and Peter Bornscheim. 2013. *Morality in Practice*. 8th ed. New York: Cengage.
- Triplett, Timm. 2002. "Bernard Gert's *Morality* and Its Application to Computer Ethics." *Ethics and Information Technology* 4, no. 1: 79–92.

Copyright © 2015, Wiley. All rights reserved.

Critical Reasoning Skills for Evaluating Disputes in Cyberethics

LEARNING OBJECTIVES

Upon completing this chapter, you will successfully be able to:

- Identify and construct alogical *argument*, and differentiate arguments from the various *claims*, or *statements*, that comprise them,
- Determine when an argument satisfies the conditions for being *valid*,
- Determine whether a valid argument is also *sound* (or *unsound*),
- Determine whether an argument is invalid,
- Determine when an invalid argument satisfies the conditions for being *inductive*,
- Determine when an invalid argument is *fallacious*,
- Apply a seven-step strategy for determining the *overall strength* of an argument,
- Identify a cluster of common “informal logical fallacies” that occur in everyday discourse and apply them in your analysis of cyberethics issues.

In this chapter, we examine some basic *reasoning* skills and concepts that aid us in our overall analysis of specific cyberethics issues examined in Chapters 4–12. We begin by briefly reflecting on a scenario that illustrates a particular kind of (conventional) reasoning process one might use in trying to determine whether it would be morally permissible to download content from a Web site.

► SCENARIO 3–1: Reasoning About Whether to Download Software from “Sharester”

You are contemplating downloading a software application that is available on a Web site called *Sharester*, a file-sharing P2P (peer-to-peer) site. Sharester is not officially designated as a “pirate site” because it mainly provides users with access to (freely available) open-source software applications. However, this site also contains a few proprietary (or copyrighted) software applications that users can download. It turns out that the particular application you are interested in downloading is proprietary; furthermore, there is no indication that the copyright holder has authorized the free downloading of that application. Although you wish to download this application for personal use (only), you are conflicted about what to do; so you discuss your concerns with a close friend, Charlie.

Charlie tries to convince you not to download a proprietary software program using the following rationale: *Downloading proprietary software (without permission from the copyright holder) is identical to stealing physical property. Stealing physical property is morally wrong. Therefore, downloading proprietary software (without permission) is morally wrong.* ■

How good is the reasoning process used by Charlie (in this scenario), and how do we determine our answer to that question? If we rely solely on our intuitions, we might be inclined to believe that Charlie's reasoning process is very solid (and in this case, the conclusion reached may indeed be true). But what about the reasoning process itself, that is, the "form of reasoning," used? Is the (assumed truth of the) evidence provided in Charlie's argument sufficient to guarantee the truth of the conclusion? While Charlie's reasoning process might seem to concur with our initial intuitions, can we always trust our intuitions (solely) when reasoning about what to do in situations similar to this one? Fortunately, there are some objective criteria that we can use to distinguish between good and bad reasoning (i.e., to determine whether Charlie's argument is valid, or whether it contains invalid/fallacious reasoning).

Later in this chapter (at the end of Section 3.8), we apply a seven-step strategy to evaluate the reasoning process used in the argument expressed in Scenario 3–1. First, however, we need to understand some basic terms and concepts that are typically used in critical reasoning.

► 3.1 WHAT IS CRITICAL REASONING?

Critical reasoning is a subfield of *informal logic*.¹ So, you may wonder why a chapter dedicated to this topic is included in a book on cyberethics.² To appreciate the important role that critical reasoning skills play in our analysis of cyberethics issues,³ recall the methodological framework that we developed in Chapter 1. There, we saw that the final step of that methodology requires that we defend or justify our position by evaluating it vis-à-vis the rules of (logical) argumentation. Also, recall that in Chapter 2 we saw how a typical cyberethics issue could be analyzed from the vantage point of one or more ethical theories, such as utilitarianism, deontology, etc. But then we must ask: How do we defend—give a (logical) reason for—our selection of one particular ethical theory over another in our analysis? And how would we convince others about which theory should be applied, for example, in a situation where we might be in a dispute with someone about how best to analyze one or more pressing cyberethics issues?⁴

Consider also that there may be occasions where someone might try to influence your thinking about which kinds of behavior are, and are not, appropriate online. As we saw in Scenario 3–1, for example, it is possible that you may be conflicted about what to do in cases where online policies and laws might seem unclear, or might not seem to be unethical (even if they are illegal). In this chapter, we will see how critical reasoning skills can help us both to clarify our own thinking/reasoning and analyze the arguments others might use to try to persuade us with respect to some claim. We will also see how the various techniques for constructing and analyzing arguments will help us to frame our positions more precisely and defend them more rigorously.

3.1.1 Some Basic Concepts: (Logical) Arguments and Claims

Critical reasoning skills provide the tools necessary for evaluating the strength of (logical) *arguments*, which comprise various *claims*. For our purposes, an argument, which contains at least two claims, can be defined as a reasoning form, or structure, that attempts to establish the truth of one claim (called a *conclusion*) based on the assumed truth of the evidence in other

claims (called *premises*) provided to support the conclusion. Thus, an argument is a *form of reasoning* that has two important characteristics or features in that it:

- i. Includes at least two claims (but can include an indefinite number of claims)
- ii. Aims at establishing a conclusion (i.e., the truth of one claim) based on evidence provided by one or more other claims called premises

We will see that whereas arguments are either *valid* or *invalid*, the claims that comprise them are either *true* or *false*. First, however, we examine an important role that arguments can play when someone is trying to support or defend a position that is being questioned or disputed.

3.1.2 The Role of Arguments

Consider a hypothetical scenario involving a claim about the development of a controversial and powerful new computer chip—code-named Chip X—in Japan. This new chip is purported to be so powerful in speed and performance that it will eclipse any computer chips that manufacturers in the United States will be capable of producing during the next several years. Chip X will also enable the manufacturer to monitor certain activities of those users whose devices contain the chip in ways that, potentially at least, pose serious threats to personal privacy.

Suppose that you are skeptical about my claim that Chip X is currently under development by Mishito Corporation and that it is planned for release sometime within the next year. There are a number of ways I could attempt to convince you that my claim is true; for example, I could try to persuade you to accompany me on a trip to Japan to see firsthand that Chip X is indeed being developed there. In this case, we could obtain direct evidence regarding the truth of my claim. But if you are unable or unwilling to accompany me to Japan, I will have to use some other, less direct, mode of verification to convince you. For example, I could show you a copy of the design specifications for Chip X extracted from a confidential Mishito Corporation document that I happened to acquire. Or perhaps I could ask a mutual colleague of ours who recently studied as an exchange student at the University of Hiroshima, where the field-testing for this new chip is being carried out, to corroborate my claim regarding Chip X. That is, I can put together various pieces of evidence to construct a logical argument that supports my claim.

Now we are in a position to debate the merits of my overall argument regarding Chip X, based on the form reasoning employed in it, without our having to travel to Japan to verify the truth of my claim. Before we debate the strength of my argument, however, we must first understand some essential features of an argument's *structure*.

3.1.3 The Basic Structure of an Argument

We noted in Section 3.1 that an argument consists of two or more claims, one of which is called the conclusion; the others are called the premises. The standard form for representing an argument is to list the premises first and then state the conclusion. The following structure represents an argument's standard form:

PREMISE 1

PREMISE 2 (optional)

PREMISE 3 (optional)

•
•
•

PREMISE *n* (optional)

CONCLUSION

To support my claim that Chip X is currently being developed in Japan, in the conclusion of my argument, I would need to list the evidence in the form of one or more premises. For example, I could use the following argument form:

PREMISE 1. When I recently visited the computer science department at the University of Hiroshima in Japan, I noticed that graduate students and professors there were field-testing a new computer chip whose code name is Chip X.

PREMISE 2. I have a copy of the design specifications for Chip X, which shows that it will be several times faster than any chip currently available in the United States.

PREMISE 3. Lee Smith, a mutual colleague of ours who was recently an exchange student in the computer science program at the University of Hiroshima and who participated in the field-testing of Chip X, will corroborate my claim.

CONCLUSION. Chip X is currently being developed in Japan.

This particular argument includes three premises and a conclusion; additional premises could be added. However, an argument requires at least one premise along with a conclusion. In this section, we are concerned only with an argument's structure and not with how strong the argument might be. An argument, however weak it may ultimately be, still qualifies as an argument if its structure (or reasoning form) includes one or more premises and a conclusion.

You might have observed that the claim expressed in the conclusion to our argument about Chip X could also be verified (i.e., determined to be either true or false) independent of the evidence provided in the argument's premises. Since the conclusion contains a statement that is descriptive, or empirical (i.e., capable of being observed through sensory experience), the truth or falsity of the conclusion could be resolved in this case simply by going to Japan to see whether such a chip was actually being developed there.

However, not all arguments have empirical or descriptive statements as their conclusions. Suppose that a friend wants to convince you that Internet users should be allowed to write a blog on how to build a bomb. (Note that this is a normative claim because it includes the word "should"; you may want to consult the distinction we drew in Chapter 1 between normative and descriptive claims.) Further, suppose that his reason for holding this view is based on the principle that people are allowed to write books on how to build bombs, and authors of blogs should have the same rights and freedoms as authors of books. And suppose your friend bases his reasoning for this claim on the right of authors to express themselves as guaranteed in the First Amendment of the U.S. Constitution. His argument could be constructed as follows:

PREMISE 1. A person's right to author a book on how to build a bomb is protected by the First Amendment (of the U.S. Constitution).

PREMISE 2. Authoring a book is similar to authoring a blog on the Internet.

CONCLUSION. A person's right to author a blog on how to build a bomb ought to be protected by the First Amendment.

Notice how this argument differs from the preceding one. For one thing, we can't simply go to Japan to determine whether this conclusion is true or false. For another, the conclusion contains a normative statement (one that includes "ought"). Unlike the previous argument, which contained a descriptive statement in its conclusion that could be verified independent of the argument, we now depend on the *form of reasoning* alone to help us determine whether the conclusion is true. In doing this, we will *assume* that the premises in this argument are true and then ask whether the conclusion would logically follow from them.

Initially, the reasoning in this argument might seem plausible: The person constructing the argument cleverly uses an analogy based on a legal right that applies in physical space. So, we might assume that any legal rights that citizens enjoy in physical space should automatically be extended to cyberspace.

In this argument, we are also asked to consider certain features or characteristics that are common to both (printed) books and blogs. Clearly, we can draw a number of analogies here. For example, both books and blogs can communicate and disseminate information to readers, each is authored by one or more persons, and so forth. However, there is a danger in pushing some of these analogies too far: Whereas traditional books are tangible items existing in physical space, blogs are not. And the scope of a blog allows it to be accessed by members of the international community, some of whom may have no access to physical books or may lack sufficient funds to purchase such books. We now begin to see dissimilarities between books and blogs, so we must be cautious about drawing conclusions when reasoning by analogy. Later in this chapter, we will see why arguments of this kind are not valid. First, however, we consider some strategies for constructing arguments.

► 3.2 CONSTRUCTING AN ARGUMENT

Think of some situations in which arguments are used by those in powerful positions, as well as by ordinary persons. Lawyers, for example, use arguments to try to persuade juries, and politicians often use arguments to convey their positions to their constituencies. All of us use arguments when we try to convince a friend, a spouse, or a boss about some point or other. If you try to convince your parents that they should buy you a new iPad, for example, you will most likely be making an argument of some sort. Ultimately, arguments will succeed or not succeed depending on (i) how well they are constructed and (ii) how strong their reasoning forms are. We refer to (ii) as *argument strength*, and we examine that concept in Section 3.3 in our discussion of valid vs. invalid arguments. In this section, we focus on how arguments are constructed.

Arguments often appear as editorials in newspapers and periodicals where they are sometimes expressed in prose forms that can obscure the argument, making it difficult to isolate and analyze. When this happens, we must locate the arguments concealed in the text before we can analyze them. Consider the political debate over the need for a new national missile defense (NMD) system, which has been controversial from both a domestic and an international perspective. A fairly straightforward argument in favor of NMD in the editorial section of a newspaper might look something like the following:

We must build a national missile defense system because without such a system we are vulnerable to nuclear attacks from rogue nations that might arise in the future. Engineers and computer scientists have testified that they can design a computer-guided missile defense system that is effective, safe, and reliable. It is our obligation as Americans to take whatever measures we can to protect the safety of our citizens.

Before we analyze this argument, however, it is perhaps worth making a few parenthetical remarks about certain events leading up to NMD. The debate in the U.S. Congress over NMD that

occurred during the George W. Bush administration, and which has been resurrected in recent years, can be viewed as an updated version of the earlier “Star Wars” debate, officially known as the (or Strategic Defense Initiative (SDI)). That debate, which took place during the Reagan administration in the 1980s, was significant for cyberethics because it was one of the first ethical controversies to catch the attention of a group of computer ethics pioneers.⁵ We will examine some specific ethical issues pertaining to Star Wars and NMD of interest to computer professionals in Chapter 4; our primary purpose in this chapter, however, is to consider the NMD controversy only insofar as it illustrates how logical arguments can be constructed and analyzed.

There has been strong support for NMD among many conservative politicians in the United States. As suggested above, a proponent of the NMD system could construct an argument for his or her case by first asserting that without such a new missile system, the United States is vulnerable to future attacks from potential “rogue” nations that might acquire nuclear weapons. The proponent might next want to assure us that there is sufficient and compelling evidence that such a missile defense system would be safe and reliable. Finally, the NMD supporter might assume the following principle: “We must do whatever is necessary to preserve the safety of America and its people.” The structure of the proponent’s argument can be represented as follows:

PREMISE 1. Without the NMD, the United States is vulnerable to nuclear attacks in the future from rogue nations.

PREMISE 2. Computer scientists and engineers have testified before Congress that they can design a computer-guided missile defense system that is both safe and reliable.

PREMISE 3. The United States must do whatever is necessary to preserve the military defense of the nation and the safety of its citizens.

CONCLUSION. The United States should build the NMD.

So far, we have considered only the structure of this argument. That is, we have described its two basic components—its premises and conclusion—and we have represented it in standard logical form. Now we ask: Is the reasoning used in the argument strong? Are there rules that will enable us to determine this? To answer these questions, we first need to understand the difference between valid and invalid arguments.

► 3.3 VALID ARGUMENTS

The first question we could ask about the sample argument described in the preceding section is whether its reasoning is strong or weak—that is, is the argument’s reasoning *valid* or is it *invalid*? “Valid” and “invalid” are technical terms in logic. Whereas claims (the individual statements that make up an argument) are either true or false, arguments will be either valid or invalid; it is incorrect to refer to an argument as either true or false, and it is incorrect to refer to a claim as either valid or invalid.

How can we determine whether a particular argument is valid or invalid? In formal systems of logic, elaborate schemes that consist of symbols, rules, and tables have been constructed for determining when arguments are valid or invalid. Alternatively, however, some “informal” systems of logic, such as the system developed by Nolt (2002), also enable us to accomplish this task. Nolt’s system does not require that we know anything about the *actual* truth or falsity of the claims in an argument’s premise(s) in order to determine whether an argument is valid or invalid. Instead, we need only to determine whether the argument’s conclusion would necessarily follow from its premises, when those premises are all *assumed* to be true.⁶ In other words, we ask:

Is the relationship between the premises and the conclusion such that if all of the premises in the argument are assumed true, it would be impossible for the conclusion to be false?

The concern here is with the relationship of truth conditions vis-à-vis the premises and the conclusion. The premises and the conclusion could be true or false (in the actual world), independent of each other, but that is not relevant for testing the argument's validity. We ask only whether the assumed truth of the premises is sufficient to guarantee the truth of the conclusion. If the answer is yes, then the argument is valid. If, however, it is logically possible for the argument's conclusion to be false at the same time that its premises are (assumed) true, then the argument is invalid.

You can apply this test for validity to the argument for the new NMD system that we considered above. Imagine that all of the argument's premises are true statements. Is it possible—that is, could you conceive of a possible instance such that—when those premises are true, the conclusion could still be false? Of course, the premises could be imagined to be false, and the conclusion could be imagined to be false as well. But the relevant questions here are: What happens when all of the premises are imagined to be true? Could the claim in the argument's conclusion (i.e., “The United States should build the new NMD.”) be false even when Premises 1–3 are assumed true? The answer is yes. Hence, the argument is invalid.

The Counterexample Strategy

To show that an argument is invalid, all we need to do is to produce one *counterexample* to the argument. A counterexample is:

A logically possible case in which the argument's conclusion could be imagined to be false while (at the same time) the argument's premises are assumed true.⁷

In the NMD argument, we can coherently imagine a logically possible situation (i.e., a situation that does not involve a logical contradiction) where the conclusion “The United States should build the new NMD” is false when the claims stated in Premises 1–3 are assumed true. For example, we can imagine a situation or case in which all three premises are true but some alternative strategy not involving the development of a new missile defense system could provide for the safety of America. So a counterexample is possible; thus, the argument is invalid.

Note, however, although this particular argument has been shown to be invalid, it does not follow that the argument's conclusion is, in fact, false. All that has been shown is that the argument is invalid because the form of reasoning it uses does not succeed in guaranteeing that the conclusion must be true. It is still possible, of course, that the conclusion could be true. But a different argument would need to be constructed to show that the inference is valid.

Suppose we added a fourth premise, “The new NMD system is necessary to preserve the defense and safety of the United States and its citizens,” to the argument. The amended argument would be as follows:

PREMISE 1. Without the NMD, the United States is vulnerable to nuclear attacks in the future from rogue nations.

PREMISE 2. Computer scientists and engineers have testified before Congress that they can design a computer-guided missile defense system that is both safe and reliable.

PREMISE 3. The United States must do whatever is necessary to preserve the military defense of the nation and the safety of its citizens.

PREMISE 4. The new NMD system is necessary to preserve the defense and safety of the United States and its citizens.

CONCLUSION. The United States should build the NMD.

This argument would now be valid. If all of its premises are assumed true, then the conclusion cannot be false. Of course, we could next ask whether all of the premises in this argument are in fact true. Premises 1 and 2 are fairly uncontroversial claims, though Premise 2 might be challenged by programmers who believe that building a completely reliable computer system is not possible. However, both Premises 3 and 4 are controversial: Premise 4 can be shown to be false if it can be demonstrated that the United States could be adequately protected without the newly proposed missile defense system; Premise 3, which is a normative statement, can also be shown to be false if, for instance, we can provide an exception to the principle included in it. For one thing, we could ask both whether indeed the United States *must* do whatever is necessary to make the United States safe and what exactly we mean by the phrase “whatever is necessary”? For example, what if making the United States safe entailed closing down all systems of transportation, all government offices, and all schools for an indefinite period of time that could go on for years? It might protect U.S. citizens but would it be an acceptable alternative? And U.S. citizens might be willing to make trade-offs rather than shut down major institutions essential to their day-to-day lives. So Premise 3, as stated, is also false. However, even if all of the premises are eventually shown to be false, the argument itself is still valid because its conclusion follows from the premises if they are assumed true.

Figure 3-1 illustrates the basic distinction between valid and invalid arguments.

Next, consider the following argument:

PREMISE 1. People who own iPhones are smarter than those who own Droids (Android phones).

PREMISE 2. My roommate owns an iPhone.

PREMISE 3. I own a Droid.

CONCLUSION. My roommate is smarter than me.

This argument meets all of the criteria for validity: *If* all three of the premises in this argument are assumed true, then the conclusion (“My roommate is smarter than me.”) must be true. In other words, no counterexample to this argument’s reasoning form is possible. However, the argument’s validity alone is not sufficient to establish that the argument

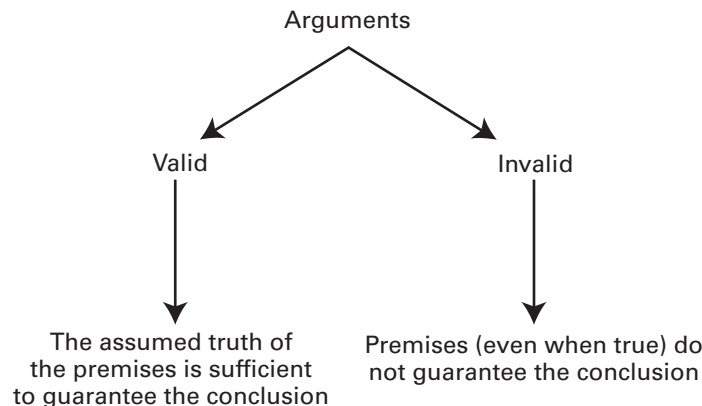


Figure 3-1 Valid and invalid arguments.

succeeds in the final analysis (as we will see in the following sections). It only shows that when all of the premises are assumed true, the conclusion would also be true.

This argument, like all valid arguments, is valid by virtue of its *logical form*; An argument's logical form, not its content, determines its validity and invalidity. An example of a valid logical form is as follows:

PREMISE 1. Every A is a B.

PREMISE 2. C is an A.

CONCLUSION. C is a B.

Any argument that has this form is valid, regardless of the content represented by A, B, or C. As long as the premises "Every A is a B" and "C is an A" are both assumed true, there is no way that the conclusion "C is a B" could be coherently conceived to be false. Even if the two premises in this particular argument turn out to be false in the actual world, the argument continues to be valid by virtue of its logical form.

We can now see that it is important to separate two distinct questions:

- a. What is the *strength of reasoning* of the argument (i.e., is it valid or invalid)?
- b. Are the argument's premises true in the actual world?

To say that an argument is valid does not necessarily mean that its premises are true in the actual world. An argument can be valid in terms of its logical form and yet still be unsound. One more step is required for an argument to qualify as a *sound argument*. To be sound, all of the premises (included in the valid argument) must be true in the real world and not merely assumed true as in the case of the test for validity. For information on how to determine whether a statement is true or false, see Appendix G (available at www.wiley.com/college/tavani).

► 3.4 SOUND ARGUMENTS

To assume that the premises of an argument are true is an important first step in the process of evaluating arguments, because doing so enables us to determine the logical relationship between the argument's premise(s) and conclusion and thus determine the argument's strength of reasoning. The reasoning strength will be either valid or invalid. If we can produce one counterexample by showing a possible case where the argument's conclusion can be false even when all of its premises are assumed true, we have shown the argument to be invalid. If the argument is shown to be invalid, we can stop here for the time being. To show that the argument was valid, all that we have to do is to show that no counterexample is possible. And to do that, we consider merely the hypothetical or assumed truth of the argument's premises vis-à-vis the argument's conclusion. If the argument is valid, then we must determine if it is also sound by going on to the next step, where we test the premises to see whether they are actually true or false in the empirical world.

Consider again the two arguments in the preceding section: one involving an NMD system and the other involving the intelligence of iMac users. Both arguments were shown to be valid. (The argument defending the need for an NMD system had to be modified, but once we modified it, it met the criteria for validity.) We can now further examine each argument to see if it is also sound. An argument will be sound if (i) it is valid, and (ii) all of the premises are actually true in the empirical world (and not merely assumed to be true).

First, consider the NMD system. If one or more of Premises 1–4 are false, then the argument will be unsound. Premise 3, “The United States must do whatever is necessary to preserve the military defense of the nation and the safety of its citizens,” is clearly questionable. Surely, the goal of national defense is one of the highest priorities of a political administration. But we have already seen that the phrase “whatever is necessary” is problematic. For example, would such a principle give the U.S. government the right to use *any* means that it happened to deem necessary to bring about some desired end?

Suppose some government officials believed that it was necessary to put all non-U.S. citizens under house arrest? Or suppose that some of those officials believed that all U.S. citizens should be subject to constant search and seizure, both within and outside their homes? Would these measures be acceptable? Perhaps under the most severe and dire circumstances, some proposals of this type might seem plausible. But it is still not exactly clear that such drastic measures would be necessary. So Premise 3 in the missile defense argument cannot be confirmed to be true, even if Premises 1, 2, and 4 can. Thus, this argument is not sound even though it is valid. Because it is unsound, the argument does not succeed. However, once again, we should be careful to note that even when an argument is unsound, or even when it is invalid, it does not necessarily follow that the argument’s conclusion is false. Rather, we can only infer that the evidence given, that is, the particular premises used to support the argument’s conclusion, is not adequate because (when used alone) the premises fail to meet certain logical requirements.

Returning to the argument involving claims about the intelligence of iMac users, we saw that when we assume the truth of all three premises of that argument, the conclusion cannot be imagined to be false; hence, the argument is valid. But is it also sound? We need to examine each premise in more detail. Premises 2 and 3—“My roommate owns an iPhone” and “I own a Droid,” respectively—are relatively easy to verify because both are descriptive claims. To see if they are true or false, we simply go to the dormitory room or to the apartment where my roommate and I live and observe whether my roommate indeed owns an iMac computer and whether I own a Droid.

Premise 1—“People who own iMac computers are smarter than those who own Droids”—however, is more controversial and hence more difficult to verify than the other premises. Clearly, more evidence would be needed to show that Premise 1 is true; in fact, it certainly seems suspect. So despite the fact that the argument is valid by virtue of its logical form, we cannot yet say that it is sound. Thus, the argument would appear, at best, to be not sound but inconclusive.

As you might suspect, sound arguments are not very common, and often they are about matters that are either trivial or uncontroversial. Consider the following argument:

PREMISE 1. CEOs of major computer corporations are high school graduates.

PREMISE 2. Bill Gates was the CEO of a major computer corporation.

CONCLUSION. Bill Gates is a high school graduate.

This argument is clearly valid because no counterexample can be constructed, that is, there is no possible case where Premises 1 and 2 could both be (assumed) true and the conclusion could be false at the same time. As it turns out, the premises are also true in the actual world, so this argument is sound; however, it is also not terribly informative. Perhaps you can now see why there are so few sound arguments that are also informative: relatively few valid arguments are sound, and relatively few sound arguments are informative or nontrivial.

Figure 3-2 illustrates the basic differences between valid arguments that are sound and those that are unsound.

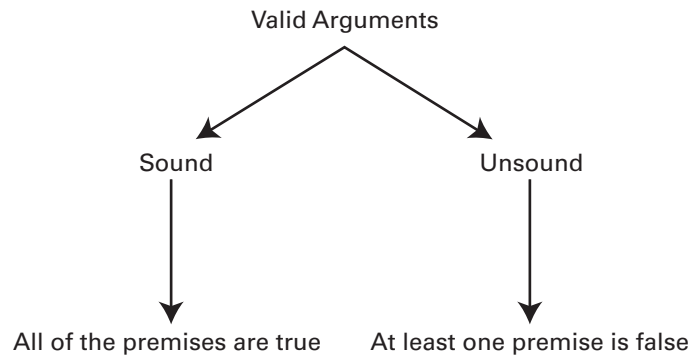


Figure 3-2 Sound and unsound (valid) arguments.

At this point, you might ask, what good is a valid argument if it contains false premises? You might also wonder whether certain types of invalid arguments whose premises are true in the actual world are stronger than valid arguments that contain one or more false premises.

► 3.5 INVALID ARGUMENTS

Consider the following argument:

PREMISE 1. All CEOs of major U.S. computer corporations have been U.S. citizens.

PREMISE 2. Steve Jobs was a U.S. citizen.

CONCLUSION. Steve Jobs was a CEO of a major computer corporation in the United States.

Even though all three of the claims included in this argument (i.e., the two premises and the conclusion) are true in the actual world, we can show that this argument is invalid by producing at least one counterexample. We can imagine a possible case where the premises in this argument are both true, but the conclusion—the late Steve Jobs was the CEO of a major computer corporation—is false. For example, we can imagine that he had been a consultant or a programmer, or he could have been employed outside the computer field. However, there is an even more serious flaw in this argument: If we substitute for “Steve Jobs” the names Hilary Clinton, LeBron James, Jessica Lawrence, or Brad Pitt, we see that although each of these persons is a U.S. citizen, none has been a CEO of a major computer corporation. Yet by the logic used in the above argument, it would follow that if any one of them is a U.S. citizen, then he or she must also be or have been a CEO. We have shown that this argument is invalid. If you had noticed that the reasoning in this argument was weak, now you know exactly why.

We next determine whether the following argument is valid or invalid:

PREMISE 1. Most CEOs of major computer corporations are college graduates.

PREMISE 2. Satya Nadella is the CEO of a major computer corporation.

CONCLUSION. Satya Nadella is a college graduate.

Notice that all of the statements included in this argument happen to be true in the empirical world. But is the reasoning valid? Clearly not! All we need is one counterexample to show why. If we substitute the name “Bill Gates” for “Satya Nadella,” the present CEO of Microsoft (as of 2015), the premises of the argument remain true but the conclusion is false. The argument is invalid, but because the premises are true, this particular invalid argument is stronger overall than either of the two arguments we considered that were valid but unsound. Overall “argument strength,” as opposed to an argument’s “strength of reasoning,” takes into account the actual truth condition of the argument’s premises. We saw that an argument’s strength of reasoning is concerned only with the hypothetical or assumed truth of those premises.

► 3.6 INDUCTIVE ARGUMENTS

Not all invalid arguments are necessarily weak arguments; in fact, some are quite strong. Hence, we should not automatically discard every invalid argument simply because it is not valid. Some invalid arguments are *inductive*. Although inductive arguments do not necessarily guarantee the truth of their conclusions in the way that valid arguments do, inductive arguments nonetheless provide a high degree of probability for their conclusions. Those invalid arguments that are not inductive are fallacious arguments; we will discuss them in the next section. In this section, we describe the criteria that must be satisfied for an argument to be inductive.

Let’s determine the strength of reasoning of the following argument:

PREMISE 1. Seventy-five percent of people who own iPods also own iMacs.

PREMISE 2. My roommate owns an iPod.

CONCLUSION. My roommate owns an iMac.

Based on the technique discussed earlier in this chapter, we can see that this argument is not valid: A counterexample to the argument is possible. For instance, we can assume that both premises are true while the conclusion (“My roommate owns an iMac computer”) is false. There is no contradiction in doing this since my roommate could be among the 25% of people who currently own iPods but who never owned iMacs. So the argument is clearly invalid.

This argument and the argument in the preceding section designed to show that Steve Jobs was the CEO of a major computer corporation are both invalid, but they are different in their strength of reasoning. The argument that tried to show that Jobs must have been a CEO because “Jobs is a U.S. citizen” and “all CEOs of major computer corporations have been U.S. citizens” has weak reasoning. On the other hand, the form of reasoning used in the argument to show that my roommate owns an iMac computer is much stronger. In fact, the conclusion (“My roommate owns an iMac computer”) is *very likely* to be true when we assume the truth of both premises. Hence, this (invalid) argument is inductive.

As suggested above, some inductive arguments, although invalid, can be stronger overall than some valid arguments. But how is that possible? We have seen examples of valid arguments that contained premises that were false in the actual world. Inductive arguments consisting of premises that are all true in the actual world are generally stronger than arguments that are valid but unsound. As you consider the various arguments involving privacy, free speech, security, etc., in Chapters 4–9, determine which ones meet the criteria of being inductive with all true premises. Such arguments will be much more successful in establishing their

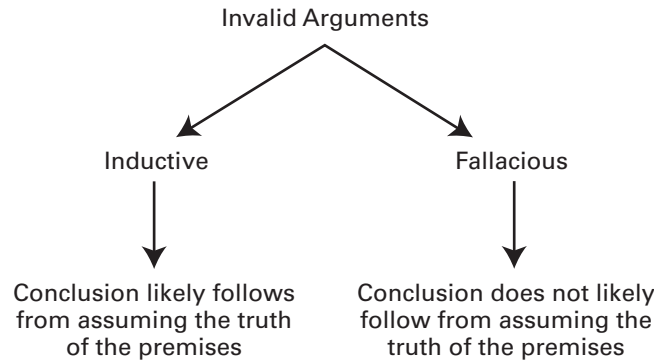


Figure 3-3 Inductive and fallacious (invalid) arguments.

positions (i.e., they will be much stronger overall) than will valid arguments that contain one or more false premises.⁸

Figure 3-3 illustrates the basic differences between invalid arguments that are inductive and invalid arguments that are fallacious.

► 3.7 FALLACIOUS ARGUMENTS

Recall the argument that we examined in Section 3.5 to show that Steve Jobs was the CEO of a major corporation. All of the statements or claims in that particular argument were true in the actual world, so the argument might have seemed fairly strong. Yet, because we could produce a counterexample (and, in fact, we saw that we could easily produce several counterexamples), clearly the argument was invalid.

We next ask whether the argument is inductive or fallacious. That is, how likely is it that the argument's conclusion, "Jobs was the CEO of a major computer corporation," would be true based simply on the assumed truth of the argument's premises? Even though the conclusion could be true—and even though it is, in fact, true—the truth or falsity would have to be established on grounds other than those given in the premises used to support the conclusion that Jobs had been a CEO. Hence, this argument is *fallacious*.

Note that an argument's being fallacious has nothing to do with the actual truth or falsity of its premises, so you have probably noticed a certain irony with respect to an argument's strength of reasoning. We have seen that an argument can be valid and yet contain one or more false premises and a false conclusion, and conversely, an argument can be fallacious despite the fact that all of its premises as well as its conclusion could be true.

Next, consider an argument in which someone tries to convince you that Internet users should not expect to retain their privacy when they engage in online activities, because the Internet is essentially a public forum or public space. Expressed in standard form, the argument reads:

PREMISE. The Internet is in public space.

CONCLUSION. Those who use the Internet should not expect to retain any personal privacy.

When we evaluate this argument’s strength of reasoning, that is, whether it is valid or invalid, we first ask if a counterexample can be produced. Let us assume that the premise—The Internet is in public space—is true. We next ask: Is it possible for that statement to be true, and for the conclusion (Internet users should not expect to retain their privacy) to be false, at the same time? The answer is “yes.” For example, a person’s backpack can be in public space, yet its owner enjoys some expectation of privacy regarding the contents enclosed in the backpack. So a counterexample can be produced, and the argument is invalid. We can next ask whether the argument is inductive or fallacious. In the argument’s current form, the conclusion does not likely follow from the premise, even when that premise is assumed true. So the argument is fallacious.

Also recall an argument that we examined in Chapter 1 for the view that at least some ethical issues involving cybertechnology must be unique. The argument had the following form:

PREMISE 1. Cybertechnology has some unique technological features.

PREMISE 2. Cybertechnology has generated some ethical concerns.

CONCLUSION. Some ethical concerns generated by cybertechnology must be unique ethical concerns.

You can no doubt see why this argument is fallacious. First, we construct a counterexample to show the argument is invalid, that is, we can imagine a case where both premises are true, but the conclusion is false. For example, we can imagine some ethical concern generated by the use of cybertechnology to be simply a variation of a traditional ethical problem that is neither new nor unique. Furthermore, we can construct a range of possible cases where both premises are assumed true, but the conclusion could be false. (In fact, in the majority of such cases, the conclusion would not likely be true merely because of what is assumed to be true in the premises.) As we noted in Chapter 1, this argument wrongly assumes that a characteristic that applies to

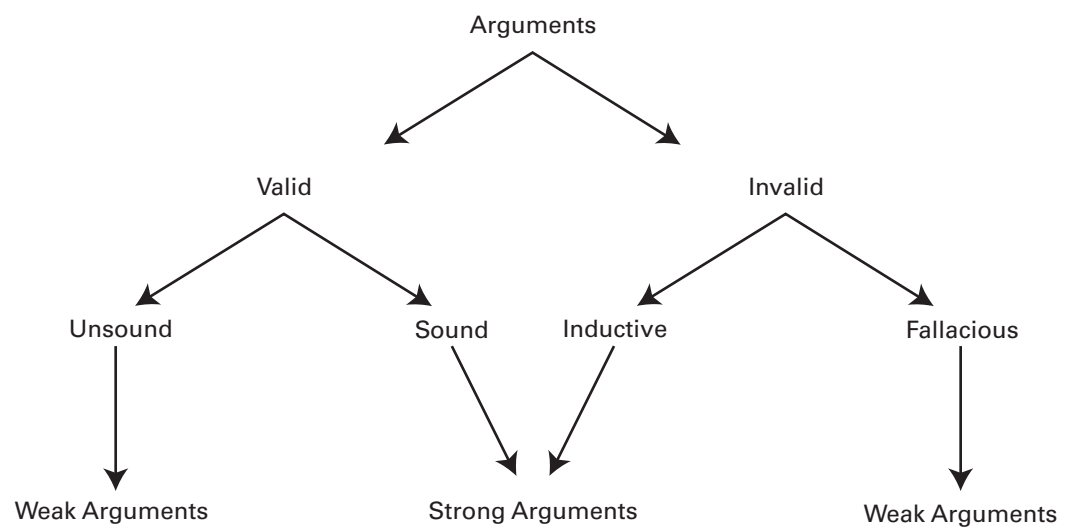


Figure 3-4 Comprehensive view of argument types.

Copyright © 2015, Wiley. All rights reserved.

the technology—namely, uniqueness—must also apply to the ethical issues generated by this technology. Of course, it is possible that some cyberethics issues may turn out to be unique ethical issues. But if that happens, it is not because of the evidence supplied in the premises of the above argument.

In Section 3.9, we will examine some “informal logical fallacies” that tend to recur in everyday reasoning. There, we will also examine techniques for spotting some of the more common fallacies, or “fallacy types,” that occur in ordinary discourse. (These techniques do not require the counterexample strategy that we used to distinguish invalid from valid arguments.) First, however, we include a scheme that summarizes the techniques we have used so far to differentiate among valid and invalid, sound and unsound, and inductive and fallacious arguments.

Figure 3-4 presents an overview of the different kinds of argument forms that we have examined in this chapter.

► 3.8 A SEVEN-STEP STRATEGY FOR EVALUATING ARGUMENTS

The following strategy, which consists of seven steps, summarizes the techniques we used in Sections 3.3–3.7 to evaluate an argument’s overall strength of reasoning:

- Step 1.** Convert the argument into standard form. (List the premises followed by the conclusion.)
- Step 2.** Test the argument for its strength of reasoning to see whether it is valid or invalid.
Strategy: Assume the premises to be true and ask yourself whether the conclusion must also be true when those premises are assumed true. Is a counterexample to the argument possible?
- Step 3.** Is the argument valid?
If yes, go to Step 4.
If no, go to Step 5.
- Step 4.** Is the (valid) argument also sound? That is, are the premises true in the actual world?
Strategy: To determine whether a claim is true or false in the actual world, see the guidelines in Appendix G (Available at www.wiley.com/college/tavani).
 - a.** If the argument is valid and if all of the premises are true in the actual world, then the argument is also sound.
 - b.** If the argument is valid, but one or more premises can be shown to be false, then the argument is unsound. (Note that if one or more premises are unable to be verified, i.e., determined to be either true or false, then the overall argument is inconclusive.)
- Step 5.** Is the (invalid) argument inductive or fallacious?
Strategy: To determine this, ask how likely the conclusion would be true when the premises are assumed true.
If the conclusion would likely be true because the premises are assumed true (i.e., the evidence for the conclusion is strong), the argument is inductive.
If the conclusion would not likely be true even when the premises are assumed true, the argument is fallacious.
Note: Keep in mind that a fallacious argument can be made up of individual claims or statements that are themselves true in the actual world.
- Step 6.** Determine whether the premises in your argument are either true or false in the actual world.

Strategy: Consult the guidelines for determining the truth or falsity of statements in Appendix G (available at www.wiley.com/college/tavani).

Step 7. Make an overall assessment of the argument by determining both (i) the argument's strength of reasoning (valid, inductive, or fallacious) *and* (ii) the truth conditions of each of the argument's premises.

Strategy: Determine, for example, whether the argument's overall strength is:

- Sound
- Valid but unsound
- Inductive with all true premises
- Inductive with some false premises
- Fallacious with a mixture of true and false premises
- Some other combination

Remember that an inductive argument with premises that are all true can be stronger overall than a valid argument with one or more false premises, which will be (valid but) unsound.

We next apply the seven-step strategy to the argument included in Scenario 3–1 at the beginning of this chapter. Applying Step 1, we convert the argument into standard form (which, in this case, it already happens to be presented):

PREMISE 1. Downloading proprietary software (without permission from the copy-right holder) is identical to stealing physical property.

PREMISE 2. Stealing physical property is morally wrong.

CONCLUSION. Downloading proprietary software (without permission) is morally wrong.

At Step 2, we examine the argument's strength of reasoning and determine that the argument is valid because if we assume the truth of both its premises (Premises 1 and 2), the conclusion cannot be false (i.e., the combination of true premises and false conclusion in this example would be a logical contradiction). Having determined that the argument is valid (Step 3), we next go to Step 4 and ask whether it is sound or unsound. Whereas Premise 2 is a true statement (and is easily verifiable), the truth or falsity of Premise 1 is less clear cut. Although there would certainly seem to be a strong analogy between stealing physical property and downloading unauthorized software, there are also some disanalogies. Thus, the two behaviors are not, strictly speaking at least, "identical." (We examine this and some other disanalogies in Chapter 8 in our discussion of intellectual property.) So, Premise 1 may be either false or indeterminate (i.e., it is not literally true, as stated), and we now see that this argument is *unsound*.

However, the argument is still valid; so we can skip Step 5, which applies only to invalid arguments. At Step 6, we note that both Premise 2 and the conclusion are true, while Premise 1 may be either false or indeterminate (since it is not literally true). So our overall evaluation of this argument, at Step 7, is *valid but unsound*. Here, we note that the conclusion happens to be true even though its truth is not logically supported by the argument's premises.

The seven-step strategy is a useful tool for evaluating a wide range of arguments. However, we should note that some less formal techniques are also available for spotting fallacious arguments that commonly occur in ordinary, everyday reasoning. In the next section, we will see that there is an informal, and arguably simpler, way of identifying and cataloging many logical fallacies that frequently appear in everyday discourse.

► 3.9 IDENTIFYING SOME COMMON FALLACIES

Contrary to what many people assume, “fallacy” does not mean false statement; rather, it means *faulty reasoning*. As we saw in Section 3.6, it is possible for an argument to contain all true statements and still be fallacious. (We also saw that an argument can contain all false statements and still be valid, solely by virtue of its logical form.)

At this point, you might be unsure about your ability to recognize a fallacious argument without using the counterexample strategy described in this chapter or without applying some of the more sophisticated rules that comprise formal systems of logic. Because so many fallacies appear in everyday reasoning, logicians have categorized them in ways that are convenient for us to recognize. We refer to these kinds of fallacious arguments as *informal logical fallacies*.

The following 8 informal fallacies, or “fallacy types,” each illustrated with one or more examples involving computers and cybertechnology, are typical of fallacious arguments that surface time and again in ordinary everyday discourse.

3.9.1 *Ad Hominem* Argument

In the *ad hominem* argument, an attack is directed at the person rather than the substance of what is being disputed. Consider an instance of this kind of attack that occurred in the Edward Snowden case of whistle-blowing, which involved the leaking of sensitive documents from the National Security Agency (NSA) in 2013 (a case that we examine in detail in Chapter 4):

Edward Snowden was not a whistle-blower, but rather a self-serving narcissist who was more interested in promoting himself than in exposing any wrong doing. Also, he is reported to have lied both to his employer and his girlfriend. So, how could anyone possibly believe that Snowden’s leaking of the sensitive NSA documents was morally justified?⁹

Why is the reasoning process used in this argument fallacious? For one thing, we could conjure up numerous counterexamples (and we include one below). But the nature of the fallacy in this particular argument is apparent in its sole focus on attacking Snowden’s character rather than providing any relevant evidence to show that Snowden’s whistle-blowing act was not morally justifiable. Consider, however, that even if Snowden is a narcissist (as asserted in the above attack), and even if Snowden had acted from purely selfish motives, it does not follow that “he is not a whistle-blower”—that is, did not engage in a genuine act of whistle-blowing. The reasoning process used in the above scenario reduces simply to an attack of Snowden’s personal attributes (i.e., his character, alleged past behavior, etc.) and says nothing about why his act of leaking the NSA documents was not morally justified.

As already noted, one can easily imagine a counterexample to the kind of *ad hominem* attack directed against Snowden in the above argument. Consider, for instance, a situation where a member of Islamic State in Iraq and Syria (ISIS) (sometimes also referred to as ISIL) betrays his organization by intentional leaking information to an American journalist about a plan to blow up a government building in the United States. Further imagine that we later find out that this individual is narcissistic and that his or her motives for leaking the information

were based solely on self-promotional grounds. Would we condemn this person's actions merely because they were the result of narcissistic/self-promotional motives? We may dislike this person for whatever reasons but that factor would be completely irrelevant in our evaluation of whether his or her actions constituted a genuine act of whistle blowing (as well as whether the whistle-blowing act was morally justified).

3.9.2 Slippery Slope Argument

The slippery slope argument has the form, “X could possibly be abused; therefore, we should not allow X.” For example, one might argue:

We should not continue to allow computer manufacturers to build computer systems that include CD burners. If we allow them to do so, young people will burn copies of copyrighted music illegally. If the rate of unauthorized copying of music continues, recording artists will lose more money. If they lose more money, the entire music industry could be in jeopardy. If the music industry in America declines, the entire U.S. economy will suffer. So, we must prohibit computer manufacturers from providing CD burners as part of the computer systems they build.

It should be fairly easy to spot the fallacy here. The author assumes that allowing computer manufacturers to continue to include CD burners as part of their computer systems will inevitably lead to an abusive slippery slope that logically implies a downturn in the overall U.S. economy. It is certainly possible that in the future, the U.S. economy does experience a downturn while computer manufacturers continue to produce computers with CD burners. And it may well be that some users abuse CD burners to make unauthorized copies of music. However, any claim that the bad economy was an inevitable result of the use of CD burners in computers cannot be substantiated merely in terms of the evidence provided in the above argument.

3.9.3 Fallacy of Appeal to Authority

Arguments that conform to the fallacy of authority (*argumentum ad verecundiam*) have the following form:

PREMISE 1. X is an authority in field Y.

PREMISE 2. X said Z.

CONCLUSION. Z.

As an example, imagine that Tim Berners-Lee, who designed the HTTP protocol that became the standard for the World Wide Web, has agreed to do an advertisement for Twitter, a social networking service (SNS). Further imagine that someone draws the following inference from the ad:

Tim Berners-Lee believes that Twitter is superior to SNSs such as Facebook and Linked-In. And Berners-Lee is clearly an expert on matters involving the Web and the Internet. So, Twitter must be superior to alternative SNSs such as Facebook and Linked-In.

Can you spot the fallacy here? It is true that Berners-Lee is an expert on Web design; it may also be true that Twitter is superior to alternative SNSs (e.g., in certain respects). However, is the argument's conclusion that Twitter is superior to Facebook, Linked-In, and other SNSs (such as Pinterest), even if true, warranted by the premises? Simply because

Berners-Lee wrote the code that became the standard protocol for the World Wide Web, and thus is an expert on some matters involving the Web, does that make him an authority on SNSs?

3.9.4 False Cause Fallacy

The false cause argument (*post hoc ergo propter hoc*—after this, therefore because of this) reasons from the fact that event X preceded event Y to the conclusion that event X is necessarily the cause of event Y. Consider the following argument about Microsoft's Windows 10 operating system:

Shortly after the release of the Microsoft Windows 10 operating system in 2015, Microsoft's stock plummeted severely. Hence, there is no doubt that the release of Windows 8 is responsible for the decline in Microsoft's loss in the stock market.

Can you identify the fallacy in the above argument? Even though it might be tempting to attribute Microsoft's decline in the price of its stock to the release of the Windows 10 operating system, the person making this argument has overlooked the possibility of other factors that might have caused this decline. For example, maybe there were factors in the economy in 2015 that affected all high tech stock prices, or maybe there were overall declines in the market during that year.

3.9.5 Fallacy of Composition/Fallacy of Division

The fallacy of composition confuses the characteristics that apply to the parts of a whole, or to the individual members of a group, with the characteristics of the whole itself. For example, consider the following form of reasoning:

The new XYZ laptop computer is the best system on the market. XYZ has the fastest processor currently available on any laptop, it comes with twice the amount of RAM than any of its competitors, and it comes equipped with a suite of office applications that are superior to those on any currently available system. Also, its monitor has the best resolution and graphic display currently available on any commercial laptop.

Here, the fallacy should be obvious. Each of the components of this laptop computer is the best that is currently available. However, it clearly does not follow that the system will necessarily be the best one available. The connections between the various parts of this system might not be well designed; we are not told how reliable the computer system is vis-à-vis its competitors. These kinds of flaws are apparent in all argument forms that commit the fallacy of composition. A film that has the best cast (Tom Hanks, Julia Roberts), one of the best directors (Steven Spielberg), and one of the best soundtrack composers (John Williams) might still be a flop. The quality of the individual parts does not necessarily guarantee the same quality in the overall product.

Next, consider the flip side of this fallacy: the fallacy of division. The fallacy of division mistakenly infers that the same attributes or characteristics that apply to the whole or to the group must also apply to every part of the whole or to every member of the group. See if you can spot the fallacy in the following argument:

Harvard University is the highest-ranked university in the United States. Thus, Harvard must have the nation's best computer science department.

Does the conclusion to the above argument follow from its premise? Clearly not! Harvard might be ranked first overall among universities in the United States, but the possibility remains that MIT, Stanford University, or some other institution has the nation's highest-ranked computer science department.

3.9.6 Fallacy of Ambiguity/Equivocation

Fallacious reasoning can occur whenever one or more terms are used either ambiguously or equivocally; ambiguous terms have more than one interpretation, and it is not always clear which interpretation the author intends. A term is used equivocally, on the contrary, when it has two (or more) different senses or meanings. Consider the following argument:

Humans can think and highly sophisticated AI (artificial intelligence) computer systems can think; therefore, highly sophisticated AI computer systems are human.

In this case, it is possible that both premises are true in the actual world. However, the sense in which AI computer systems are said to think is not necessarily the same sense in which humans, in fact, think. Here, the term “think” is used equivocally. So even if it is true that computers can think in one sense of that term, it doesn’t necessarily follow that computers can think in the same sense that humans do. Even if computers and humans could both think in the same sense, it doesn’t follow that computers must be human. Consider that some animals, such as orangutans, do many things in ways that are similar to humans, yet we do not infer that orangutans are humans. (One might object to the above argument being called a fallacy by noting that if thinking is defined as something that only humans could do, then, by definition, computers (that “think”) would have to be human. While this kind of move might avoid the fallacy of ambiguity/equivocation, it introduces other problems since the premise “computers can think” would now be more controversial and require further analysis.)

Another example of the fallacy of ambiguity/equivocation can be found in the following argument:

Computers have memory. Humans have memory. Having memory enables humans to recall some of their childhood experiences. Therefore, computers can recall experiences from their childhood.

Notice that “memory” is used in two different senses in the above argument.

Although both examples of the fallacy of ambiguity/equivocation illustrate exaggerated cases, which might seem implausible, we should note that many arguments used in everyday discourse commit variations of this fallacy.

3.9.7 The False Dichotomy/Either–Or Fallacy/All-or-Nothing Fallacy

One fallacy type—frequently referred to by labels such as the False Dichotomy, the Either/Or Fallacy, or the All-or-Nothing Fallacy—typically presents us with two options that might initially seem to be mutually exclusive. For example, some political leaders assert, if you are not with us, you are against us. In this case, there would seem to be no possible neutral ground. Or, you might hear a politician in the United States claim, either you endorse the Patriot Act or you hate America. But often times, assertions of this kind can be very misleading. Of course, there are genuine situations where only two (mutually exclusive) options are possible, for example, a light switch is either in the on *or* off position. However, many claims, especially one’s involving controversial political issues, appeal to a strict either/or rhetorical strategy in cases where additional options are available.

Solove (2011) notes that a variation of the All-or-Nothing Fallacy can be found in the reasoning process frequently used to defend what he calls the “false trade-off between privacy and security.” Some people who use this line of fallacious reasoning assert, “we must give up our privacy if we want to be secure from terrorist attacks.” This and similar kinds of refrains have been used by government leaders, national defense agencies, and conservative politicians to influence the enactment of controversial laws such as the USA Patriot Act. But are the choices that one must make in these kinds of situations always reducible to “all-or-nothing”

(or to an “either/or”) option? Or have we instead been presented with false dichotomies by those who use such fallacious reasoning?

Solove correctly points out that our surrendering privacy would not necessarily make us more secure; similarly, he notes that not every gain in privacy necessarily entails a loss in security. Consider that there are ways of increasing security that do not intrude on privacy. For example, Solove notes that making cockpits in airplanes more secure enhances the security of airline passengers, but it does not reduce their personal privacy.¹⁰ So we can see that the concepts of privacy and security are not mutually exclusive and thus do not necessarily present us with either/or scenarios or false dichotomies, despite what many of those who use variations of the all-or-nothing fallacy in the privacy vs. security debate would have us believe. We revisit Solove’s insight in more detail in our analysis of privacy in Chapter 5.

3.9.8 The Virtuality Fallacy

The *virtuality fallacy*, coined by James Moor, has the following form:

PREMISE 1. X exists in cyberspace.

PREMISE 2. Cyberspace is virtual.

CONCLUSION. X (or the effect of X) is not real.¹¹

Those who defend questionable forms of online behavior, such as launching viruses and engaging in unauthorized entries into computer systems, sometimes suggest that these activities cause no real harm to people. Some reason that because these activities are carried out in the virtual world, their effects are only virtual and thus not real. You should be able to spot the fallacy in this line of reasoning. Imagine that someone has posted an insulting remark about you on Facebook or some other online forum. Arguably, the locus of offense is in cyberspace, or virtual space, as opposed to real (physical) space. Does it follow that the particular harm caused to you is any less real than it would have been had it occurred in a physical setting? In Chapter 11, we examine some arguments used by members of virtual communities who reason that no real harm can result in these forums because they are only in virtual space.¹² There, we will see how the virtuality fallacy applies.

We have considered some techniques for identifying fallacy types in ordinary language. The common fallacy types identified in this section represent only a small sample of those identified and labeled by logicians and philosophers. Yet, all have one feature in common: Their reasoning is so weak that even when the premises are assumed true, the conclusion would not likely be true because of the weak or insufficient evidence supporting it.

You can devise your own labels for some of the fallacious arguments we have encountered that have not yet been given names. For example, consider the fallacious argument we examined in Section 3.7 that tried to show that computer ethics issues must be unique because (i) computers have some unique technological features and (ii) computers raise ethical issues. We could call this the “Computer Ethics is Unique Fallacy.” Also consider the fallacy we examined earlier in this chapter, which reasoned that Internet users should have no expectation of personal privacy while they are online because the Internet is in public space. We could label this the “Privacy in Public Fallacy.” You can no doubt come up with additional labels for fallacious arguments that you encounter in your analysis of cyberethics issues. (For the names of some standard logical fallacies not covered in this chapter, see <http://yourlogicalfallacyis.com/>.)

These techniques for spotting logical fallacies in ordinary, everyday discourse are useful in helping us to evaluate some of the arguments we will examine in the remaining chapters in this textbook. We should also note that some authors, including Artz (2000), have suggested that “narrative reasoning” in the form of stories, as opposed to strict logical reasoning (even in the informal sense described in this chapter), can be very useful in helping us to understand and evaluate some cyberethics issues. However, we will not pursue this claim here, because the aim of this chapter has been to show how conventional critical reasoning skills for analyzing informal logical arguments can be applied in the context of cyberethics.

► 3.10 CHAPTER SUMMARY

In this chapter, we considered some basic critical reasoning skills, and we examined both the structure and the strength of logical arguments. We considered key criteria for differentiating among arguments that were valid and invalid, sound and unsound, and inductive and fallacious. We then considered a seven-step strategy for determining an argument’s overall strength. We also identified eight common fallacies, or “fallacy types,” that illustrated flaws in reasoning in the context of cybertechnology and cyberethics, and we considered some techniques that will help us to spot additional informal logical fallacies that occur in everyday reasoning. (Note that some additional material on critical reasoning skills is included in Appendix G, Available at www.wiley.com/college/tavani.) In the remaining chapters of this textbook, we will see how the critical reasoning skills introduced in this chapter can be used to evaluate arguments affecting many cyberethics disputes.

► REVIEW QUESTIONS

1. What is critical reasoning, and how can it be applied to ethical issues involving cybertechnology?
2. What is a logical argument, and how is it different from a claim or a statement?
3. Identify and briefly describe the two important characteristics or features of an *argument*.
4. What role or purpose do arguments play in helping us to resolve issues in cyberethics?
5. Describe the basic structure of an argument.
6. What is the essential difference between an argument that is valid and one that is invalid? Construct an example of each.
7. What is a counterexample, and how can it be used to show that an argument is invalid?
8. How is it possible for an argument to be valid but not successful?
9. What is required for an argument to be sound? Construct an example of a sound argument.
10. When is a valid argument unsound? Construct an example of an unsound argument.
11. What differentiates invalid arguments that are inductive from invalid arguments that are fallacious? Construct an example of each.
12. What is an “informal logical fallacy”?
13. What is the *Ad Hominem Fallacy*? Provide an example of this fallacy in the context of cyberethics.
14. What is the *Slippery Slope Fallacy*? Provide an example of this fallacy in the context of cyberethics.
15. What is the *Fallacy of Appeal to Authority (Ad Verecundiam)*? Provide an example of this fallacy in the context of cyberethics.
16. What is the *False Cause Fallacy*? Provide an example of this fallacy in the context of cyberethics.
17. What is the difference between the *Fallacy of Composition* and the *Fallacy of Division*? Provide an example of each fallacy in the context of cyberethics.
18. What is the *Fallacy of Ambiguity/Equivocation*? Provide an example of this fallacy involving either an issue in cyberethics or an aspect of cybertechnology.
19. What is the *False Dichotomy/Either–Or Fallacy/All-or-Nothing Fallacy*? Provide an example of this fallacy in the context of cyberethics.
20. What is the *Virtuality Fallacy*? Provide an example of this fallacy in the context of cyberethics.

► DISCUSSION QUESTIONS

21. Construct an argument for the view that your university should/should not have the right to monitor student e-mail. Next, analyze your argument to see if it is either valid or invalid. If it is valid, determine whether it is also sound. If it is invalid, determine whether it is inductive or fallacious.
22. Identify some of the arguments that have been made on both sides in the debate about sharing copyrighted MP3 files on the Internet. Evaluate the arguments in terms of their strength of reasoning. Can you find any valid arguments? Can you find any inductive arguments?
23. Based on what you have learned in this chapter, construct an argument to support or to refute the view that all undergraduate students should be required to take a course in cyberethics. Next, apply the seven-step strategy (in Section 3.8) to your argument.
24. Construct an argument for the view that privacy protection should be improved for ordinary users who conduct searches on Google. Next, evaluate your argument via the rules for validity versus invalidity. If your argument is invalid, check to see if it also includes any of the informal fallacies we examined in Section 3.9.

Scenarios for Analysis

1. Apply the seven-step strategy (in Section 3.8) to your evaluation of the argument in the following scenario:

A major association representing the music industry in the United States has determined that 4,000 copies of a new album featuring a popular rock group, called DUO, had been illegally downloaded last month. The cost of this album for those who elect to download it legally from online music stores is \$10 per copy. So the association concludes that the music company that holds the copyright to this album lost \$40,000 dollars in revenue last month (on that album alone).

2. Determine whether the strength of reasoning used in the argument in the following scenario is valid or invalid. If it is invalid, does it commit any of the fallacies we examined in Section 3.9?

You are engaged in an intense discussion with your friend, Bill, who works in the IT department at your university. Bill complains that many students are using P2P (peer-to-peer) file-sharing applications on the university's network to download excessive amounts of unauthorized copyrighted material. He also claims that the most effective solution to this problem would be to disable student access to all (existing) P2P sites and to prevent students at your institution from setting up their own P2P sites for any reason whatsoever (even to include noncopyrighted material). You convey to Bill your belief that this measure is too drastic. However, Bill argues that the only way to eliminate unauthorized file sharing among students at your institution is to disable access to all P2P software on the university's network.

► ENDNOTES

1. Philosophers and logicians often contrast “informal logic,” comprising the fields of critical reasoning and critical thinking, with “formal logic.” The latter is a much more rigorous system based on a framework of elaborate rules and symbols. Informal logic, on the contrary, focuses on the kind of reasoning that occurs in ordinary language, that is, in everyday discourse. As such, the critical reasoning skills described in this chapter can help us to analyze cyberethics-related arguments without having to master the rules in a formal system of (symbolic) logic.
2. Following Boyd (2003), Cederblom and Paulsen (2012), Thomson (2009), and others, I use the expression “critical reasoning” instead of “critical thinking” to describe the reasoning skills examined in this chapter. In my view, “critical thinking” has become far too broad a term, and it can now be interpreted to mean very different things in different academic disciplines. For our purposes, “critical reasoning” better captures the kind of precision and rigor required in the strategies and techniques used in this textbook to evaluate informal logical arguments.
3. See Søraker (2006) for an interesting account of the role that critical reasoning skills, especially in the context of “pragmatic arguments,” can play in our analysis of cyberethics issues.
4. Many ethics instructors now recognize the importance of rigorous critical reasoning skills for analyzing issues examined in ethics courses, and an increasing number of introductory ethics and applied ethics textbooks include a separate chapter on logical arguments and critical reasoning. See, for example, Bowyer (2001), Vaughn (2010), and Waller (2010). See also Ruggiero (2012) who devotes an entire book to critical thinking in the context of ethics.

5. An analysis of key issues and arguments surrounding the classic SDI controversy, as well as NMD, is included in Bowyer (2001, 2002) and Yurcik and Doss (2002).
6. Both the methodology and the terminology that I use for argument evaluation in Sections 3.3–3.7 follow and build upon standard models of informal logic, such as those used by Nolt (2002) and others. In those sections, I also include several examples from my (Wiley custom) textbook on reasoning (Tavani 2010).
7. The notion of “counterexample” used here follows a classic definition of the term used in informal logic; see, for example, the description in Nolt (2002). Whereas I use the term “valid” to label the argument form for which no counterexample is possible, other authors (including Nolt) use “deductive” to refer to this argument form.
8. Whereas I use the term “strong” to describe the “overall reasoning strength” of arguments that are either sound or inductive with all true premises, and the term “weak” to refer to the overall reasoning strength of arguments that are either fallacious or valid with one or more false premises, Nolt uses “rational” and “irrational” to distinguish between the two kinds of “overall reasoning strengths.”
9. See Tavani and Grodzinsky (2014) for an expanded discussion of how *ad hominem* attacks have been used in arguments involving the Edward Snowden controversy.
10. Solove (2011, p. 34). An example of the Either/Or Fallacy that applies in the controversial whistle-blowing controversy involving Edward Snowden is included in Tavani and Grodzinsky (2014).
11. Moor described this fallacy in a talk entitled “Just Consequentialism and Computing” at the 2000–2001 *Humanities Lecture Series*, Rivier University: Nashua, NH, February 2001.
12. For an interesting discussion of “rationalist” vs. “emotivist” accounts of reasoning about issues involving virtual environments, see Søraker (2010).

► REFERENCES

- Artz, John M. 2000. “Narrative vs. Logical Reasoning in Computer Ethics.” In R. Baird, R. Ramsower, and S. Rosenbaum, eds. *Cyberethics: Social and Moral Issues in the Computer Age*. Amherst, NY: Prometheus, pp. 73–79.
- Boyd, Robert. 2003. *Critical Reasoning and Logic*. Upper Saddle River, NJ: Prentice Hall.
- Bowyer, Kevin W. 2001. *Ethics and Computing: Living in a Computerized World*. 2nd ed. New York: IEEE Press.
- Bowyer, Kevin W. 2002. “Star Wars Revisited: Ethics and Safety Critical Software.” *IEEE Technology and Society* 21, no. 1: 13–26.
- Cederblom, Jerry and David W. Paulsen. 2012. *Critical Reasoning: Understanding and Criticizing Arguments and Theories*. 7th ed. Boston, MA: Wadsworth.
- Nolt, John E. 2002. *Informal Logic: Possible Worlds and Imagination*. New York: McGraw Hill.
- Ruggiero, Vincent R. 2012. *Thinking Critically About Ethical Issues*. 9th ed. New York: McGraw-Hill.
- Solove, Daniel J. 2011. *Nothing to Hide: The False Tradeoff between Privacy and Security*. New Haven, CT: Yale University Press.
- Søraker, Johnny Hartz. 2006. “The Role of Pragmatic Arguments in Computer Ethics.” *Ethics and Information Technology* 8, no. 3: 121–30.
- Søraker, Johnny Hartz. 2010. “The Neglect of Reason: A Plea for Rationalist Accounts of the Effects of Virtual Violence.” In C. Wankel and S. Malleck, eds. *Emerging Ethical Issues of Life in Virtual Worlds*. Charlotte, NC: Information Age Publishing, pp. 15–32.
- Tavani, Herman T. 2010. *The Elements of Reasoning: A Short Introduction to Informal Logic*. Hoboken, NJ: John Wiley and Sons (Custom Learning Solutions).
- Tavani, Herman T. and Frances Grodzinsky. 2014. “Trust, Betrayal, and Whistle-Blowing: Reflections on the Edward Snowden Case.” *Computers and Society* 44, no. 3: 8–13.
- Thomson, Anne. 2009. *Critical Reasoning: A Practical Introduction*. 3rd ed. New York: Routledge.
- Vaughn, Lewis. 2010. *Doing Ethics: Moral Reasoning and Contemporary Issues*. 2nd ed. New York: W. W. Norton.
- Waller, Bruce N. 2010. *Consider Ethics*. 3rd ed. New York: Pearson.
- Yurcik, William and David Doss. 2002. “Software Technology Issues for a U.S. National Missile Defense System.” *IEEE Technology and Society Magazine* 21, no. 2: 36–46.

► FURTHER READINGS

- Bergmann, Marie, James H. Moor, and Jack Nelson. 2013. *The Logic Book*. 6th ed. New York: McGraw Hill.
- Copi, Irving M. and Carl Cohen. 2011. *Introduction to Logic*. 14th ed. Upper Saddle River, NJ: Prentice Hall.
- Hurley, Patrick. 2015. *A Concise Introduction to Logic*. 12th ed. Belmont, CA: Wadsworth Publishing.
- Kahane, Howard and Nancy Cavender. 2010. *Logic and Contemporary Rhetoric: The Use of Reason in Everyday Life*. 11th ed. Belmont, CA: Wadsworth Publishing.
- Moore, Brooke Noel and Richard Parker. 2015. *Critical Thinking*. 11th ed. New York: McGraw-Hill.
- Munson, Ronald and Andrew Black. 2012. *The Elements of Reasoning*. 6th ed. Boston, MA: Wadsworth.
- Ruggiero, Vincent R. 2012. *Beyond Feelings: A Guide to Critical Thinking*. 9th ed. New York: McGraw-Hill.

Professional Ethics, Codes of Conduct, and Moral Responsibility

LEARNING OBJECTIVES

Upon completing this chapter, you will successfully be able to:

- Define *professional ethics* and differentiate professional ethics issues from other kinds of applied ethics issues affecting cybertechnology,
- Determine whether computing/IT professionals have any special moral responsibilities (by virtue of their specific professional roles),
- Articulate the roles/purposes of *professional codes of ethics* and identify their strengths and weaknesses,
- Understand the kinds of conflicts of professional responsibility that can arise where an employee may be permitted, or possibly even required, to blow the whistle,
- Articulate the key distinctions that differentiate *moral responsibility*, *legal liability*, and *accountability* (in the context of the computing/IT profession),
- Determine whether some computer companies might have special moral responsibilities (because of the nature of the products they develop or the services they provide).

In this chapter, we examine a range of issues often categorized under the general heading “professional ethics.” We begin by briefly reflecting on a scenario that raises questions about professional responsibility in the context of a serious malfunction in a semiautonomous, computerized weapon system.

► SCENARIO 4-1: Fatalities Involving the Oerlikon GDF-005 Robotic Cannon

In 2007, the South African army conducted a series of training exercises (in the Republic of South Africa) using the Oerlikon GDF-005, a semiautonomous, robotic cannon. The weapon was designed to use radar and laser technology to “lock on” to its targets, which include unmanned aerial vehicles (UAVs), cruise missiles, and helicopters. The robotic cannon was also designed to reload automatically when the magazines in its pair of 35 mm guns become empty. During a routine shooting exercise in October 2007, the cannon malfunctioned, resulting in 9 deaths and 14 injuries. Following an official investigation into the fatal incident, the South African National Defense Ministry issued a report that critics found vague and

inconclusive. While some analysts suspected that a software glitch was responsible for the malfunction, others believed that the cannon's malfunctioning could have been caused by a hardware failure.¹ So, it was not clear whether the deaths and injuries resulting from the Oerlikon GDF-005's malfunctioning were due to a software flaw, a hardware problem, or possibly both. ■

To what extent can/should the professionals, especially the software and hardware engineers, involved in developing this robotic cannon be held accountable for its malfunctions? Determining the locus of responsibility for malfunctions of this kind is not only important; it is urgent. Consider that many of the advanced and sophisticated technologies developed by computing/information technology (IT) professionals, which include more and more robotic and autonomous components, are increasingly used in military defense systems and warfare; these technological systems could also malfunction in ways similar to the Oerlikon GDF-005. Furthermore, malfunctions in these kinds of systems increasingly threaten the lives of ordinary civilians, in addition to those of the military personnel who use them (and their intended enemy combatants). In 2010, for example, the U.S. military lost control of a helicopter drone for more than 30 minutes during a test flight; the drone veered toward Washington, D.C., violating air space intended to protect the White House and other official government buildings (Lin, 2012). Although no one was injured during this mishap, the outcome could easily have been quite different. So, we can see why there is good reason to be concerned with some of the broader societal implications of malfunctions involving these kinds of computerized/robotic systems.

We should note that malfunctions in computerized systems with “safety-critical” and “life-critical” applications (defined in Section 4.2) are by no means new. Consider an incident, now viewed by many as a classic case of a computer malfunction in the medical field, which occurred in the 1980s. Between 1985 and 1987, a series of malfunctions involving the Therac-25, a computerized radiation therapy machine, resulted in six serious accidents and three deaths. The problem was eventually traced to a bug in a software program that, in certain instances, caused massive overdoses of radiation to be administered to patients.² The Therac-25 case is examined in Section 4.5.2, in our analysis of moral accountability and the problem of “many hands.” Before examining specific cases involving unreliable computer systems in the context of responsibility issues for computing/IT professionals, we first address some foundational questions in professional ethics.

► 4.1 WHAT IS *PROFESSIONAL ETHICS*?

Recall that in Chapter 1 we described professional ethics as one of the three main perspectives through which cyberethics issues can be identified and analyzed. We saw that, when applied to computing, professional ethics is a field of applied ethics concerned with moral issues that affect computer professionals. You may also recall from our discussion of professional ethics in Chapter 1 that Don Gotterbarn suggested that professional ethics is the principal, perhaps even exclusive, perspective through which ethical issues involving the computing field should be examined.³ Although this claim is controversial and will not be further considered here, we have devoted this chapter to an analysis of cyberethics issues from the vantage point of professional ethics.

Why have a category called “professional ethics?” After all, one could reasonably argue that independent of whether a particular moral issue happens to arise in either a professional or a nonprofessional context, ethics is ethics; the same basic ethical rules apply to professionals as to ordinary individuals. In response, many ethicists argue that some moral issues affecting professionals are sufficiently distinct and specialized to warrant a separate field of study. Some ethicists also argue that, at least in certain cases, professionals have special moral obligations,

which exceed those of ordinary individuals. To grasp the essential points in the arguments advanced by these ethicists, it is useful first to understand what is meant by “profession” and “professional.”

4.1.1 What Is a Profession?

The meaning of “profession” has evolved significantly over time. Harris, Pritchard, and Rabins (2009) point out that while a profession was once associated with people “professing a religious or monastic life,” that concept took on a more secular meaning in the late seventeenth century when it was first used to refer to one who “professed to be duly qualified.” According to Davis (2015a), the term “profession” has at least four senses in its everyday usage; that is, it can mean:

1. a vocation (or calling),
2. an occupation,
3. an “honest occupation” (that one may “openly admit to profess”),
4. a “special kind of honest occupation.”⁴

Davis claims that (4), which rules out dishonest occupations (e.g., such as in organized crime) and also emphasizes special skills/abilities/knowledge, is the primary sense in which we think of a profession today.

Some believe that one way of distinguishing a profession from many “ordinary occupations” has to do with certain kinds of characteristics that apply to the former, but not the latter. Having a code of ethics would be one such distinguishing factor. Consider that the field of computing/IT has a number of professional societies with ethical codes (see Section 4.3), as do professions such as medicine and law. However, the computing/IT profession also differs from some of the better-known traditional professions such as medicine and law in at least one key respect. While many doctors and lawyers work in private practice, most computer/IT professionals are not self-employed; even though some work as independent consultants, most are employed by corporations.

Based on criteria used to demarcate traditional professions, one might question whether the computer/IT field is a “true profession.” Barger (2008) argues that it is; in his view, the field qualifies as a genuine profession because it satisfies two key criteria that have traditionally characterized a profession: (i) expert knowledge, which he describes as a “special technical knowledge that is certified by some authority and is not possessed by the layperson” and (ii) autonomy with respect to “independence in conducting one’s professional practice.”⁵

4.1.2 Who Is a Professional?

As in the case of “profession,” the meaning of the term “professional” has also evolved. Traditionally, professionals included lawyers, medical doctors, and professors. In our current and expanded use of that term, we refer to a broader range of professional categories, such as real estate professionals, marketing professionals, and so forth. A defining attribute of traditional professions, such as medicine and law, is that members often find themselves in situations in which their decisions and actions can have significant social effects; for example, medical doctors can prescribe the use of certain drugs for their patients, who otherwise would have no legal access to them, and lawyers are bound by special obligations such as client confidentiality that would not apply if they were acting as ordinary citizens. In these cases, a professional’s roles and responsibilities can exceed those of ordinary individuals. Sometimes, these roles and responsibilities are said to *differentiate* professionals from nonprofessionals.

Buchanan (2004) believes that the roles and responsibilities of professionals are differentiated from ordinary individuals because:

... professionals are experts in a field, which provides them an advantage over the lay person and that professional's work has the potential to impact—either positively or negatively—the general public at large.⁶

Buchanan goes on to note that “information professionals” have the potential to adversely affect an “increasingly large and diverse clientele by failing to act responsibly, fairly, timely, and appropriately.” So it would seem that these roles and responsibilities differentiate professionals working in the fields of computing and IT from ordinary individuals. The extent to which a computer/IT professional's roles and responsibilities are “highly differentiated,” however, is a matter of some dispute. To understand why this is so, it would first help to understand what, exactly, is meant by the expression “computer/IT professional.”

4.1.3 Who Is a Computer/IT Professional?

Broadly speaking, a computer/IT professional is anyone employed in the computing and IT fields—from software and hardware engineers to specialists such as support personnel, network administrators, and computer repair technicians. Computer professionals can also include faculty and instructors who teach in computer science and information management departments at universities, as well as in industry settings. We could possibly even extend “computer/IT professional” to include professionals who are responsible for providing access to electronic information in libraries, although they may prefer to describe themselves as “information professionals” or “information science professionals.”⁷

A computer/IT professional might also be thought of in more narrow terms, in which case only software engineers would be included. Of course, there are various gradients in between the two ends of this spectrum. A computer/IT professional could be defined in a way that would exclude professionals in the fields of communications and library science yet still include professionals whose computer-specific job descriptions extend beyond software engineering per se, such as software technical writers, software quality analysts, and managers and supervisors who play key roles in the software development process and make up a software engineering team.⁸

For purposes of this chapter, we will consider *computer/IT professionals* to include software/hardware engineers and software/hardware engineering teams, as well as computer science instructors in colleges, universities, and industry settings who are responsible for educating and training the members of software engineering teams. We will also include IT professionals in end-user support roles (e.g., network administrators and computer support personnel) in our definition of computer/IT professionals. However, we will not include lawyers, accountants, nurses, or other professionals who are employed by computer companies or who work closely with computers as part of their regular employment.

► 4.2 DO COMPUTER/IT PROFESSIONALS HAVE ANY SPECIAL MORAL RESPONSIBILITIES?

Some ethicists believe that all professionals, regardless of their practice, have special moral obligations *as professionals*. But are there also specific moral obligations that apply to computer/IT professionals in the sense in which we have defined the term? As noted earlier, many computer/IT professionals include software engineers and members of software engineering teams. An important question for us to consider is whether this subgroup of professionals have any special moral obligations that differentiate them from other professionals who work in the computing/IT fields. Gotterbarn (2001) believes that because software engineers and their teams are

responsible for developing safety-critical systems, they have significant opportunities to (i) do good or cause harm, (ii) enable others to do good or cause harm, and (iii) influence others to do good or cause harm. Thus, Gotterbarn suggests that the roles and responsibilities involved in the development of safety-critical systems constitute a differentiating factor.

Safety-Critical Software

What, exactly, is safety-critical software? Bowyer (2001) points out that the phrase “safety-critical system” is often used to refer to computer systems that can have a “direct life-threatening impact.” He notes that, under this definition, examples of safety-critical software applications typically include:

- Aircraft and air traffic control systems
- Mass transportation systems
- Nuclear reactors
- Missile systems
- Medical treatment systems

However, Bowyer believes that an understanding of safety-critical systems that includes only these examples is too narrow. He suggests that a broader definition be adopted in which safety-critical systems are also understood as software applications used in the design of physical systems and structures whose failures can also have an impact that is life threatening. Here, the range of safety-critical applications can be expanded to include software used in the:

- Design of bridges and buildings
- Selection of water disposal sites
- Development of analytical models for medical treatment⁹

We use the expanded sense of “safety-critical software” in this chapter. And we examine two important cases of computer malfunctions involving safety-critical software in our discussion of moral accountability in Section 4.5. A related question is whether the computing/IT profession itself (as opposed to the individuals comprising that profession) has any special moral obligations. Also, we could ask whether certain kinds of computer corporations might have these kinds of moral obligations because of the significant social impact of their products and services. In Section 4.6, we briefly consider whether corporations that develop *autonomous systems*, that is, “machines” capable of making life-critical decisions independent of human oversight, might have some special moral obligations.

► 4.3 PROFESSIONAL CODES OF ETHICS AND CODES OF CONDUCT

We have already seen that many professions have established professional societies, which in turn have adopted codes of conduct. For example, the medical profession established the American Medical Association (AMA), and the legal profession established the American Bar Association (ABA). Both associations have formal codes of ethics/conduct for their members. How do the ethical guidelines/requirements included in professional codes differ from “ordinary morality?” Davis (2015b) argues that these codes, which are “morally binding” on those professionals to whom they apply, can “impose new moral obligations and requirements” on the members of a profession; to illustrate this point, he notes that an employee can be required to sign a contract to uphold an employer’s code of ethics as a condition of employment. So, Davis believes that codes of ethics should not be understood as simply a restatement of “ordinary” or conventional morality.¹⁰

As in the case of the medical, legal, and other well-known professions, the computing profession has also established a number of professional societies, the largest of which are the Association for Computing Machinery (ACM) and the Institute of Electrical and Electronics Engineers Computer Society (IEEE-CS). Both organizations have adopted professional codes of ethics; the full texts of these two codes are included in Appendices A–E (www.wiley.com/college/tavani), as well as at the ACM and IEEE sites.

Both the ACM and IEEE codes contain general statements about what is expected, and in some cases what is required, to be a member in good standing. The IEEE Code of Ethics contains 10 general directives; the first four instruct members to:

1. accept responsibility in making engineering decisions consistent with the safety, health, and welfare of the public . . . ;
2. avoid real or perceived conflicts of interest wherever possible . . . ;
3. be honest . . . ;
4. reject bribery in all its forms . . .¹¹

The ACM Code of Ethics and Professional Conduct, on the other hand, is more complex. It contains 24 imperatives, formulated as statements of personal responsibility. Like the IEEE Code, the ACM Code also lists general moral imperatives:

- 1.1 Contribute to society and human well-being.
- 1.2 Avoid harm to others.
- 1.3 Be honest and trustworthy.
- 1.4 Be fair and take action not to discriminate.
- 1.5 Honor property copyrights and patents.
- 1.6 Give proper credit for intellectual property.
- 1.7 Respect the privacy of others.
- 1.8 Honor confidentiality.

From these general imperatives, a list of more specific professional responsibilities follows. These include the directives:

- 2.1 Strive to achieve the highest quality . . . in . . . work.
- 2.2 Acquire and maintain professional competence.

These directives are then followed by six “organizational leadership imperatives,” which include:

- 3.1 Articulate social responsibilities of members . . .
- 3.2 Manage personnel and resources to design and build information that enhance the quality of working life.

The fourth component of the ACM Code stresses “compliance with the code,” and consists of two imperatives:

- 4.1 Uphold and promote the principles of this code.
- 4.2 Treat violations of this code as inconsistent with membership of the ACM.¹²

4.3.1 The Purpose of Professional Codes

Professional codes of ethics serve a variety of functions. For example, these codes can “regulate” the members of a profession, as Gotterbarn and Miller (2009) point out. Bynum and Rogerson (2004) identify five important functions that professional codes serve: inspiration,

education, guidance, accountability, and enforcement. The authors point out that codes *inspire* members of the profession by “identifying values and ideals” to which members can aspire. Bynum and Rogerson also note that codes *educate* by informing members about the profession’s values and standards and can *guide* members by specifying “standards of good practice.” Additionally, codes describe the level of *accountability* and responsibility that is expected and demanded by a professional society. Finally, codes have an *enforcement* function with regard to behavior deemed to be “ethically unacceptable” in a society.¹³ Martin and Schinzinger (2004) describe this feature as a “disciplinary or penal function” that codes can serve when members violate one or more policies stipulated by the professional code.

To be effective, a professional code must be broad, yet specific. Perlman and Varma (2002) point out that a code must be broad enough to cover the ethical conflicts and concerns likely to arise in its professional field (such as computing/IT), but, at the same time, a code cannot be so broad that it covers extraneous incidents. It must also be sufficiently specific to serve as a guide to making sound decisions for practical action in actual circumstances that are likely to arise in the computing field. To accomplish the first objective, Perlman and Varma believe that a code needs to encompass the principles that guide professions in general and ethics in particular. And to satisfy the second objective, a code is measured by the degree to which its rules serve as “effective guides” for computer/IT professionals and practitioners. Perlman and Varma also note that in engineering contexts, professional codes face a special challenge because the practice of engineering often dictates secrecy, whereas the ethics of engineering requires transparency, or openness.

4.3.2 Some Criticisms of Professional Codes

One might be surprised to learn that professional codes in general, as well as specific codes developed for computer/IT professionals, have been criticized. But we should note that some critics have pointed out that the ethical codes adopted by professional computer societies have no “teeth.” For example, violations of the ACM or IEEE codes, unlike violations of professional codes in the fields of medicine and law, do not necessarily threaten the employment of those who violate them. Also, computer/IT professionals are not usually required to be members of the ACM, IEEE, or any other professional society to be employed in the computing field or to practice as computer/IT professionals.

Critics have pointed out numerous ways in which professional codes in general can be dismissed. For example, Davis (1995) notes that these codes are often perceived as “self-serving, unrealistic, inconsistent, mere guides for novices, too vague, or unnecessary.”¹⁴ To Davis’ list, we could add yet another criticism—professional codes tend to be “incomplete.” Fairweather (2004) believes that codes of conduct for computing professionals have been influenced by a conception of computer/information ethics that is limited to four traditional areas of concern: privacy, accuracy, property, and accessibility. He argues that a professional code built on a foundation that includes such a narrow range of ethical concerns can provide certain loopholes for unethical behavior in an organization.

Ladd (1995) has criticized ethical codes on slightly different grounds, arguing that these codes rest on a series of confusions that are both “intellectual and moral.” His complex arguments can, for our purposes, be summarized in three main points. First, Ladd notes that ethics is basically an “open ended, reflective, and critical intellectual activity.” Because ethics is a field of study that consists of issues to be examined, explored, discussed, deliberated, and argued, it requires a process of deliberation and argumentation. (Recall our definition of ethics in Chapter 2.) Directives listed in a professional code may give an employee the mistaken notion that all he or she needs to do is to locate a directive and then blindly follow it. More importantly, however, Ladd notes that individuals are not told what to do in a situation where two or more principles or directives (in a code) conflict with one another. Here, the individual

needs to deliberate; yet professional codes do not typically provide any hierarchy of principles or any mechanism for choosing one directive over another.

Second, Ladd is critical of codes because of the confusions they introduce with respect to responsibilities involving microethics vs. macroethics issues (i.e., confusions about which responsibilities apply to individual professionals and which responsibilities apply to the profession itself). Recall that we briefly discussed the microethical/macroethical distinction in Chapter 2. In the context of professional ethics, microethical issues apply to personal relationships between individual professionals and other individuals, such as clients. Macroethical issues, on the contrary, apply to social problems that confront members of a profession collectively or as a group. As such, most microethical issues involve the application of ordinary moral notions (such as honesty and civility) that would also hold when dealing with other individuals in nonprofessional contexts. Macroethical issues, however, are more complex, since they involve the formulation of policies at the level of social organizations. Ladd believes that we need to distinguish between questions such as “Which responsibilities do I, as a computer professional, have in such and such a situation?” and “Which responsibilities does the computing profession, as a profession, have in such and such a situation?” He concludes that professional codes of ethics cannot help us to make this important distinction.

Third, Ladd believes that attaching disciplinary procedures and sanctions to codes effectively turns them into legal rules or “authoritative rules of conduct” rather than ethical rules. The role of ethics in general, he argues, is to appraise, criticize, and even defend the principles, rules, and regulations, but it is not to dictate or to punish. Also, when individuals are compelled to obey directives, they are deprived of their autonomy, that is, their ability to choose, which is crucial in moral deliberation. So Ladd argues that professional codes rest on some mistaken notions about both the nature and the purpose of morality, which ultimately result in a series of intellectual and moral confusions.

4.3.3 Defending Professional Codes

It is very important to point out at this stage of our discussion of professional codes that not everyone has been critical of them. In fact, even some critics who have identified specific limitations or weaknesses in professional codes have also defended and praised them. For example, Barger (2003), who concedes that the hope of finding a single professional code that everyone would accept might seem “dim,” also notes that we should not infer that an effort to do so is futile. And Davis, whose criticisms of professional codes were described in the previous section, has argued that codes are extremely important for engineering professionals because they are central to guiding individual engineers in how to behave morally as professionals. Since an engineer cannot always rely on his or her own private conscience when making moral decisions, Davis believes that codes play an essential role in “advising individual engineers in how to conduct themselves.”¹⁵ He also believes that codes of conduct can help individual engineers to better understand “engineering as a profession.” Bynum and Rogerson (2004), who also describe some positive functions that codes serve, point out that we should acknowledge the “limitations of codes.” For example, they note that professional codes should not be viewed simply as laws or algorithms or as “exhaustive checklists.”

Gotterbarn (2000) has suggested that some of the criticism leveled against professional codes might be eliminated if we think of them as serving three important, but distinct, functions, that is,

- Codes of ethics
- Codes of conduct
- Codes of practice

Gotterbarn describes codes of ethics as “aspirational,” because they often serve as mission statements for the profession and can thus provide vision and objectives. Codes of conduct, on the other hand, address the professional and the professional’s attitude and behavior. Finally, codes of practice relate to operational activities within a profession. Gotterbarn points out that the degree of enforcement possible with respect to specific violations of a professional code is dependent on the type of code violated. For example, he notes that violations involving codes of ethics, which are primarily aspirational, are often considered no more than “light concerns.” Consequently, violations of these codes may not have any serious consequences for individuals. Violations involving codes of conduct, on the other hand, can range from warnings given to an individual to the possibility of exclusion from practicing in a profession. Violations of codes of practice go one step further, however, in that they may also lead to legal action.¹⁶

Gotterbarn notes that the hierarchy in the three types of codes parallels the three levels of obligation owed by professionals. The first level includes a set of ethical values, such as integrity and justice, which professionals share with other humans by virtue of a shared humanity. The second level of responsibility is shared by all professionals, regardless of their fields of specialization. The third (and deeper) level comprises several obligations that derive directly from elements unique to a particular professional practice, such as software engineering. This threefold distinction is incorporated in a professional code developed by a joint task force of the IEEE-CS/ACM, which we examine in the next section.

Table 4-1 lists some of the strengths and weaknesses of professional codes.

4.3.4 The IEEE-CS/ACM Software Engineering Code of Ethics and Professional Practice

In the 1990s, the ACM and IEEE approved a joint code of ethics for software engineers: the IEEE-CS/ACM (SECEPP). Gotterbarn (2001) believes that SECEPP, as a professional code for software engineers, is unique for two reasons: First, it is intended as the code for the software engineering profession, unlike the individual codes (such as those of the ACM and IEEE) designed for particular professional societies within the computing profession. Second, Gotterbarn believes that SECEPP is distinctive in that it has been adopted by two international computing societies: ACM and IEEE-CS.

TABLE 4-1 Some Strengths and Weaknesses of Professional Codes

Strengths	Weaknesses
Codes inspire the members of a profession to behave ethically	Codes include directives that tend to be too general and too vague
Codes guide the members of a profession in ethical choices	Codes are not always helpful when two or more directives conflict
Codes educate the members about their professional obligations	Codes comprise directives that are neither complete nor exhaustive
Codes discipline members when they violate one or more directives	Codes are ineffective (have no “teeth”) in disciplinary matters
Codes inform the public about the nature and roles of the profession	Codes sometimes include directives that are inconsistent with one another
Codes “sensitize” members of a profession to ethical issues and alert them to ethical aspects they otherwise might overlook	Codes do not always distinguish between microethics issues and macroethics issues
Codes enhance the profession in the eyes of the public	Codes can be self-serving for the profession

SECEPP is organized into two main parts: a short version and a longer, or full, version. Each version has its own preamble, and the full text for each version is included in Appendix E (www.wiley.com/college/tavani). SECEPP comprises eight core principles:

1. **PUBLIC:** Software engineers shall act consistently with the public interest.
2. **CLIENT AND EMPLOYER:** Software engineers shall act in a manner that is in the best interests of their client and employer, consistent with the public interest.
3. **PRODUCT:** Software engineers shall ensure that their products and related modifications meet the highest professional standards possible.
4. **JUDGMENT:** Software engineers shall maintain integrity and independence in their professional judgment.
5. **MANAGEMENT:** Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.
6. **PROFESSION:** Software engineers shall advance the integrity and reputation of the profession consistent with the public interest.
7. **COLLEAGUES:** Software engineers shall be fair to and supportive of their colleagues.
8. **SELF:** Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.¹⁷

Does SECEPP Provide an Appropriate Balance between Generality and Specificity?

We noted earlier that professional codes are often criticized for being too vague and too general to be useful, yet there is also a danger in being too specific. If a professional code is too specific, it might fail to instruct members about general principles regarding ethical behavior and ethical decision making. SECEPP does include some very specific language, but it also has general prescriptions found in most professional codes. And, at the general level, SECEPP emphasizes the profession's obligation to the public at large, including concern for the public's health, safety, and welfare. For example, in the preamble to the full version of the code, software engineers are encouraged to:

- Consider broadly who is affected by their work
- Examine if they and their colleagues are treating other human beings with due respect
- Consider how the public, if reasonably well informed, would view their decisions
- Analyze how the least empowered will be affected by their decisions

The preamble to the short version of SECEPP summarizes aspirations at a high level of abstraction. The specific clauses included in the full version, on the other hand, give examples of how these aspirations change the way that software engineering professionals act as professionals. The code's principal authors note that “without the aspirations, the details can become legalistic and tedious; without the details, the aspirations can become high sounding but empty; together, the aspirations and the details form a cohesive code.”¹⁸ SECEPP's supporters believe that this code achieves an ideal balance between the general and the specific.

Can SECEPP Avoid Criticisms of Earlier Professional Codes?

One criticism often directed at professional codes is that they are incomplete; recall Ben Fairweather's argument in Section 4.3.2. Another criticism notes that most codes provide no mechanism for choosing between principles when two or more of them conflict; recall

John Ladd's critique, included in Section 4.3.2. Gotterbarn (2000) believes that SECEPP has overcome both difficulties.

Regarding the charge of incompleteness, Gotterbarn is willing to concede that the principles included in SECEPP are not intended to be exhaustive. He also acknowledges that no code could reasonably anticipate every possible moral controversy that can arise. However, he believes that SECEPP addresses the problem of completeness by providing "general guidance for ethical decision making." Gotterbarn argues that ethical tensions that arise can best be addressed by "thoughtful consideration of fundamental principles, rather than blind reliance on detailed regulations." He also points out that SECEPP should not be viewed as a simple algorithm that generates ethical decisions. And Gotterbarn notes that the individual principles that comprise SECEPP are not intended to be used in isolation from one another.

With respect to the second major criticism of professional codes, Gotterbarn points out that SECEPP has a "hierarchy of principles." This hierarchy enables engineers to prioritize their roles and responsibilities and to determine which ones are overriding when two or more conflict. Recall the list of eight principles that make up SECEPP: The ordering of these principles is intended to offer some guidance in cases where two or more rules conflict. SECEPP's hierarchy of principles states that concern for the health, safety, and welfare of the public is primary in all ethical judgments.

In concluding Section 4.3, we acknowledge that professional codes clearly have some limitations. However, we should be careful not to underestimate the important contributions that well-developed professional codes, such as SECEPP, have made to the computing profession so far. Critics may be correct in pointing out that merely following the directives of a professional code can never be a substitute for the kind of careful moral deliberation that is needed in certain controversial cases. Nevertheless, well-developed codes of conduct provide professionals with an important first step in the overall ethical deliberation process.

► 4.4 CONFLICTS OF PROFESSIONAL RESPONSIBILITY: EMPLOYEE LOYALTY AND WHISTLE-BLOWING

What, exactly, is employee loyalty? Do employees and employers have a special obligation of loyalty to each other? Should loyalty to one's employer ever preclude an employee's "blowing the whistle" in critical situations? In which cases can whistle-blowing be justified? Each of these questions is examined in this section.

4.4.1 Do Employees Have an Obligation of Loyalty to Employers?

Many ethicists believe that while loyalty may not be an obligation that is absolute, we nonetheless have a *prima facie* obligation of loyalty in employment contexts. In other words, all things being equal, an employee should be loyal to his or her employer and vice versa. For our purposes, however, we ask, What is the origin of the concept of employee loyalty in an engineering context? Carl Mitcham points out that, historically, engineers believed that they had a basic obligation to be loyal to "institutional authority."¹⁹ He also notes that, originally, an engineer was a "soldier who designed military fortifications or operated engines of war, such as catapults." Mitcham further notes that early civil engineering could be viewed as peacetime military engineering in which engineers were "duty bound to obey their employer," which was often some branch of the government.²⁰ So it may be the case that this historical precedent has indeed contributed to the sense of loyalty that many engineers currently have for their institutions and employers.

Does employee loyalty still make sense in the context of employment in a large computer corporation in the twenty-first century? Skeptics might argue that loyalty makes sense only in

employment contexts where there is a mutual commitment on the part of both parties involved—employer and employee. In good economic times, many employees seem to assume that their employers are very loyal to them. And in the past, it was not uncommon for employees in many professional fields to expect to be able to spend their entire career working for one employer, provided he or she was a loyal employee. But downturns in the U.S. economy, especially during the past three decades, have caused many employers to reassess their obligations to employees. As a result of significant downsizing in many corporations, beginning in the 1980s and continuing into the second decade of this century, employees can reasonably question the degree of loyalty that their employers offer them. In fairness to corporations, however, CEOs also have an obligation of loyalty to stockholders to ensure that the company remains viable and profitable. This obligation can sometimes conflict with a corporation's obligations to its employees.

In one sense, the skeptic's arguments seem plausible. Consider, for instance, how many corporations have gone through downsizing and restructuring phases in which loyal employees who have served a company faithfully for several years have been laid off or dismissed as part of restructuring plans. Also consider that many computer programming jobs in the United States are now being "outsourced" by major corporations to countries where programmers are willing to write the software code for much lower wages.

On the contrary, however, some employers have shown what would certainly appear to be a strong sense of loyalty to employees. For example, there have been instances where an employer will keep an employee on the payroll even though that employee has a chronic illness that causes him/her to miss several months of work. There have also been cases in which several employees have been kept on by a company despite the fact that their medical conditions have caused the corporation's health insurance costs to increase significantly, thereby reducing the company's overall profits. Also, you may have heard of a case involving the owner of Malden Mills whose physical plant in Massachusetts was destroyed by fire. The mill's proprietor, Aaron Feuerstein, could have been excused from any future obligations to his employees, and he could have chosen to rebuild his facility in either a different state (within the U.S.) or a different country where employees would work for lower wages. Instead, Feuerstein continued to pay and provide benefits for his employees while a new facility was being built in Massachusetts. So there have been instances where employers have been very loyal to their employees.

How should the notion of employee loyalty apply in computing/IT contexts? Do computer/IT professionals necessarily have an obligation to be loyal to their employers? Let us assume for the sake of argument that, all things being equal, computer/IT professionals should be loyal to their employers. However, sometimes, an employee's obligation of loyalty can conflict with other obligations, such as those to society in general, especially where health and safety considerations are at stake.

How are computer/IT professionals supposed to balance their obligation of loyalty to an employer against their obligations of loyalty that lie elsewhere? Even if a computer/IT professional has a *prima facie* obligation (i.e., all things being equal) of loyalty to his or her employer, it does not follow that he or she has an absolute obligation of loyalty! Consider that computer/IT professionals, because of the nature of the critical-safety projects they work on, can also have an obligation to society as a whole, especially since safety and health issues can arise. Divided loyalties of this type, of course, can result in serious conflicts for employees, and in certain cases, the moral dilemmas they generate are so profound that an employee must determine whether to blow the whistle.

4.4.2 Whistle-Blowing

What, exactly, is whistle-blowing? We first note that there is no "standard" or universally agreed-upon definition of this controversial activity. However, a plausible definition has been put forth by John Boatright, who describes whistle-blowing as

the voluntary release of nonpublic information, as a moral protest, by a member or former member of an organization outside the channels of communication to an appropriate audience about illegal and/or immoral conduct in the organization that is opposed in some significant way to the public interest.²¹

Boatright's description identifies three key requirements regarding the nature of the *information* revealed in a whistle-blowing act, noting that this information must be:

1. Nonpublic
2. Voluntarily disclosed by a member, or former member, of an organization
3. Presented to an "appropriate" audience

Additionally, Boatright's description requires that the information revealed to the appropriate source is about conduct deemed to be (i) illegal and/or immoral in nature and (ii) harmful to the general public (if not revealed). Each point deserves further development and elaboration; to do that, however, would take us beyond the scope of this section and chapter. For our purposes, we will accept Boatright's description as our working definition of whistle-blowing. We should also note, however, that this definition does not specifically mention any of the kinds, or distinct categories, of whistle-blowing that can be articulated. To that end, Luegenbiehl (2015) and De George (2010) both draw helpful distinctions for differentiating some relevant categories of whistle-blowing. For example, Luegenbiehl distinguishes between what he calls "open" and "anonymous" whistle-blowing, noting that the identity of the whistle-blower is known in the former type but not in the latter. And De George articulates some helpful distinctions that he bases on three contrasting sets of concepts:

- Internal vs. external
- Personal vs. impersonal
- Governmental vs. nongovernmental²²

We briefly consider examples of each. De George's category of "internal whistle-blowing" includes the typical kinds of disclosures made by employees to their supervisors, or by students to their teachers or administrators, that is, *within* their respective organizations. Situations where an employee or a student reports an incident of harassment affecting oneself would fall under De George's category of "personal whistle-blowing," as opposed to what we think of as more typical instances of "impersonal whistle-blowing" that are directed to broader kinds of concerns (i.e., beyond merely the interests of oneself). His category of "governmental whistle-blowing" includes examples in which government employees (as well as employees of firms that have government contracts) divulge abuse (e.g., involving the misuse of power, money, etc.) in one or more government agencies. Instances in which one or more employees—either governmental or nongovernmental employees—go outside the organization to make their revelations fall under De George's category of external whistle-blowing.²³ Note that our focus in this chapter will be mainly on instances of ("open") whistle-blowing that qualify both as impersonal and external, but which also span governmental and nongovernmental contexts.

Having articulated some distinct categories of whistle-blowing, we next ask, Who, exactly, is—i.e., who qualifies as—a legitimate *whistle-blower*? Bok (2003) defines a whistle-blower as an individual who makes "revelations meant to call attention to negligence, abuses, or dangers that threaten the public interest."²⁴ Bok also points out that because whistle-blowers make public their disagreement with their employers or with some authority, their activities can be viewed as a form of "dissent." As Brenkert (2010) notes, however, the whistle-blower's act of (voluntary) disclosure or revelation must also be *deliberate*; in other words, it cannot be accidental.

Are whistle-blowers heroes or are they villains, or perhaps something in between? Although they are sometimes perceived as traitors and snitches, whistle-blowers have also

been viewed by others as brave and heroic individuals. In 2002, for example, three whistle-blowers—Cynthia Cooper (a WorldCom employee), Coleen Rowley (an FBI employee), and Sherron Watkins (an Enron employee)—were named “Persons of the Year” by *Time* magazine. But many whistle-blowers have been maligned by their supervisors and coworkers, and they have sometimes been described as “enemies” of business or of the public good (even in cases where the disclosures they made would seem morally justified in light of the harm that otherwise could have resulted). In the past, many whistle-blowers have put their professional careers on the line, and in some cases, they have even put their lives in danger. So, it would seem prudent for prospective whistle-blowers to be well aware of the risks involved before following through with their actions.

Determining When to Blow the Whistle

When should an employee blow the whistle? Consider an incident in which Colleen Rowley came forth to describe how critical messages had failed to be sent up the Federal Bureau’s chain of command in the days immediately preceding the tragic events of September 11, 2001. Was it appropriate for her to blow the whistle on her supervisor, or was she being disloyal to her supervisor and her fellow employees in doing so? Also, consider employees in positions of authority in corporations and major financial institutions who were in positions to blow the whistle about their companies’ illegal accounting practices, as Cynthia Cooper (at WorldCom) and Sherron Watkins (at Enron) had both done. One could argue that failure to blow the whistle on key individuals and corporations involved in deceptive practices leading up to the 2008–2009 downturn in the U.S. economy resulted in thousands of people losing their retirement savings, and in some cases, their entire lifetimes savings. For example, if someone had blown the whistle on Bernie Madoff (and other notorious financiers who used “Ponzi,” or Ponzi-like, schemes to deceive investors), the financial losses suffered by many investors, as well as the damage to the U.S. economy (in general) during that period, might have far less severe.

In other cases, an employee’s blowing could result in saving human lives. Consider, for example, the Space Shuttle Challenger disaster (in 1986), which resulted in the deaths of the seven crew members. Subsequent reports of this incident in the press revealed that the engineers who designed the space shuttle were well aware of the safety risks in launching the shuttle in cooler temperatures. In fact, some engineers, when learning that the Challenger was scheduled for launch on a cool January morning, went to their supervisors to express their concerns. Despite the worries of these engineers, however, NASA officials elected to stick with the original launch date. Given that the engineers received no support from their supervisors, should they have gone directly to the press? Many believe that the engineers involved had a professional responsibility to blow the whistle on their management team.

We have not yet considered a case of whistle-blowing in a computing/IT context per se. One that occurred in the early 1980s, which some now view as a classic whistle-blowing case in cyberethics, involved a U.S. military proposal called the Strategic Defense Initiative (SDI). SDI, which was more informally referred to in the media during that time as “Star Wars,” was a national missile defense (NMD) system whose purpose was to provide a “defense shield” in the United States against incoming ballistic missiles. While supporters argued that SDI was essential for America’s national security and defense, some critics countered that the system’s software was unreliable and thus could not be counted on to perform adequately. One critic, David Parnas, went public with his concerns about SDI, one of which was based on his conclusion that there would not be sufficient time during an attack to repair and reinstall software that failed.²⁵

Some SDI supporters accused Parnas of being disloyal, as well as acting out of his own self-interest. With respect to the latter charge, however, Parnas’ defenders pointed out that he had walked away from a very lucrative consulting contract. (As a consultant on the SDI project in the early 1980s, he was paid \$1,000 per day for his expertise.) Yurcik and Doss (2002)

believe some of the concerns expressed by Parnas also apply in the controversy surrounding the most recent proposal for an SDI-like initiative—namely, the NMD proposed by the George W. Bush administration in 2001. (Recall that in Chapter 3, we briefly described that version of NMD in connection with our sample arguments illustrating both valid and invalid logical forms.) We next consider a more recent case of whistle-blowing that also involves a national security-related concern.

► SCENARIO 4-2: NSA Surveillance and the Case of Edward Snowden

In May 2013, Edward Snowden, a former employee of the National Security Agency (NSA) contractor Booz Allen Hamilton, leaked sensitive national security-related material both to *Washington Post* and *The Guardian*. The leaked reports revealed that the NSA used at least three Internet surveillance programs, known internally as Tempora, PRISM, and XKeyscore. The reports also revealed that the NSA collected “metadata” from telephone communications that it had intercepted both in the United States and Europe. Many of Snowden’s critics have since claimed that the sheer scale of the material involved in this incident makes it the “most significant leak” in U.S. history. Whereas some of Snowden’s critics have described him as a dissident and a “traitor” (who has also caused grave damage to U.S. intelligence capabilities), at least some of his defenders view him as a “hero” and a “patriot” (in part, at least, because they believe that Snowden’s disclosure showed how the U.S. government had gone too far in its surveillance practices on its own citizens as well as on many of the leaders of closely allied nations).²⁶ ■

Does Snowden’s voluntary and deliberate act of leaking this information to the press qualify as a genuine instance of whistle-blowing? As we saw in Chapter 3, some of Snowden’s critics claimed that he was “not a whistle-blower” but instead a traitor, a liar, and a narcissist. But as we also noted in Chapter 3, some of these attacks commit logical fallacies (such as the *Ad Hominem* fallacy and the False Dichotomy). We saw, for example, that even if Snowden were a narcissist (an *ad hominem* attack on Snowden as a person), that would not necessarily preclude his having been a (genuine) whistle-blower as well. And one’s being both a traitor and a whistle-blower is not a contradiction, since all whistle-blowers (by definition) betray the trust of an organization on which they blow the whistle. So, criticisms to the effect that someone is *either* a whistle-blower *or* an organizational betrayer commit the False Dichotomy Fallacy.

Although the kinds of arguments described in preceding paragraph may fail to establish the conclusions drawn by Snowden’s critics, we can still ask whether Snowden’s actions meet the conditions for being a (genuine) whistle-blower. Following the criteria put forth by Bok (described previously), it would appear that Snowden had indeed made “revelations” that were meant to call attention to abuses that “threaten the public interest,” and that his act could also be viewed as a form of “dissent.” And in compliance with Brenkert’s insight, Snowden’s act was “deliberate” (and not accidental). So, Snowden’s actions seem to meet the relevant criteria. As to questions about which kind or category of whistle-blowing Snowden engaged in, we can see that it complies with Luegenbiehl’s criterion for “open” (as opposed to anonymous). And, finally, Snowden’s actions would seem to satisfy De George’s criteria for *external, impersonal, and governmental* whistle-blowing. However, whether Snowden was also morally justified in blowing the whistle is a question that has been and continues to be hotly disputed.

We should note that Snowden, in defending his actions (as an American citizen), pointed out that he took an oath to support and defend the U.S. Constitution, whose fourth and fifth Amendments forbid “systems of massive, pervasive surveillance.”²⁷ Snowden also claimed that his “sole motive” for leaking the documents was “to inform the public as to that which is done in their name and that which is done against them.”²⁸ Nevertheless, in June 2013, the U.S. government officially charged Snowden with “theft of government property, unauthorized communication of national defense information” and “willful communication of classified communications intelligence information to an unauthorized person.”²⁹

Can Snowden's actions be defended on moral grounds, even if they violated one or more U.S. laws? Unfortunately, much of the highly charged rhetoric on both sides of this debate has failed to address the key question of whether Snowden's behavior was morally justifiable. (In fact, we have already noted some of rhetoric in the *ad hominem* attacks against Snowden.) To analyze the moral aspects of this case, we need a viable framework or model of whistle-blowing. While numerous frameworks have been advanced, Brenkert (2010) notes the most popular have been variations of what he calls the "harm theory" of whistle-blowing. He also points out that the most well-known version of this theory is the model articulated by De George (1999, 2010). We next examine that model, which draws an important distinction between whistle-blowing actions that are morally permissible and those that are morally obligated.³⁰

De George's Model for Evaluating Whistle-Blowing

De George proposes a series of conditions for determining when an employee is (i) morally permitted to blow the whistle and (ii) morally obligated to do so. He believes that employees are morally permitted to go public with information about the safety of a product, or a policy, when three key conditions are satisfied. For our purposes, we can summarize De George's three conditions as follows:

1. The "policy" will do "serious and considerable harm" to the public.
2. The employee(s) have reported the "serious threat" to their immediate supervisor and have made "their moral concern known."
3. The employee(s) have exhausted the "internal procedures and possibilities" within the organization, including going to the board of directors, if necessary.³¹

While De George argues that one is morally permitted to blow the whistle when conditions 1–3 are satisfied, he does not believe that a person is yet morally required to do so. To have a strict moral obligation to blow the whistle, De George believes that two additional conditions must be satisfied:

4. The employee(s) have "accessible, documented evidence that would convince a reasonable, impartial, observer that one's view of the situation is correct."
5. The employee(s) have "good reasons to believe that by going public the necessary changes will be brought about."³²

Applying De George's Model in the Snowden Case

We should note that De George includes a caveat with Condition 5, pointing out that the whistle-blower's chance of "being successful must be worth the risk one takes and the danger to which one is exposed."³³ We should also note that De George intends his model to be applied in *nongovernmental* cases of whistle-blowing, which would seem to rule out the Snowden case (since Snowden was an employee of a company that had contracts with the U.S. government). But De George also suggests that his model might work in some instances of external whistle-blowing involving governmental organizations/agencies as well.³⁴ So, it would be interesting to see whether his model could apply in the Snowden controversy (which on De George's criteria would be an instance of "governmental, external, impersonal" whistle-blowing).

In extending his model to this case, we begin by asking whether Snowden's actions satisfy De George's first three criteria, which pertain to "moral permissibility." Arguably, Snowden had satisfied De George's first criterion, or condition, insofar as Snowden showed how the NSA's massive surveillance techniques targeting the phone uses and Internet practices of U.S. citizens caused "serious and considerable harm" to the public. Even if one objects that no American citizens were "physically harmed" because of the NSA's surveillance techniques, the psychological harm caused to U.S. citizens as well as the international fallout resulting from the disclosure that "friendly" heads of states were among those being monitored (in the NSA

surveillance scheme) would clearly seem to have caused “considerable harm” to American citizens as well as to the U.S.’s reputation abroad.

It is more difficult to show that Snowden had also satisfied De George’s second and third conditions. But we should note that Snowden has claimed (in various interviews, including a televised *NBC News* interview with Brian Williams) that he had indeed gone to his immediate supervisors first.³⁵ So, he may have satisfied Condition 2. However, it is less clear that Snowden satisfied De George’s third condition, that is, by going up the appropriate chain of command. And if Snowden had failed to do that, it would seem to follow (on De George’s criteria) that Snowden’s act of whistle-blowing was not morally permissible. Yet, some of Snowden’s supporters believe that he had good reasons for not pursuing the chain of command.³⁶

If, however, Snowden failed to satisfy the third condition in De George’s model, we could infer that Snowden was not morally permitted to blow the whistle. Yet, we also noted that if some of Snowden’s supporters are correct, there may have been very good reasons for why Snowden did not (and could not successfully) comply with Condition 3. So, given these mitigating circumstances, it still might be useful for us to go ahead and ask whether Snowden’s actions satisfied De George’s fourth and fifth conditions. Consider that Snowden’s actions seem to comply with Condition 4, in that Snowden arguably had “documented evidence that would convince a reasonable, impartial, observer that [his] view of the situation is correct.” Also, consistent with Condition 5 in De George’s framework, Snowden had “good reasons to believe that by going public the necessary changes w[ould] be brought about.” So, it is possible that Snowden may well have satisfied De George’s criteria for being “morally required” to go public, even if he failed to satisfy one (or more) of De George’s first three conditions for being “morally permitted” to blow the whistle.

Of course, we can question whether De George’s criteria are too stringent, as some critics have suggested, or too lenient, as others have argued. We could also question whether De George’s model puts too much emphasis on the criterion of “serious harm,” at the expense of other important criteria that also apply in evaluating whistle-blowing cases; this critique has also been made by some of De George’s critics.³⁷ However, we will not pursue those criticisms here. And, in fairness to De George, we should also note that he intends to limit the application of his model to nongovernmental cases whistle-blowing. Nevertheless, we chose to apply it in the Snowden case because his model is both well known and well respected and because De George himself has also suggested that it might be extended to some governmental cases as well.³⁸ We should also note that a virtue of De George’s model is that it has been very helpful in analyzing instances of whistle-blowing in engineering contexts (including the classic Ford Pinto case). His model also provides engineers with a fairly clear and straightforward set of guidelines to consider in morally complex cases, without also requiring them to be what some call “moral saints” and what De George calls “moral heroes.”³⁹

We conclude this section by noting that much more could be said about whistle-blowing and about the Snowden case. We also note that other prominent controversial cases associated with whistle-blowing were not examined here. For example, one might ask why the WikiLeaks controversy was not considered. That controversy is examined instead in Chapter 7, in Section 7.9.

► 4.5 MORAL RESPONSIBILITY, LEGAL LIABILITY, AND ACCOUNTABILITY

So far, our examination of issues involving the moral responsibility of computing/IT professionals has focused mainly on questions concerning employee loyalty and whistle-blowing. We have seen that some of these questions have centered on the responsibilities of computing/IT professionals as individuals, whereas others have dealt with responsibilities facing the computing/IT

profession. We have also noted that these questions illustrate essential differences between issues of responsibility involving microethics and macroethics. However, we have not yet fully considered the concept of moral responsibility itself.

Philosophers often describe the concept of moral responsibility in terms of two conditions that must be satisfied: causality and intent. In other words, some agent, X, is held morally responsible for an act, Y, if X caused Y. Here, a person could be held responsible even if he or she did not intend the outcome. Consider a scenario in which a camper, whose careless behavior in failing to put out a camp fire properly resulted in a major forest fire that caused millions of dollars in damage, claims that he did not intend the damage that resulted. Nonetheless, the camper can be held responsible as a moral “agent” for the outcome caused by his careless behavior. Agents can also be held morally responsible when they intend for something to happen, even if they ultimately fail to cause (or to bring about) the intended outcome. For example, consider a scenario in which a disgruntled student intends to blow up a college’s computer lab, but at the last minute is discovered and prevented from doing so. Even though the student failed to carry out his objective—cause the bomb to detonate in the computer lab—we hold the student morally culpable because of his intentions.

4.5.1 Distinguishing Responsibility from Liability and Accountability

It would be helpful at this point to distinguish responsibility from the related notions of liability and accountability. Responsibility differs from liability in that the latter is a legal concept, sometimes used in the narrow sense of “strict liability.” To be strictly liable for harm is to be liable to compensate for it even though one did not necessarily bring it about through faulty action. Here, the moral notion of blame may be left out. A property owner may be legally liable for an injury to a guest who falls in the property owner’s house, but it does not necessarily follow that the property owner was also morally responsible for any resulting injury. (We elaborate on the notion of liability in Section 4.5.3, where we examine some legal liability issues affecting producers of defective computer software.)

Nissenbaum (2007) distinguishes between responsibility and accountability by suggesting that responsibility is only part of what is covered by the “robust and intuitive notion of accountability.” In Nissenbaum’s scheme, accountability is a broader concept than responsibility, and means that someone, or some group of individuals, or perhaps even an entire organization, is “answerable.” In a computing context, she notes that accountability means:

there will be someone, or several people to *answer* . . . for malfunctions in life-critical systems that cause or risk grave injuries and cause infrastructure and large monetary losses . . .⁴⁰

Table 4-2 summarizes the elements we have used to differentiate moral responsibility, legal liability, and accountability.

TABLE 4-2 Responsibility, Legal Liability, and Accountability

Moral Responsibility	Legal Liability	Accountability
Attributes blame (or praise) to individuals	Does not attribute blame or fault to those held liable	Does not necessarily attribute blame (in a moral sense)
Usually attributed to individuals rather than collectivities or groups	Typically applies to corporations and property owners	Can apply to individuals, groups of individuals, and corporations
Notions of guilt and shame apply, but no legal punishment or compensation need result	Compensation can be required even when responsibility in a formal sense is not admitted	Someone or some group is answerable (i.e., it goes beyond mere liability)

Copyright © 2015, Wiley. All rights reserved.

Nissenbaum believes that the notion of accountability has been “systematically undermined” in the computer era, despite the fact that we are increasingly dependent on safety-critical and life-critical systems controlled by computers. In Section 4.2, we saw that safety-critical applications included software used in aircraft and air traffic control systems, in nuclear reactors, in missile systems, and in medical treatment systems, as well as in the selection of water disposal sites and in the design of bridges and buildings. Nissenbaum argues that a major barrier to attributing accountability to the developers of safety-critical software is the problem of “many hands.”

4.5.2 Accountability and the Problem of “Many Hands”

Computer systems are typically developed in large organizational settings. Because these systems are the products of engineering teams or of corporations, as opposed to the products of a single programmer working in isolation, Nissenbaum notes that “many hands” are involved in their development. Thus, it is very likely that no single individual grasps all of the code used in developing a particular safety-critical system.⁴¹ As a result, it is difficult to determine who exactly is accountable whenever one of these safety-critical systems results in personal injury or harm to individuals; it is not always clear whether the manufacturer of the system hardware (the machine itself) or the engineering teams that developed the system’s software, or both, should be held accountable.

When thinking about the problem of many hands from the perspective of strict moral responsibility, as opposed to accountability, two difficulties arise: First, we tend to attribute moral responsibility for an accident to an individual, but not to groups or “collectivities.” Thus, we sometimes encounter difficulties when we try to attribute blame to an organization. Nissenbaum suggests that by using “accountability,” we can avoid the tendency to think only at the level of individuals in matters typically associated with assigning moral responsibility.

The second difficulty arises because the concept of moral responsibility is often thought of as exclusionary, as Ladd (1995) points out. In other words, if we can show that A is responsible for C, then we might infer that B cannot be held responsible for C. Ladd believes that moral responsibility should be viewed as nonexclusionary because both A and B (and possibly others) can be responsible for C. Nissenbaum suggests that we can avoid this confusion if we use “accountability” instead of “responsibility”; in her scheme, holding Sally accountable for making unauthorized copies of proprietary software does not necessarily preclude holding Harry accountable as well (e.g., if Harry pays Sally for making copies of the pirated software).

So Nissenbaum believes that holding one individual accountable for some harm need not necessarily let others off the hook, because several individuals may be accountable. Nor does it mean letting organizations off the hook, because they too may be accountable. As Nissenbaum puts the matter, “We should hold each fully accountable because many hands ought not necessarily lighten the burden of accountability.” The following scenario (involving the Therac-25 machine, which we described briefly in the opening section of this chapter) illustrates how the involvement of “many hands” can obscure the process in determining accountability for accidents affecting some safety-critical systems.

▶ SCENARIO 4-3: The Case of the *Therac-25 Machine*

The Therac-25 was a computer-controlled radiation treatment machine built by Atomic Energy of Canada Limited (AECL). Between 1985 and 1987, six reported incidents occurred in which patients were severely overdosed by the Therac-25. As a result of the severe burns caused by the excessive radiation dosages, three individuals died, and three others had irreversible injuries.⁴² ■

Eventually, the Therac-25 malfunction was traced not to a single source but to numerous faults, including two significant software coding errors (“bugs”) and a faulty microswitch. One bug involved radiation dosage errors: If a subsequent request to change a previously entered radiation dosage was entered in a certain sequence through the user interface, a software bug caused the new entry to be ignored, and the entry instructing the original dosage was used. Instead of receiving 200 RADs, one radiation patient received 20,000 RADs; this patient died shortly afterward from excessive radiation (Leveson and Turner 2001).

Several interesting questions regarding the locus of responsibility were raised in the wake of the Therac-25 accidents. Who, exactly, should be held accountable for the deaths and injuries that resulted from the computer malfunction? Is the hospital, which may be found legally liable, also accountable in a larger sense? Should the company that manufactured and sold the Therac system to the hospital be held accountable? Were the engineers and programmers who worked on the design and development of this system ultimately responsible for the injuries and deaths caused to the radiation patients? Should they be held legally liable?

Nissenbaum argues that guidelines for producing safer and more reliable computers should include a standard of care that incorporates (i) a formal analysis of system modules (as well as of the entire system), (ii) meaningful quality assurance and independent auditing, and (iii) built-in redundancy.⁴³ She believes that a standard comprising these criteria would provide a nonarbitrary means for determining accountability; it would offer a tool for distinguishing between malfunctions due to inadequate practices and those that occur in spite of a programmer or designer’s best efforts. Nissenbaum also suggests that if such standard had existed at the time Therac-25 was produced, the software developers of the system could have been held accountable for the deaths and injuries that resulted.

4.5.3 Legal Liability and Moral Accountability

In Section 4.5.1, we saw that legal liability can be distinguished from both moral responsibility and accountability. Nissenbaum believes that in computer/IT contexts, it is important to keep “accountability” distinct from “liability to compensate.” She concedes that liability offers a partial solution to problems resulting from computer malfunctions, because at least it addresses the needs of the victims; however, she also notes that accepting liability as a substitute for accountability can further obscure the process of determining who is accountable for computer malfunctions.

Is it reasonable to hold computer corporations legally liable for products they sell that are either unreliable or defective? Supporters of liability law believe that holding owners legally liable makes sense because owners are typically in the best position to directly control their property. Nissenbaum believes that because ownership implies a “bundle of rights,” it should also imply responsibilities such as being liable. Software owners (who are also usually the software’s producers) are in the best position to directly affect the quality of the software they release to the public. Yet, ironically, the trend in the software industry, Nissenbaum points out, is to “demand maximal property protection while denying, to the extent possible, accountability.” Software manufacturers frequently include disclaimers of liability on their products such as “This software is sold as is.”

Nissenbaum also suggests that strict liability would shift the accountability to the producers of defective software, thereby addressing an anomaly (perhaps even a paradox) with respect to our current understanding of overall accountability: While liability laws protect the public against potential harm, most producers of software deny accountability for software errors. Producers sometimes base their argument on the notion that software is prone to error in ways that other technologies are not. Nissenbaum concludes that strict liability laws can send a message cautioning software producers to take extraordinary care to produce safe and reliable systems.

► 4.6 DO SOME COMPUTER CORPORATIONS HAVE SPECIAL MORAL OBLIGATIONS?

Before concluding our chapter on professional responsibility, we return to a question raised in Section 4.2 concerning whether computer/IT professionals in particular or the computing profession in general has any special moral responsibilities. There, we focused on (microlevel) professional responsibility-related issues for those individuals (e.g., software engineers) responsible for developing life-critical and safety-critical applications. However, we postponed our consideration of whether any computer corporations themselves, by virtue of the social impact of their products and services, might also have some special moral obligations to society (a macrolevel, professional responsibility concern). In Section 4.5, we saw that it indeed made sense to attribute moral accountability to computer corporations (as well as to the individuals who comprise them). So, we are now in a much better position (than we were in Section 4.2) to ask whether any of those corporations might also have some special moral obligations that differentiate them from other kinds of corporations.

Among the kinds of computer companies that might have special moral obligations, at least two would seem to be plausible candidates: (i) major search engine companies and (ii) companies that develop autonomous systems. Regarding (i), many have pointed out the critical roles that search engines play in our current information society, including the dissemination of knowledge. Hinman (2005), for example, has argued that because of their privileged place in our society as “gatekeepers of the Web,” major search engine companies should shoulder significant social responsibility. However, we will not analyze his argument here.⁴⁴ Instead, we will focus on (ii) in the remainder of this section.

Why should companies that develop autonomous systems have special responsibilities to society? We begin by noting that an influential report published in the United Kingdom by The Royal Academy of Engineering (2009) pointed out that *autonomous systems*—from “unmanned vehicles and robots on the battlefield, to autonomous robotic surgery devices, applications for technologies that can operate without human control, learn as they function and ostensibly make decisions”—will soon be available and that these systems raise a number of ethical, legal, and social issues. To this list, we could also add professional responsibility issues for the companies that develop these systems.

In Scenario 4–1, we examined an actual case that closely mirrors one kind of concern described in the Royal Academy’s report—that is, the malfunction of a semiautonomous cannon designed to make some decisions “independent of human oversight.” Recall that this incident, involving a routine test of a military weapon system in the Republic of South Africa, resulted in human casualties. We can easily imagine similar kinds of incidents occurring in the near future where other autonomous systems are involved.

However, it is not only with life-critical applications that we need to worry about autonomous systems malfunctioning. For example, Wallach and Allen (2009) present a hypothetical scenario (set in the near future) in which autonomous software agents have been designed to make financial recommendations for their human clients about when to buy and sell stock. A cluster of financial transactions initiated by these autonomous agents then triggers a series of cascading events in which other agents make decisions that ultimately result in an international catastrophe where (i) hundreds of people lose their lives and (ii) billions of dollars are lost in the financial sector. Wallach and Allen ask whether disasters of this kind could be avoided if autonomous systems are embedded with the appropriate kinds of ethical decision-making procedures (software code) that will enable them to make “good moral decisions.”

Should the companies that develop these systems be held responsible for “moral-decision-making software code” that they build into them? Wallach, Allen, and others suggest that they should.⁴⁵ But to whom, exactly, should these companies be responsible in the event that the

autonomous machines they develop result in accidental human deaths or in severe economic loss? The lack of a clear answer to this question would seem to suggest that explicit policies and laws are needed to anticipate, rather than have to react to, this and related concerns in light of the profound societal impact that autonomous systems will likely have.

In Chapter 12, we examine some broader ethical issues involving the development and use of autonomous machines. There, for example, we consider some implications that the future development of these machines will likely have for our notions of autonomy and trust. In this chapter, however, we simply pose the question of whether the companies that develop autonomous machines may have some special moral responsibilities to society. It would seem that a plausible case can be made for the view that they do.

We conclude this section by noting that there may indeed be other kinds of computer corporations that also have special responsibilities to society in light of the significant social impacts of their products and services. In some ways, concerns of this type are more appropriately analyzed under the category “business ethics.” To the extent that these concerns particularly affect computer/IT professionals, however, they also warrant discussion within the context of cyberethics (and professional responsibility) as well.

► 4.7 CHAPTER SUMMARY

In this chapter, we examined some ethical problems that confront computer/IT professionals, and we focused on some specific issues and challenges that software engineers and their team members sometimes face in the development of safety-critical software. We saw that professional codes of ethics/conduct are useful insofar as they inspire and educate professionals entering and working in the fields of computing and IT. We also saw, however, that many professional codes have serious limitations. We noted that the IEEE-CS/ACM SECEPP was designed to avoid many of the shortcomings of earlier professional codes of ethics.

We also considered questions having to do with employee loyalty, especially in cases where computer/IT professionals have conflicting obligations involving their employers and the public good. We then considered criteria for determining when these professionals would be morally justified in blowing the whistle; in particular, we examined conditions for distinguishing between when professionals may be “permitted” to blow the whistle vs. situations in which they might also be “required” to blow the whistle. Next we saw that because the notion of accountability in the context of computing has become diffused by the “problem of many hands,” it is not always easy to determine where accountability and responsibility for computer malfunctions and errors in safety-critical software systems ultimately lie. We also saw why it was important to differentiate among the concepts of accountability, responsibility, and liability. Finally, we asked whether some computer corporations may have special moral responsibilities due to the significant societal impact of the technologies they develop.

Among the relevant professional ethics-related concerns not considered in this chapter are issues involving the notions of risk and risk analysis in the context of computing/IT systems; these are examined in Chapter 6. Also not examined in this chapter are professional ethics issues surrounding “open” vs. “closed” software development methodologies. Some ethical aspects of methodologies used in software development are examined in Chapter 8 in our discussion of open-source software (OSS) and free software foundation (FSF) development. An ethical issue that also has implications for computer professionals but not examined in this chapter involves the global outsourcing of programming jobs. This controversy is briefly considered in Chapter 10 in connection with our discussion of globalization and the

transformation of work. Another professional-ethics-related question not considered in this chapter is whether computer/IT professionals should participate in nanocomputing research and development, given certain kinds of controversies surrounding nanotechnology. This question is briefly examined in Chapter 12.

► REVIEW QUESTIONS

1. What is professional ethics?
2. What is a profession, and who is a professional?
3. Who is a computer/IT professional?
4. Do computer/IT professionals have special moral responsibilities that ordinary computer users do not have?
5. How do Don Gotterbarn, Keith Miller, and Simon Rogerson propose that we define the profession of software engineering?
6. According to Gotterbarn, what responsibilities do software engineers and their teams have that differentiate them from other professionals working in the computer field?
7. How is “safety-critical software” defined by Kevin Bowyer?
8. What are professional codes of ethics, and what functions do they serve?
9. Briefly describe John Ladd’s criticisms of professional codes.
10. Explain Gotterbarn’s threefold distinction: codes of ethics, codes of conduct, and codes of practice.
11. Do Gotterbarn’s distinctions help to eliminate any of the criticisms that have been raised against professional codes?
12. How does the IEEE-CS/ACM SECEPP improve on earlier professional codes affecting software engineers?
13. What, exactly, is whistle-blowing? What are the key elements in John Boatright’s definition of whistle-blowing?
14. How does Richard De George differentiate the following whistle-blowing categories: internal vs. external, personal vs. impersonal, and governmental vs. nongovernmental?
15. According to Sissela Bok, what conditions must an employee satisfy to qualify a legitimate whistle-blower?
16. Can professional codes of conduct adequately guide engineers in determining when it would be appropriate to blow the whistle and when it would not?
17. What does Helen Nissenbaum mean by “accountability” in a computing context?
18. How is Nissenbaum’s notion of accountability different from the concepts of liability and responsibility?
19. What does Nissenbaum mean by “the problem of many hands” in a computing/IT context?
20. Why do some computer corporations, including companies that develop autonomous systems, have special societal obligations?

► DISCUSSION QUESTIONS

21. Evaluate Richard De George’s criteria for when it is morally permissible, as opposed to when it is morally required, for an engineer to blow the whistle (described in Section 4.4.2). Apply these criteria to a recent controversy where you believe that blowing the whistle would have been morally permissible or perhaps even morally required.
22. Describe some virtues of the ethical codes of conduct adopted by professional societies such as the ACM and IEEE-CS, and list some shortcomings of these professional codes as well. In the final analysis, do the advantages of having a code outweigh the prospects of not having one? Use either an actual or a hypothetical case to establish the main points in your answer. Do you believe that a coherent and comprehensive code of conduct for the computing/IT profession is possible? Does SECEPP satisfy those conditions?
23. Recall the various arguments that we examined as to when it is appropriate, and sometimes mandatory, for software engineers and IT professionals to blow the whistle. The criteria for when whistle-blowing is permissible, at least for those working in some federal government agencies, changed following the September 11, 2001, terrorist attacks. In November 2002, the Homeland Security Act was passed in both houses of Congress and was signed into law by former President George W. Bush. On one interpretation of the revised law, whistle-blowing acts similar to that of Colleen Rowley—who blew the whistle on her FBI superiors who failed to act on information they received in the days preceding September 11—would be illegal and thus a punishable offense. What implications could this have for software engineers and other computer professionals whose employment comes under the auspices of the Homeland Security Department? In this case, what set of rules should computer/IT professionals follow?

Scenarios for Analysis

1. Describe the process of ethical deliberation that you would use in trying to resolve the following dilemma.

You have been working for the XYZ Computer Corporation as an entry-level software engineer since you graduated from college last May. You have done very well so far; you are respected by the management team, are well liked by your fellow engineers, and have been assigned to a team of engineers that has consistently worked on the most critical and valued projects and contracts that XYZ Corp. has secured. XYZ's most recent contract is for a U.S. defense project involving the missile defense system, and again you have been assigned to the engineering team that will develop the software for this project. However, you are staunchly opposed to the project's objectives, so you ask to be reassigned. Your supervisor and coworkers, as well as upper management, are disappointed to learn of your strong feelings about this project. You are asked to reconsider your views, and you are promised a bonus and a substantial pay increase if you agree to work on this project during the next year. You also discover from a colleague that refusing to work on this project would greatly diminish your career advancement at XYZ and may even make you vulnerable in the event of future layoffs. To compound matters, you and your spouse are expecting your first child in about three months, and you recently purchased a home. What would you do in this case?

2. In analyzing the following scenario, what course of action would you take? Would you be willing to blow the whistle? Explain your position.

For the past six months, you have worked on a project to develop a transportation-related soft-

ware program for the city of Agropolis, a project designed to make some much needed improvements to Agropolis's system of public transportation. You and your team of programmers have worked very hard on this project, but you have encountered difficulties that could not possibly have been anticipated in the original design plan; these difficulties have put your project significantly behind schedule. The city transportation planners are nervous, because they depend on the software from your company to get the new transportation system up and running. And the management at your company is very uncomfortable because it signed a contract to deliver the required software on time. Although the software is not yet foolproof, testing so far reveals that it works about 99% of the time. The few glitches that remain apply only to the transportation system's backup code, which arguably would be needed in only the most severe emergencies. Residents of the city are also eager to have the new transportation system in place.

A decision is made by the management of your company and by the managers of the city transportation system to go ahead and implement the software as it is. They base their decision on the probability that a backup system would not be needed for several months (at which time the remaining bugs should be fixed). A decision was also made by the management groups on both sides not to announce publicly that the software still has a few bugs. You and a few of your coworkers believe that the bugs are more dangerous than the managers are willing to admit. What would you do in this case?

► ENDNOTES

1. See Shachtman (2007) for an interesting account of this incident, which is also briefly described in Wallach and Allen (2009).
2. See Leveson and Turner (2001).
3. See, for example, Gotterbarn (1995).
4. See Davis (2015a, pp. 489–90) for more details on these distinctions.
5. Barger (2008, p. 85).
6. Buchanan (2004, p. 620).
7. Buchanan suggests that an “information professional” can be conceived of along these lines.
8. Gotterbarn, Miller, and Rogerson (1999) suggest that a “software engineering team” can be thought of in terms of as those who contribute by direct participation to “the analysis, specification, design, development, certification, maintenance and testing of software systems.”
9. See Bowyer (2001, p. 165).
10. See Davis (2015b, pp. 283–9).
11. The full text of IEEE Code of Ethics is available at http://www.ieee.org/web/membership/ethics/code_ethics.html.
12. The full text of the ACM Code of Ethics is available at <http://web.mit.edu/afs/athena.mit.edu/course/2/2.95j/Codes-of-Ethics/ACM-Code-of-Ethics.html>.
13. For a full description of these five features, see Bynum and Rogerson (2004, pp. 135–36).
14. Davis (1995, p. 586).
15. *Ibid.*
16. See Gotterbarn (2000, pp. 209–10) for more detail on these three distinctions.
17. The text of both versions of the SECEPP Code (short and full) is available at <http://seeri.etsu.edu/Codes/TheSECode.htm>.

18. See Gotterbarn, Miller, and Rogerson (1999).
19. Mitcham (1997, p. 262).
20. *Ibid.*
21. Boatright (2000, p. 109).
22. De George (2010, p. 300).
23. *Ibid.*
24. Bok (2003, p. 53). We should note that one virtue of Bok's definition is that it explicitly acknowledges the role that negligence can play independently of any specific and overt abuses.
25. See Parnas (1990).
26. Much has been written about the Edward Snowden controversy in the context of whistle-blowing. For an analysis of this (whistle-blowing) case from the perspectives of trust and betrayal, see Tavani and Grodzinsky (2014).
27. See Snowden (2013) for more detail. See also Griggs and Gross (2014).
28. Rieder (2013).
29. See Finn and Horwitz (2013).
30. In this section, we focus on the whistle-blowing model articulated in De George (2010). An earlier version of that model, which focuses on the classic "Pinto Case" in the context of an engineering environment, is included in De George (1999).
31. The full text for these three conditions is included in De George (2010, pp. 306–8). In Condition 1, De George also includes the notion of a "product," as well as a policy, that can cause serious harm. And he uses the expression "the firm," where we use "organization."
32. De George's fourth and fifth conditions are also presented in summarized form here. See De George (pp. 310–11) for the full text of his two conditions for being "morally required" to blow the whistle.
33. De George, p. 311.
34. *Ibid.*, p. 312.
35. See Esposito, Cole, and Schone (2014).
36. For example, Daniel Ellsberg (a former whistle-blower who leaked the notorious "Pentagon Papers" in the early 1970s) believes that the reason Snowden did not pursue the appropriate chain of superiors in his organization was because he saw what had happened to those employees who had done so in previous incidents (Democracynow.org, 2014). So Ellsberg suggests that Snowden had reasonable grounds for not complying with the kind of requirement stated in De George's third condition.
37. For example, Brenkert (2010, pp. 567–70) has criticized De George's whistle-blowing model for placing too much emphasis on the criterion of "harm."
38. See, for example, De George (2010, p. 312).
39. See De George (1999, p. 184).
40. Nissenbaum (2007, p. 274). [Italics Added]
41. *Ibid.*, p. 275.
42. For more detail, see Leveson and Turner (2001).
43. See Nissenbaum, p. 279. Note that she also includes some additional guidelines, such as "independent auditing" and "excellent documentation."
44. For a more detailed discussion of Hinman's argument in this context, see Tavani (2012).
45. See, for example, the views expressed on this topic by various contributors in Anderson and Anderson (2011).

► REFERENCES

- Anderson, Michael, and Susan Leigh Anderson, eds. 2011. *Machine Ethics*. Cambridge: Cambridge University Press.
- Barger, Robert N. 2003. "Can We Find a Single Ethical Code?" In M. David Ermann and Michele S. Shauf, eds. *Computers, Ethics, and Society*. New York: Oxford University Press, pp. 42–7.
- Barger, Robert N. 2008. *Computer Ethics: A Case-Based Approach*. New York: Cambridge University Press.
- Boatright, John R. 2000. *Ethics and the Conduct of Business*. 3rd ed. Upper Saddle River, NJ: Prentice Hall.
- Bok, Sisela. 2003. "The Morality of Whistle Blowing." In M. David Ermann and Michele S. Shauf, eds. *Computers, Ethics, and Society*. 3rd ed. New York: Oxford University Press, pp. 47–54.
- Bowyer, Kevin, ed. 2001. *Ethics and Computing: Living Responsibly in a Computerized World*. 2nd ed. New York: IEEE Press.
- Brenkert, George G. 2010. "Whistle-Blowing, Moral Integrity, and Organizational Ethics." In George G. Brenkert and Tom L. Beauchamp, eds. *The Oxford Handbook of Business Ethics*. New York: Oxford University Press, pp. 563–601.
- Buchanan, Elizabeth A. 2004. "Ethical Considerations for the Information Professions." In R. A. Spinello and H. T. Tavani, eds. *Readings in CyberEthics*. 2nd ed. Sudbury, MA: Jones and Bartlett Publishers, pp. 613–24.
- Bynum, Terrell Ward, and Simon Rogerson, eds. 2004. *Computer Ethics and Professional Responsibility*. Malden, MA: Blackwell.
- Davis, Michael. 1995. "Thinking Like an Engineer." In D. G. Johnson and H. Nissenbaum, eds. *Computing Ethics and Social Values*. Englewood Cliffs, NJ: Prentice Hall, pp. 586–97.
- Davis, Michael. 2015a. "Profession and Professionalism." In J. Britt Holbrook and Carl Mitcham, eds. *Ethics, Science, Technology, and Engineering: A Global Resource*. Vol. 3, 2nd ed. Farmington Hills, MI: Macmillan Reference, pp. 489–93.
- Davis, Michael. 2015b. "Codes of Ethics." In J. Britt Holbrook and Carl Mitcham, eds. *Ethics, Science, Technology, and Engineering: A Global Resource*. Vol. 1, 2nd ed. Farmington Hills, MI: Macmillan Reference, pp. 283–9.
- De George, Richard T. 1999. "Ethical Responsibilities of Engineers in Large Corporations: The Pinto Case." In D. G. Johnson, ed. *Ethical Issues in Engineering*. Englewood Cliffs, NJ: Prentice Hall, pp. 175–86.
- De George, Richard T. 2010. *Business Ethics*. 7th ed. Upper Saddle River, NJ: Pearson Prentice-Hall.
- Democracynow.org. 2014. "Debate: Was Snowden Justified? Former NSA Counsel Stewart Baker vs. Whistleblower Daniel Ellsberg." Available at http://www.democracynow.org/2014/2/14/debate_was_snowden_justified_former_nsa#. Accessed 3/19/14.
- Esposito, Richard, Matthew Cole, and Mark Schone. 2014. "Exclusive, Edward Snowden Gives Wide-Ranging Interview to Brian Williams." <http://www.nbcnews.com/storyline/nsa-snooping/exclusive-edward-snowden-gives-wide-ranging-interview-brian-williams-n110351>. Retrieved July 6.
- Fairweather, N. Ben. 2004. "No PAPA: Why Incomplete Codes of Ethics are Worse than None at All." In T. W. Bynum and S. Rogerson, eds. *Computer Ethics and Professional Responsibility*. Malden, MA: Blackwell, pp. 142–56.

- Finn, Peter and Sari Horwitz. 2013. "U.S. Charges Snowden with Espionage." *The Washington Post*, June 21. Available at http://www.washingtonpost.com/world/national-security/us-charges-snowden-with-espionage/2013/06/21/507497d8-dab1-11e2-a016-92547bf094cc_story.html. Accessed 1/19/15.
- Gotterbarn, Don. 1995. "Computer Ethics: Responsibility Regained." In D. G. Johnson and H. Nissenbaum, eds. *Computers, Ethics and Social Values*. Englewood Cliffs, NJ: Prentice Hall, pp. 18–24.
- Gotterbarn, Don. 2000. "Computer Professionals and YOUR Responsibilities." In D. Langford, ed. *Internet Ethics*. New York: St. Martin's Press, pp. 200–19.
- Gotterbarn, Don. 2001. "The Ethical Software Engineer." In K. W. Bowyer, ed. *Ethics and Computing: Living Responsibly in a Computerized World*. 2nd ed. New York: IEEE Press, p. 67.
- Gotterbarn, Don and Keith Miller. 2009. "The Public is Priority: Making Decisions Using the Software Code of Ethics." *IEEE Computer* 42, no. 6: 66–73.
- Gotterbarn, Don, Keith Miller, and Simon Rogerson. 1999. "Software Engineering Code of Ethics Approved." *Communications of the ACM* 42, no. 10: 102–7.
- Griggs, Brandon and Doug Gross. 2014. "Edward Snowden Speaks at SXSW, Calls for Public Oversight of U.S. Spy Programs." Available at <http://www.cnn.com/2014/03/10/tech/web/edward-snowden-sxsw/>. Accessed 1/19/15.
- Harris, Charles E., Michael S. Pritchard, and Michael J. Rabins. 2009. *Engineering Ethics: Concepts and Cases*. 4th ed. Belmont, CA: Wadsworth.
- Hinman, Lawrence M. 2005. "Esse Est Indicato in Google: Ethical and Political Issues in Search Engines." *International Review of Information Ethics* 3: 19–25.
- Ladd, John. 1995. "The Quest for a Code of Professional Ethics: An Intellectual and Moral Confusion." In D. G. Johnson and H. Nissenbaum, eds. *Computers, Ethics and Social Values*. Englewood Cliffs, NJ: Prentice Hall, pp. 580–5.
- Leveson, Nancy G. and Clark S. Turner. 2001. "An Investigation of the Therac-25 Accidents." In K. W. Bowyer, ed. *Ethics and Computing: Living Responsibly in a Computerized World*. 2nd ed. New York: IEEE Press, pp. 200–24. Reprinted from *IEEE Computer* 26, no. 7 (1993): 18–41.
- Lin, Patrick. 2012. "Introduction to Robot Ethics." In P. Lin, K. Abney, and G. Bekey, eds. *Robot Ethics: The Ethical and Social Implications of Robotics*. Cambridge, MA: MIT Press, pp. 3–16.
- Luegenbiehl, Heinz. 2015. "Whistleblowing." In J. Britt Holbrook and Carl Mitcham, eds. *Ethics, Science, Technology, and Engineering: A Global Resource*. Vol. 4, 2nd ed. Farmington Hills, MI: Macmillan Reference, pp. 529–31.
- Martin, Michael W. and Roland Schinzinger. 2004. *Ethics in Engineering*. 4th ed. New York: McGraw-Hill.
- Mitcham, Carl. 1997. "Engineering Design Research and Social Responsibility." In K. Schrader-Frechette and L. Westra, eds. *Technology and Values*. Lanham, MD: Rowman and Littlefield, pp. 261–78.
- Nissenbaum, Helen. 2007. "Computing and Accountability." In J. Weckert, ed. *Computer Ethics*. Aldershot, UK: Ashgate, pp. 273–80. Reprinted from *Communications of the ACM* 37 (1994): 37–40.
- Parnas, David Lorge. 1990. "Professional Responsibility to Blow the Whistle on SDI." In M. D. Ermann, M. B. Williams, and C. Guiterrez, eds. *Computers, Ethics, and Society*. New York: Oxford University Press, pp. 359–72.
- Perlman, Bruce and Roli Varma. 2002. "Improving Engineering Practice." *IEEE Technology and Society* 21, no. 1: 40–7.
- Rieder, Rem. 2013. "Snowden's NSA Bombshell Sparks Debate." *USA Today*, June 13. Available at <http://www.usatoday.com/story/money/columnist/rieder/2013/06/12/rem-rieder-surveillance/2415753/>. Accessed 1/19/15.
- Shachtman, Noah. 2007. "Robotic Cannon Kills 9, Wounds 14." *Wired*, 10.18.07. Available at <http://www.wired.com/2007/10/robot-cannon-ki/>.
- Snowden, Edward. 2013. "Statement to Human Rights Group (in Full)." *The Daily Telegraph*, July 12. Available at <http://www.telegraph.co.uk/news/worldnews/europe/russia/10176529/Edward-Snowdens-statement-to-human-rights-groups-in-full.html>. Accessed 11/30/13.
- Tavani, Herman T. 2012. "Search Engines and Ethics." *Stanford Encyclopedia of Philosophy*. Available at <http://plato.stanford.edu/entries/ethics-search/>.
- Tavani, Herman T. and Frances Grodzinsky. 2014. "Trust, Betrayal, and Whistle-Blowing: Reflections on the Edward Snowden Case." *Computers and Society* 44, no. 3: 8–13.
- The Royal Academy of Engineering. 2009. *Autonomous Systems: Social, Legal and Ethical Issues*. London, UK. Available at www.raeng.org.uk/autonomoussystem.
- Wallach, Wendell and Colin Allen. 2009. *Moral Machines: Teaching Robots Right from Wrong*. New York: Oxford University Press.
- Yurcik, William and David Doss. 2002. "Software Technology Issues for a U.S. National Missile Defense System." *IEEE Technology and Society* 21, no. 2: 36–46.

► FURTHER READINGS

- Allen, Colleen, and Wendell Wallach. 2012. "Moral Machines: Contradiction in Terms or Abdication of Human Responsibility." In P. Lin, K. Abney, and G. Bekey, eds. *Robot Ethics: The Ethical and Social Implications of Robotics*. Cambridge, MA: MIT Press, pp. 55–68.
- Birsch, Douglas. 2004. "Moral Responsibility for Harm Caused by Computer System Failures." *Ethics and Information Technology* 6, no. 4: 233–45.
- Buchanan, Elizabeth A. and Kathrine A. Henderson. 2009. *Case Studies in Library and Information Science Ethics*. Jefferson, NC: McFarland.
- Friedman, Batya, Peter H. Kahn, Jr., and Alan Borning. 2008. "Value Sensitive Design and Information Systems." In K. E. Himma and H. T. Tavani, eds. *The Handbook of Information and Computer Ethics*. Hoboken, NJ: John Wiley and Sons, pp. 69–101.
- Moor, James H. 2000. "If Aristotle Were a Computing Professional." In R. Baird, R. Ramsower, and S. E. Rosenbaum, eds. *Cyberethics: Social and Moral Issues in the Computer Age*. Amherst, NY: Prometheus, pp. 34–40.
- Rashid, Awais, John Weckert, and Richard Lucas. 2009. "Software Engineering Ethics in a Digital World." *IEEE Computer* 42, no. 6: 34–41.

Privacy and Cyberspace

LEARNING OBJECTIVES

Upon completing this chapter, you will successfully be able to:

- Describe the ways in which personal privacy is threatened in the digital era and determine whether cybertechnology has introduced any new or unique privacy issues,
- Explain why *privacy* is a difficult concept to define and describe the key elements of a comprehensive privacy theory that helps us to distinguish between a *loss* of privacy and a *violation* of privacy,
- Explain why privacy is valued and why it is an important social value as well as an individual good,
- Describe how one's privacy is impacted by *data-gathering* techniques such as RFID technologies and (Internet) cookies,
- Assess privacy-related concerns generated by *data-analysis* techniques involving Big Data, such as data mining and Web mining,
- Explain what is meant by the problem of protecting “privacy in public,” especially in light of challenges posed by routine uses of online search facilities such as Google and social networking sites such as Facebook,
- Evaluate the debate between proponents of stronger privacy legislation and those who advocate for industry self-regulation practices as an alternative,
- Describe and assess the arguments for whether people should, in certain cases, have a right to have online personal information about them “erased,” or at least “delinked” from search engine indexes.

In this chapter, we examine a wide range of privacy issues, including concerns that affect the day-to-day activities of ordinary individuals carried out in both online and offline contexts. We begin by reflecting on a scenario that illustrates some privacy concerns surrounding the opening of a controversial data center by the National Security Agency (NSA) in 2014.

► SCENARIO 5–1: A New NSA Data Center

The NSA, officially formed in 1952 and headquartered at Ft. Meade, Maryland, is one of the largest intelligence organizations in the United States (and in the world). Although the NSA's original charter was to conduct surveillance on “foreign” sources—that is, non-U.S. governments, organizations, and individuals—many critics, including reputable journalists, claim that the agency's mission has since been significantly expanded to include surveillance on American citizens as well. Sensitive documents that Edward

Snowden leaked to the press in May 2013 allegedly revealed some of the controversial surveillance techniques the NSA uses to collect data on U.S. citizens. These revelations have resulted in increased public awareness of the NSA, as well as closer media scrutiny of that organization's activities, especially in light of some embarrassing details about foreign government officials that were included in the leaked documents. (See Scenario 4–2 in Chapter 4 for more detail about controversies resulting from the documents leaked by Snowden.)

In the aftermath of the Snowden leaks, the NSA not only continued to defend its activities but has also recently opened a controversial data center, or “data farm” as some refer to it, in Utah. The new data center, which is able to gather significantly more information than was previously possible, reportedly collects phone records, text messages, e-mails, and other forms of electronic data. Privacy advocates in the United States are concerned that the new center also provides NSA with enhanced tools to analyze much of the electronic data generated by ordinary American citizens. Many NSA critics believe that the organization would have collected more data in the past but that it was hampered from doing so by limitations affecting both (i) storage capacity and (ii) the technology-related resources needed to conduct more extensive searches. But the new NSA data center (whose contents are officially classified) is reported to have 100,000 square feet of computers (to harvest data) and five zettabytes of storage capacity to house it.¹ So, critics worry that the NSA's activities now pose an even greater threat than before to the privacy of ordinary American citizens. ■

Are privacy advocates justified in their concerns about the NSA's increased ability to “spy” on American citizens and to collect vast amounts of data about them? The NSA has argued that its surveillance and data-collection techniques simply follow the organization's charter, which is to keep Americans safe. That organization's defenders suggest that in an era of global terrorism, U.S. citizens should be less concerned about their privacy and more worried about their safety. Arguably, this tension underlies one of the crucial dilemmas facing American citizens, as well as citizens in many countries around the world—namely, how much individual privacy should be we willing to trade off for increased security? But is the dichotomy between privacy and security, as articulated here, a “genuine dilemma”?² (Recall our analysis of the False Dichotomy/Either–Or fallacy in Chapter 3.) We examine key issues at the heart of the privacy-versus-security debate in detail in Chapter 6, and we briefly describe some concerns affecting governmental surveillance and data-collection techniques in Section 5.4.4.

The main purpose of Scenario 5–1 has been to get us to begin thinking about the value of privacy in general, especially in light of the serious challenges it faces in the second decade of the twenty-first century. For example, we will see that our privacy is threatened not only by governmental surveillance but by many of the data-collection and data-analysis techniques currently used in the commercial sector as well. This is especially apparent in the case of major search engine companies like Google, which collect vast amounts of personal information on ordinary users. So, our privacy is currently threatened on many different levels and in many different sectors.

► 5.1 PRIVACY IN THE DIGITAL AGE: *WHO* IS AFFECTED AND WHY SHOULD WE WORRY?

Of all the ethical and social issues associated with the digital era, perhaps none has received more media attention than concern about the loss of personal privacy. As we shall see, however, cybertechnology is not the first technology to threaten personal privacy. Nevertheless, that technology now threatens privacy in ways that were not previously possible.

5.1.1 Whose Privacy Is Threatened by Cybertechnology?

Virtually no one living today in a developed nation is exempt or immune from some kind of cybertechnology-related privacy threat. In fact, people who have never owned or never even used a networked computer are still affected. Consider that in carrying out many of our day-to-day activities, we supply information to organizations that use computers and electronic devices to record, store, and exchange those data. These activities can include information we provide in filling out various forms, or they can include information acquired from our commercial transactions in a bank or a store.

Some might assume that using the Internet only for noncommercial activities will help them to avoid many of the typical privacy threats associated with the online world, such as privacy risks associated with shopping at e-commerce sites. However, even users who navigate the Web solely for recreational purposes are at risk with respect to their privacy. For example, if you use a major search engine (such as Google) or interact in a social media site (such as Facebook), personal data about your interests and preferences can be acquired by those organizations, whose methods for collecting this information are not always obvious to ordinary users. Furthermore, personal data about us collected via our online activities can be sold to third parties.

Also consider that applications such as Google Street View (a feature of Google Earth and Google Maps) make use of satellite cameras and GPS software that enable Internet users to zoom in on your house or place of employment and potentially record information about you. Additionally, closed-circuit televisions (CCTVs) located in public places and in shopping malls record many of your daily movements as you casually stroll through those environments. So even if you have never owned or used a computer, cell phone, (Internet-enabled) electronic device, and so forth, your privacy is threatened in ways that were not possible in the past.

Concerns about privacy now affect many aspects of one's life—from commerce to health-care to work to recreation. So, some analysts organize these concerns into categories such as consumer privacy, medical/healthcare privacy, employee/workplace privacy, etc. Although some cyber-related privacy concerns are specific to one or more spheres or sectors—that is, employment, healthcare, and so forth—others cut across multiple dimensions of our lives and thus affect each of us regardless of our employment or health status.

Unfortunately, we cannot examine all of the current privacy concerns in a single chapter. So, we will have to postpone our analysis of certain kinds of privacy issues until the later chapters in the book. For example, we will examine some cyber-related privacy concerns that conflict with cybersecurity issues and national security interests in Chapter 6, where privacy-related concerns affecting “cloud computing” are also considered. Some specific employee/workplace privacy-related issues are examined in our discussion of workplace surveillance and employee monitoring in Chapter 10. And in our analysis of emerging and converging technologies (such as nanotechnology and ambient intelligence) in Chapter 12, we examine some controversies surrounding a relatively new category of privacy called “location privacy.”

Despite the significant impact that cybertechnology has had and continues to have for our privacy, one still might ask: Do any of the current privacy concerns differ in kind from privacy issues that arose in the predigital era? In other words, is there anything new, or even special, about cyber-related privacy issues? We next propose a strategy for addressing that question.

5.1.2 Are Any Privacy Concerns Generated by Cybertechnology Unique or Special?

We begin by noting that concerns about personal privacy existed long before the advent of computers and cybertechnology. Prior to the digital era, for example, technologies such as the camera and the telephone presented challenges for privacy. So we can ask: What, if anything,

is special about the privacy concerns that are associated with cybertechnology? Consider the impact that changes involving this technology have had on privacy with respect to the:

- Amount of personal information that can be collected
- Speed at which personal information can be transmitted
- Duration of time that the information can be retained
- Kind of information that can be acquired and exchanged

Cybertechnology makes it possible to collect and store much more information about individuals than was possible in the predigital era. The *amount* of personal information that could be collected in that era was determined by practical considerations, such as the physical space required to store the data and the time and difficulty involved in collecting the data. Today, of course, digitized information that can be stored electronically in computer databases takes up very little storage space and can be collected with relative ease. As we will see in Section 5.5, many people are now worried about the sheer volume of personal data that can be collected and analyzed by various techniques associated with “big data” and data mining.

Next, consider the *speed* at which information is exchanged and transferred between databases. At one time, records had to be physically transported between filing destinations; the time it took to move them depended upon the transportation systems—for example, motor vehicles, trains, airplanes, and so forth—that carried the records. Now, of course, records can be transferred between electronic databases in milliseconds through wireless technologies, high-speed cable lines, or even ordinary telephone lines.

With so much information being collected and transferred so rapidly, many have expressed concerns about its accuracy as well as the difficulties in tracking down and correcting any inaccuracies that might have been transferred. In an interview conducted for the BBC TV series *The Machine that Changed the World*, Harvard law professor Arthur Miller points out that trying to correct such information is like “chasing a greased pig”—you may get your hands on the pig, but it is very difficult to keep the pig firmly in your grip.³ Although issues concerning the accuracy of personal information are clearly distinguishable from those concerning privacy per se, accuracy issues are frequently associated with privacy issues, and both are impacted by cybertechnology.

Also, consider the *duration* of information—that is, how long information can be kept. Before the digital era, information was manually recorded and stored in file cabinets and then in large physical repositories; it is unlikely that report cards my parents received as high school students still exist somewhere as physical records in file cabinets, for at that time report cards were not computerized but instead existed, literally, as ink marks on paper. But the report cards my daughter received when she was a high school student were both generated and stored using computer technology. As an electronic record, her report card can be kept indefinitely, and the grades she received as a high school student (as well as the grades she received in elementary school and in college) can follow her throughout her life.

In the past, practices involving the retention of personal data were perhaps more “forgiving.” Because of practical limitations, such as physical storage space, that affected how long personal data could be kept on file, much of the personal information collected and stored had to be destroyed after a certain number of years. Since information could not be archived indefinitely, people with blemished records sometimes had the opportunity to start over again by physically relocating. Today, however, one’s electronic dossier would likely follow, making it very difficult, if not impossible, for that person to start over with a clean slate. We can argue whether the current means of data retention is a good thing, but it is difficult to dispute the claim that now, because of cybertechnology, most of us have what Arthur Miller calls a “womb-to-tomb dossier.” It is also worth noting, however, that a 2014 court ruling by the European Court of Justice (ECJ) gave citizens in European nations the right, in certain cases, to have

some kinds of online personal information about them deleted or “erased.” We examine this principle, commonly referred to as the right to “be forgotten” or “to erasure,” in detail in Section 5.8.

Cybertechnology has also generated privacy concerns because of the *kind* of personal information that can now be collected. For example, every time you engage in an electronic transaction, such as making a purchase with a credit card or withdrawing money from an ATM, transactional information is collected and stored in several computer databases; this information can then be transferred electronically across commercial networks to agencies that request it. Personal information, retrieved from transactional information that is stored in computer databases, has been used to construct electronic dossiers containing detailed information about an individual’s commercial transactions, including purchases made and places traveled—information that can reveal patterns in a person’s preferences and habits.

Additionally, we should note that cybertechnology raises privacy concerns because of the myriad ways in which it enables our personal information to be manipulated once it has been collected. For example, unrelated pieces of information about us that reside in separate databases can be combined to construct electronic personal dossiers or profiles. Also, information about us included in one database can be matched against records in other databases that contain information about us. Furthermore, our personal information can be mined (from databases, as well as from our activities on the Web) to reveal patterns in our behavior that would have been very difficult to discern in the predigital era. Of course, our personal data could have been, and in some instances was, manipulated in the predigital era as well. But there were practical limitations to the amount of data collection and analysis that could be done manually by humans.

Although the privacy concerns that we now associate with cybertechnology may not be totally new, or even altogether different in kind, from those we associate with earlier technologies, few would dispute the claim that cybertechnology has exacerbated them. In Sections 5.4–5.5, we examine some controversial uses of cybertechnology that raise specific concerns for personal privacy. First, however, we examine the concept of personal privacy to better understand what privacy is and why we value it.

► 5.2 WHAT IS PERSONAL PRIVACY?

Although many definitions have been put forth, there is no universally agreed upon definition. To understand why this concept has been so difficult to define, consider the diverse range of metaphors typically associated with privacy. Sometimes, we speak of privacy as something that can be lost or diminished, suggesting that privacy can be understood in terms of a repository of personal information that can be either diminished altogether or gradually eroded. Contrast this view with descriptions of privacy as something that can be intruded upon or invaded, where privacy can be understood in terms of a spatial metaphor, such as a zone, that deserves protection. Alternatively, privacy is sometimes described as something that can be violated or breached, when we think of it in terms of either a right or an interest that deserves legal protection. Because of these different conceptions of privacy, we will see that it is useful to distinguish between the notions of one’s having privacy (in a descriptive sense) and one’s having a (normative) right to privacy. We will say more about this distinction in Section 5.2.4.

Privacy analysts have pointed out that in the United States, the meaning of privacy has evolved since the eighteenth century. Initially, privacy was understood in terms of freedom from (physical) intrusion. Later, it became associated with freedom from interference into one’s personal affairs, including one’s ability to make decisions freely. Most recently,

privacy has come to be closely identified with concerns affecting access to and control of personal information—a view that is also referred to as “informational privacy.” Although the main emphasis in this chapter is on informational privacy, we also briefly describe the other two views.

5.2.1 Accessibility Privacy: Freedom from Unwarranted Intrusion

In a seminal paper on privacy, Samuel Warren and Louis Brandeis suggested that privacy could be understood as “being let alone” or “being free from intrusion.” Appearing in the *Harvard Law Review* in 1890, the Warren and Brandeis article made the first explicit reference to privacy as a legal right in the United States. Many Americans are astonished to find out that there is no explicit mention of privacy in either the Constitution or its first ten amendments, the Bill of Rights. However, some legal scholars believe that a right to privacy can be inferred from the Fourth Amendment, which protects citizens against unreasonable searches and seizures of personal affects (i.e., papers, artifacts, etc.) by the government. Some legal scholars suggest that the Fourth Amendment may also provide legal grounds for a right to privacy protection from nongovernmental intrusion as well.

Warren and Brandeis also suggested that our legal right to privacy is grounded in our “right to inviolate personality.” In part, they were responding to a certain use of a new technology—not the computer, of course, but rather the camera—which had begun to threaten individual privacy in new ways.⁴ Photographs of people began to appear in newspapers, for example, in gossip columns, along with stories that were defamatory and sometimes even false. Warren and Brandeis believed that individuals have a (legal) right not be intruded upon in this manner. Because this definition of privacy as freedom from unwarranted intrusion focuses on the harm that can be caused through physical access to a person or to a person’s possessions, DeCew (1997, 2006) and others have described this view as *accessibility privacy*.

5.2.2 Decisional Privacy: Freedom from Interference in One’s Personal Affairs

Privacy is also sometimes conceived of as freedom from interference in one’s personal choices, plans, and decisions; some refer to this view as decisional privacy. This kind of privacy has also been associated with reproductive technologies having to do with contraception. In *Griswold v. Connecticut* (1965), the court ruled that a person’s right to get counseling about contraceptive techniques could not be denied by state laws. The view of privacy as freedom from external interference into one’s personal affairs has since been appealed to in legal arguments in a series of controversial court cases, such as those involving abortion and euthanasia. For example, this view of privacy was appealed to in the landmark Supreme Court decision on abortion (*Roe v. Wade* 1973), as well as in a state court’s decision involving Karen Ann Quinlan’s right to be removed from life-support systems and thus her “right to die.”⁵ Because it focuses on one’s right not to be interfered with, decisional privacy can be distinguished from both accessibility privacy and informational privacy.

5.2.3 Informational Privacy: Control over the Flow of Personal Information

Because of the increasing use of technology to gather and exchange personal information, many contemporary analysts view privacy in connection with one’s ability to restrict access to and control the flow of one’s personal information. Privacy concerns are now often framed in terms of questions such as: Who should have access to one’s personal information? To what extent can individuals control the ways in which information about them can be gathered,

TABLE 5-1 Three Views of Privacy

Accessibility privacy	Privacy is defined as one's (physically) being let alone, or being free from intrusion into one's physical space
Decisional privacy	Privacy is defined as freedom from interference in one's choices and decisions
Informational privacy	Privacy is defined as control over the flow of one's personal information, including the ways in which that information is collected and exchanged

stored, mined, combined, recombined, exchanged, and sold? These are our primary concerns in this chapter, where we focus on informational privacy.

Table 5-1 summarizes the three views of privacy.

5.2.4 A Comprehensive Account of Privacy

James Moor has put forth a privacy framework that incorporates important elements of the nonintrusion, noninterference, and informational views of privacy. According to Moor,

An individual [has] privacy in a *situation* with regard to others if and only if in that situation the individual [is] protected from intrusion, interference, and information access by others.⁶

An important element in this definition is Moor's notion of "situation," which he deliberately leaves broad so that it can apply to a range of contexts, or zones, that can be "declared private." For example, a situation can be an "activity" or a "relationship," or it can be the "storage and access of information" in a computer (Moor 2000).

Central to Moor's theory is a distinction between naturally private and normatively private situations, enabling us to differentiate between the conditions required for (i) having privacy and (ii) having a right to privacy. This distinction, in turn, enables us to differentiate between a loss of privacy and a violation of privacy. In a naturally private situation, individuals are protected from access and interference from others by natural means, for example, physical boundaries such as those one enjoys while hiking alone in the woods. In this case, privacy can be lost but not violated, because there are no norms—conventional, legal, or ethical—according to which one has a right, or even an expectation, to be protected. In a normatively private situation, on the other hand, individuals are protected by conventional norms (e.g., formal laws and informal policies) because they involve certain kinds of zones or contexts that we have determined to need normative protection. The following two scenarios will help us to differentiate between normative and natural (or descriptive) privacy.

► SCENARIO 5-2: Descriptive Privacy

Mary enters her university's computer lab at 11:00 P.M. to work on a research paper that is due the next day. No one else is in the lab at the time that Mary arrives there, and no one enters the lab until 11:45 P.M., when Tom—the computer lab coordinator—returns to close the lab for the evening. As Tom enters, he sees Mary typing on one of the desktop computers in the lab. Mary seems startled as she looks up from her computer and discovers that Tom is gazing at her. ■

Did Mary lose her privacy when Tom entered the lab and saw her? Was her privacy violated? Before Tom noticed her in the lab, we could say that Mary had privacy in the descriptive, or natural, sense of the term because no one was physically observing her while she was in the lab. When Tom entered and noticed that Mary was typing on a computer, Mary lost her natural (or descriptive) privacy in that situation. However, we should not infer that her privacy was violated in this incident, because a university's computer lab is not the kind of situation or zone that is declared normatively private and thus protected.

► **SCENARIO 5-3: Normative Privacy**

Tom decides to follow Mary, from a distance, as she leaves the computer lab to return to her (off-campus) apartment. He carefully follows her to the apartment building, and then stealthily follows Mary up the stairway to the corridor leading to her apartment. Once Mary is safely inside her apartment, Tom peeps through a keyhole in the door. He observes Mary as she interacts with her laptop computer in her apartment. ■

Has Mary's privacy been violated in this scenario? In both scenarios, Tom observes Mary interacting with a computer. In the first scenario, the observation occurred in a public place. There, Mary may have lost some privacy in a descriptive or natural sense, but she had no expectation of preserving her privacy in that particular situation. In the second scenario, Mary not only lost her privacy but her privacy was violated as well, because apartments are examples of zones or "situations" that we, as a society, have declared normatively private.

We have explicit rules governing these situations with respect to privacy protection. Note that it was not merely the fact that Tom had observed Mary's interactions with a computer that resulted in her privacy being violated in the second scenario. Rather, it was because Tom had observed her doing this in a normatively protected situation. So, there was nothing in the information per se that Tom acquired about Mary that threatened her privacy; it was the situation or context in which information about Mary was acquired that caused her privacy to be violated in the second scenario.

5.2.5 Privacy as "Contextual Integrity"

We have seen the important role that a situation, or context, plays in Moor's privacy theory.⁷ But some critics argue that the meaning of a situation or context is either too broad or too vague. Helen Nissenbaum elaborates on the notion of a context in her model of privacy as "contextual integrity," where she links adequate privacy protection to "norms of specific contexts." She notes that the things we do, including the transactions and events that occur in our daily lives, all take place in some context or other. In her scheme, contexts include "spheres of life" such as education, politics, the marketplace, and so forth (Nissenbaum 2004a, 2010).

Nissenbaum's privacy framework requires that the processes used in gathering and disseminating information (i) are "appropriate to a particular context" and (ii) comply with norms that govern the flow of personal information in a given context.⁸ She refers to these two types of informational norms as follows:

1. Norms of appropriateness
2. Norms of distribution

Whereas norms of appropriateness determine whether a given type of personal information is either appropriate or inappropriate to divulge within a particular context, norms of distribution restrict or limit the flow of information within and across contexts. When either norm has been "breached," a violation of privacy occurs; conversely, the contextual integrity of the flow of personal information is maintained when both kinds of norms are "respected."⁹

As in the case of Moor's privacy model, Nissenbaum's theory demonstrates why we must always attend to the context in which information flows, and not to the nature of the information itself, in determining whether normative protection is needed. To illustrate some of the nuances in her framework of privacy as contextual integrity, consider the following scenario in which a professor collects information about students in his seminar.

► **SCENARIO 5-4: Preserving Contextual Integrity in a University Seminar**

Professor Roberts teaches a seminar on social issues in computing to upper-division, undergraduate students at his university. Approximately half of the students who enroll in his seminar each semester are computer science (CS) students, whereas the other half are students majoring in humanities, education, business, etc. At the first class meeting for each seminar, Professor Roberts asks students to fill out an index card on which they include information about their major, their year of study (junior, senior, etc.), the names of any previous CS courses they may have taken (if they are non-CS majors), their preferred e-mail address, and what they hope to acquire from the seminar. Professor Roberts then records this information in his electronic grade book. ■

Has Professor Roberts done anything wrong in requesting and collecting this information? For the most part, it is information that he could have gathered from the registrar's office at his university—for example, information about which CS courses and which general education courses the students have previously taken, and so forth. But Roberts finds it much more convenient to collect information in the classroom, and he informs the students that he uses that information in determining which kinds of assignments he will decide to give to the class in general and which kinds of criteria he will use to assign students to various group projects.

Because Professor Roberts has informed the students about how the information they provided to him will be used in the context of the classroom, and because the students have consented to give him the information, no privacy violation seems to have occurred. In fact, the process used by Professor Roberts satisfies the conditions for Nissenbaum's norm of appropriateness with respect to contextual integrity.

Next, suppose that Professor Roberts has lunch a few weeks later with a former student of his, Phil, who recently graduated and now has a job as a software engineer for a publishing company. Phil's company plans to release its first issue of a new magazine aimed at recent CS graduates, and it has launched an advertising campaign designed to attract undergraduate CS majors who will soon graduate. Phil asks Professor Roberts for the names of the CS majors in the seminar he is teaching. Professor Roberts is initially inclined to identify some students that Phil would likely know from classes that he had taken the previous year at the university. But should Professor Roberts reveal those names to Phil?

If he did, Professor Roberts would violate the privacy norm of distribution within the context of the seminar he is teaching. Consider that the students gave information about themselves to Professor Roberts for use in the context of that seminar. While his use of that information for purposes of the seminar is context appropriate, passing on (i.e., distributing) any of that information to Phil is not, because it would violate the integrity of that context. Even though the information about the students that Professor Roberts has collected is neither sensitive nor confidential in nature, it was given to him for use only in the context of the seminar he is teaching. Insofar as Professor Roberts uses the information in that context, he preserves its integrity. But if he elects to distribute the information outside that context, he violates its integrity and breaches the privacy of his students.

► **5.3 WHY IS PRIVACY IMPORTANT?**

Of what value is privacy? Why does privacy matter and why should we care about it? In 1999, Scott McNealy, then CEO of Sun Microsystems, uttered his now famous remark to a group of reporters: "You have zero privacy anyway. Get over it." And in 2013, Facebook CEO Mark Zuckerberg proclaimed that privacy "is no longer a social norm." So, should we infer from these remarks that the idea of personal privacy is merely a relic of the past? Although Froomkin (2000), Garfinkel (2000), and others speak of the "death of privacy," not everyone has been

willing to concede defeat in the battle over privacy. Some privacy advocates staunchly believe that we should be vigilant about retaining what little privacy we may still have. Others note that we do not appreciate the value of privacy until we lose it, and by then, it is usually too late. They point out that once privacy has been lost, it is difficult, if not impossible, to get back. So perhaps, we should heed their warnings and try to protect privacy to the degree that we can.

We might also question whether the current privacy debate needs to be better understood in terms of differences that reflect generational attitudes. For many so-called Millennials, who are now college aged, privacy does not always seem to be of paramount importance. Most Millennials, as well as many members of Generations X and Y, seem all too eager to share their personal information widely on social networking services such as Facebook, and many also seem willing to post “away messages” on AIM or Skype that disclose their whereabouts at a given moment to a wide range of people. But for many older Americans, including Baby Boomers, privacy is something that is generally still valued. So the relative importance of privacy may vary considerably among the generations; however, we will proceed on the assumption that privacy has value and thus is important.

Is privacy universally valued? Or is it valued mainly in Western, industrialized societies where greater importance is placed on the individual? Solove (2008) notes that privacy is a “global concern,” which suggests that it is valued universally. However, it has also been argued that some non-Western nations and cultures do not value individual privacy as much as we do in the West. Alan Westin believes that countries with strong democratic political institutions consider privacy more important than do less democratic ones.¹⁰ Nations such as Singapore and the People’s Republic of China seem to place less importance on individual privacy and greater significance on broader social values, which are perceived to benefit the state’s community objectives. Even in countries such as Israel, with strong democratic systems but an even stronger priority for national security, individual privacy may not be as important a value as it is in most democratic nations. So, even though privacy has at least some universal appeal, it is not valued to the same degree in all nations and cultures. As a result, it may be difficult to get universal agreement on privacy laws and policies in cyberspace.

5.3.1 Is Privacy an Intrinsic Value?

Is privacy something that is valued for its own sake—that is, does it have intrinsic value? Or is it valued as a means to an end, in which case it has only instrumental worth? Recall our discussion of intrinsic and instrumental values in Chapter 2. There, we saw that happiness has intrinsic value because it is desired for its own sake. Money, on the other hand, has instrumental value since it is desired as a means to some further end or ends.

While few would argue that privacy is an intrinsic value, desired for its own sake, others, including Fried (1990), argue that privacy is not merely an instrumental value or instrumental good. Fried suggests that unlike most instrumental values that are simply one means among others for achieving a desired end, privacy is also essential, that is, necessary to achieve some important human ends, such as trust and friendship. We tend to associate intrinsic values with necessary conditions and instrumental values with contingent, or non-necessary conditions; so while privacy is instrumental in that it is a means to certain human ends, Fried argues that it is also a necessary condition for achieving those ends. Solove also believes that privacy has aspects that cut across the intrinsic-instrumental divide, and he argues “intrinsic and instrumental value need not be mutually exclusive.”¹¹

Although agreeing with Fried’s claim that privacy is more than merely an instrumental value, and with Solove’s insight that privacy is a value that spans the intrinsic-instrumental divide, Moor (2004) takes a different approach to illustrate this point. Like Fried, Moor argues that privacy itself is not an intrinsic value. But Moor also believes that privacy is an articulation, or “expression” of the “core value” security, which in turn is essential across cultures, for